BayOff: A feasibility study on the application of Bayesian methods of inference to the treatment and analysis of Official Statistics data

The BayOff team [1]
Final Report

November 2000

[1]Supported by EUROSTAT under program SUPCOM 98-Lot 8

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Preface

BayOff is the acronym for **Bayesian Techniques in Official Statistics**, a research project under the framework of the *SUPCOM Calls for Tenders* (OJEC 98/s 119-76952/EN, SUPCOM 98-Lot 8) of the European Commission. The objective of BayOff was to carry out a prospective study on the feasibility of applying Bayesian methods and techniques to the problems encountered in Official Statistics.

The Bayesian approach to inference and decision making has been traditionally criticised for the heavy computational demands it places, the dependence on subjective inputs and its descriptive inadequacies, among other reasons. However, recent developments in areas such as Bayesian computation, specially Markov chain Monte Carlo simulation, prior elicitation, hierarchical modelling, nonlinear modelling, dynamic modelling and sensitivity analysis, together with the advent of cheap computational power has put at the hands of statistical practicioners an impressive range of tools, to the point that in complex statistical problems, the Bayesian approach may be the only feasible one. Indeed the range of applications of Bayesian methods are expanding enormously from image processing to financial forecasting, medical decision making and energy demand modelling.

Interestingly enough, the impact of these methods is yet to be seen within Official Statistics. Partly for that reason, the BayOff project was started. This study might have influence in at least two directions: the adoption by European Union Statistical Offices of Bayesian methods and the promotion of Bayesian research in the European Union.

Specific tasks comprised in BayOff are:

- Stage I
  - Research to follow up recent references in journals and proceedings.
  - Contacts with leading figures in Bayesian Statistics.
  - Identification of interesting applications in the field of practical data analysis.
  - Identification of existing software.
  - Setting up of a Web site on the subject
- Stage II

- Contacts with National Statistical Institutes (NSIs) and Eurostat to communicate Stage I results.
- Identification of possible applications in NSIs and Eurostat.
- Assessment of the willingness of the NSIs and Eurostat to apply Bayesian techniques.
- Planning of a future strategy to apply this work to Official Statistics.

A team ('the BayOff team') lead by David Rios Insua (U. Rey Juan Carlos, Madrid, Spain) has carried out the study. Other members of the research team are:

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This report describes the findings of the project and is structured in four chapters and six appendices. The first chapter provides a state-of-the-art survey of the field; after a brief introduction to Bayesian analysis, we summarise a literature search and describe a questionnaire sent to some experts in the field, both from Academia and NSIs, together with important recent developments. The current state may be briefly summarised as follows: *although Bayesian Analysis has experienced enormous growth in the last few years, it has permeated very little the field of Official Statistics.*

The second chapter tries to explain such status by describing generic criticisms (and counterarguments) to Bayesian analysis, specific criticisms when applied to official data and, finally, potential advantages of using Bayesian methods in Official Statistics. Chapter 3 further expands on such advantages by describing important developments for these activities: Bayesian sampling; non-informative priors; Bayesian computation and Bayesian model selection. We then illustrate some ideas and advantages with several small relevant examples: unemployment forecasting (through dynamic linear models), mortality prediction (through nonlinear models), industrial accidentability prediction (through hierarchical models) and institutional evaluation (through generalised linear models). Chapter four describes the challenge ahead, summarising conclusions and outlining a strategy to foster the use of Bayesian methods in Official Statistics.

Appendix A describes the questionnaire sent to academic statisticians and the detailed answers of experts consulted. Appendix B describes the questionnaire sent to official statisticians and their answers. Appendix C reviews

Bayesian software. Appendix D provides some views on the Web site under development related with BayOff. Appendix E includes slides of the presentation given at the International Society for Bayesian Analysis (ISBA) meeting, held at Herakleion (Crete), where the BayOff members had the opportunity to present the results before a privileged audience, within a special session on Official Statistics. Another appendix outlines possible methodological-applied projects that may enhance the adoption of Bayesian methods within NSIs. Finally, the last appendix provides relevant references following the NORIS classification.

We are grateful to the persons and institutions who have supported this study. It is then the occasion of acknowledging the institutional, technical and financial support of the Commission of the European Communities (EUROSTAT) and of many official and academic statisticians, in particular, of all the respondents to the various questionnaires submitted for their extremely clarifying views. Finally, we are grateful to the support of our students Luis Pena, Javi Vallejo, Manolo Urea, and Javi Palacian.

The BayOff Team

Madrid and Athens, November 2000

Chapter 1

Bayesian Analysis in Official Statistics

Statistical information is a key input to decision-making by the economic and social agents: Governments, firms, trade unions, but also households and individuals. Official Statistics are central to the Information Society. As Mr. Yves Franchet, Director General of Eurostat, stated

> *Statistics are vitally important for EU institutions, Member State governments and all those responsible for political economic and social decisions, as well as Europe's citizens, need factual information, both to check the advisability of European political initiatives and to monitor how projects are implemented. Statistics are essential to a democratic society.*

Academic research has produced a considerable amount of theoretical and practical advances in the field of decision-making. The Bayesian framework for Decision Analysis has been recognised as the scientific approach *par excellence*. Statistics are also linked to Decision Theory: making inferences about unknown quantities is a decision of the statistician. As a consequence, a Bayesian approach to Statistical Inference has been developed.

In this chapter, after providing an introductory review of Bayesian Analysis, we present questionnaires on its relations with Official Statistics sent to leading Bayesian statisticians and official statisticians. Recent developments in Bayesian Statistics with emphasis on issues related to Official Statistics are extracted from questionnaire answers and the results of a literature search.

1.1 An introduction to Bayesian analysis

The basic formalism of Bayesian analysis[1] does not take long to state. A *decision maker*, henceforth DM, has to choose a single action a from a set A of possible actions. Features that are unknown about the external world are parameterised by a state of nature, θ, which is known to lie in a set Θ of possible states. The consequence $c(a,\theta) \in C$ of her choice depends both on the action chosen and the unknown state of nature. Before choosing the action, she may observe an outcome $X = x$ of an experiment, which depends on the unknown state θ. Specifically, the observation X is drawn from a distribution $P_X(\cdot|\theta)$. The DM's objectives are encoded in a real valued loss function $l(a,\theta)$ which measures the loss of the consequence $c(a,\theta)$, i.e. in everyday terms it measures the value or worth of the consequence to her. Thus her problem is: observe $X = x$ and then choose and action $d(x) \in A$, using the information she has gathered from the observation $X = x$ to minimise, in some sense, $l(d(x),\theta)$. Since θ is unknown, this is not straightforward and several definitions of appropriate senses of minimisation have been offered. Here we describe the Bayesian solution.

This approach suggests encoding her prior knowledge of $\theta \in \Theta$ through her prior distribution $P_\theta(\cdot)$. This knowledge is updated through Bayes' Theorem in the light of the observation to give her posterior distribution (in terms of density functions):

$$p_\theta(\theta|x) \underset{\theta}{\propto} p_X(x|\theta) \times p_\theta(\theta).$$

Then, the DM should choose the action $d(x) = a$ to minimise her posterior expected loss:

$$\text{Choose} \quad d(x) = \underset{a}{\operatorname{argmin}} \left(E_\theta\left[l(a,\theta)|x\right]\right)$$

The optimal action is called Bayes action. We call Bayes rule to the function associating with each x the corresponding minimum expected loss alternative. Interestingly enough, most standard statistical problems may be framed according to the above approach. This does not mean that, in practice, we should follow that process, but that, conceptually, such framework allows us to treat in a coherent and unified manner most statistical problems.

We start with *point estimation* problems. We may describe them as statistical problems in which the spaces of alternatives and parameters coincid and the loss function is an increasing function of a certain distance between θ and a. The final aim is to provide *point estimators* of θ, which, somehow, summarise all information available about θ in one value. For example, under *quadratic loss function*, $l(\theta,a) = (\theta - a)^2$, the Bayes estimator is the posterior mean, assuming the posterior variance is finite. Similarly, under absolute loss $l(\theta,a) = |\theta - a|$, the Bayes estimator is the posterior median, assuming the posterior mean is

[1] We shall review here a decision analytic framework which allows the integration of both inference and decision making tasks.

finite. We may also justify the adoption of the posterior mode as a summary of the posterior distribution.

We turn now our attention to *set estimation.* A set estimator of θ is a function from the sample space to a collection of subsets in Θ. A natural approach to the problem is to specify a probability α and find a set with such posterior probability. However, there may be many sets with such property. Assume first that the space of alternatives is $\mathcal{A} = \{[\theta_1, \theta_2] : \theta_1 \leq \theta_2\}$. As loss function, we consider $L(\theta, [a_1, a_2]) = \theta_2 - \theta_1 + \begin{cases} 2/\alpha(a_1 - \theta) & \text{si } \theta < a_1 \\ 0 & \text{si } a_1 \leq \theta < a_2 \\ 2/\alpha(\theta - a_2) & \text{si } a_2 < \theta \end{cases}$

We then have that the Bayes rule is the interval between the $\alpha/2$ quantile and the $1 - \alpha/2$ quantile of the posterior distribution of θ. We may also justify the use of highest posterior regions from a decision theoretical point of view.

Consider now *hypothesis testing* problems. They may be naturally viewed as decision making problems. We assume here that we have to test two hypotheses. Suppose we partition Θ in two subsets Θ_0 and Θ_1. The statement $\theta \in \Theta_0$ is the hypothesis and $\theta \in \Theta_1$ is the alternative. The space of actions is $\{0, 1\}$, action $1(0)$ meaning rejection (acceptance) of the hypothesis. The loss function $l(\theta, a)$ will satisfy $l(\theta, 1) > l(\theta, 0)$ for $\theta \in \Theta_0$ and $l(\theta, 1) < l(\theta, 0)$ for $\theta \in \Theta_1$. If we reject H_0 when it is true, we incurr in a type I error; when we accept it, being false, we incurr in a type II error.

A classic example of loss function is the 0-1-c, for which

$$l(\theta, 1) = \begin{cases} c & \text{if } \theta \in \Theta_0 \\ 0 & \text{if } \theta \in \Theta_1 \end{cases}$$

$$l(\theta, 0) = \begin{cases} 0 & \text{if } \theta \in \Theta_0 \\ 1 & \text{if } \theta \in \Theta_1 \end{cases}$$

When $c = 1$, we have the 0-1 loss function. We have that the Bayes rule consists of choosing $a = 1$ when

$$Pr(\theta \in \Theta_0 | x) < \frac{1}{1 + c},$$

i.e., we reject the null when its posterior probability is small enough, specifically, when is smaller than $1/(1 + c)$. Observe that, when $c = 1$, the lower limit is $\frac{1}{2}$.

We consider, as a final stylised problem, that of prediction. We aim at forecasting one (or more) future observations of a quantity y of interest, possibly after having observed such quantity and/or other related quantities $x = (x_1, x_2, \ldots, x_n)$. Typically, we assume that y and x are conditionally independent given θ and we would use the predictive density

$$p(y|x) = \int p(y|\theta) p(\theta|x) d\theta$$

for forecasting. Such density may be later used to provide point or set forecasts and test hypotheses on future observations, much as we did before.

As we see, we may view most statistical problems under a common framework, which facilitates integration within a decision making structure. This report describes how we may or might apply this framework within the field of Official Statistics.

1.2 Questionnaires on Bayesian Analysis and Official Statistics

To put in context Bayesian Analysis within Official Statistics, we produced a couple of questionnaires trying to ascertain important recent developments in Bayesian Analysis, their actual and potential impacts on Official Statistics and possible ways of enhancing the role of Bayesian methods in Official Statistics. The first questionnaire was sent to academic statisticians; the second one, once we ellaborated the Stage I report, was sent to official statisticians.

In this Section, we present and justify such questionnaires. Detailed answers appear in Appendices A and B, whereas various comments on the answers appear all throughout this report.

1.2.1 The questionnaire for academic statisticians

The heading and questionnaire prepared were the following:

We have been asked by EUROSTAT (the European Statistical Office) to undertake a prospective study on the feasibility and potentiality of Bayesian methods in the field of Official Statistics (BayOff project). The study might have great impact in the adoption by European Union Statistical Offices of Bayesian methods (currently almost nil) and the promotion of Bayesian research in the European Union.

As part of the study, we have to interview some twelve-fourteen leading figures in the field on prospects and we would like you to be included within that group. If, as we hope, you are willing to take part, we shall be grateful if you could answer the enclosed questionnaire, if possible, within two weeks. Should you not be in a position of doing so, we would appreciate if you could suggest alternative leading figures that might take part in this survey.

With best wishes

QUESTIONNAIRE

1. Which are the most relevant developments within the field of Bayesian Statistics over the last ten years?

2. Which are, according to you, important open problems that Bayesian Statistics still needs to face?

3. Bayesian Analysis has had little impact within Official Statistics. Could you give reasons why?

4. Could you mention any important work concerning Bayesian Analysis within Official Statistics?

5. Could you describe why Bayesian Analysis could be relevant within Official Statistics?

6. Could you suggest ways in which Bayesian Analysis might permeate the field of Official Statistics?

7. May we include your name together with your answers, or do you prefer to keep them anonymous?

The questionnaire was e-mailed to twenty leading figures in the field of Bayesian Statistics, most of whom had had some experience within Official Statistics. Fifteen of them replied, as detailed in Appendix A.

The objective of the first two questions was to place Bayesian Analysis in context, i.e. identify important recent developments and open problems; questions 3 and 4 were aimed at placing Bayesian Analysis within the field of Official Statistics, i.e. identify reasons for its relatively little success and relevant applications; finally, questions 5 and 6 were aimed at identifying a strategy to promote the use of Bayesian Methods within Official Statistics.

1.2.2 The questionnaire for official statisticians

The second stage of the BayOff project started by submitting by email a brief note together with a questionnaire and some background to NSI representatives at the EPROS/DOSIS Working group. Eurostat provided the research team with the addresses of contact persons, selected by their participation in the EPROS/DOSIS Working Group on Research in Official Statistics. As other Working Groups of the European Statistical System, it is composed by representatives of the 15 EU Member States' National Statistical Institutes plus the EEA countries. Even if the profile of the representative is quite general, they can act as 'entry points' to the NSIs. Generally, they are aware of research and development activities in their respective NSI, as well as of the current Eurostat's policy of fostering R&D activities in the field of Official Statistics.

The contact persons received a questionnaire that we used to get a picture of Official Statisticians' view on the feasibility of applying Bayesian techniques in Official Statistics. To avoid response biases, the BayOff stage I report was not distributed at this point. The questionnaire, submitted electronically, introduced some background on the project, as reflected in Appendix B. It included the following six questions:

1. Are you familiar with the Bayesian methodology for Statistical Inference? If not, please forward this questionnaire to a colleague familiar with it.

2. Do you know whether your organisation has ever applied Bayesian methods in the production of statistics? If yes, in which areas and which methods?

3. In the case that you could not provide examples of use of Bayesian techniques in the production of official statistics, could you mention some possible reasons for it?

4. Are there in your organisation regular presentations of statistical methodology?

5. Is there any methodological or research unit that evaluates new techniques and methods to improve statistical estimations? Otherwise, how are new methods tested?

6. May we include your name together with your answers, or do you prefer to keep them anonymous?

The first question was aimed at ascertaining familiarity of Bayesian methods; the second one was aimed at finding out which Bayesian methods had been actually used at their Institutes, if any, and if not, they should explain in the third one the reasons for not using these methods. The fourth and fifth questions explored the possibility of spreading Bayesian methods through seminars at NSIs or collaborative work with their methodological units, whereas, finally, the sixth question referred to anonymity. Note that some of the questions had already been answered by leading figures of Bayesian Statistics.

The questionnaire were sent up to three times, and even a fourth time, during the second phase of the second stage, as discussed later.

1.3 Literature search: description

As a companion to the previous questionnaire, we charted the current state-of-the-art in the field through a literature search, based on the CIS database, over the period 1992-1997. We limited the search to some of the main journals, to wit:

1. Journal of the American Statistical Association.

2. Journal of the Royal Statistical Society, Series A, General.

3. Journal of the Royal Statistical Society, Series B, Methodological.

4. Statistical Science.

5. Biometrika.

6. Journal of Official Statistics.

7. International Statistical Review.

8. Scandinavian Journal of Statistics.

9. Statistics and Computing.

10. The American Statistician.

The search was also completed through various important conference proceedings, mainly the Valencia Bayesian Statistics series and the International Statistics Institute Sessions.

We initially conducted a generic search, which lead to a huge number of Bayesian entries in the mainstream statistical literature, well beyond the maximum legally allowed in the CIS contract. By using a keyword search, we identified some of the most active research areas within the field, in the last decade, which we corroborated with the answers of the experts. To further concentrate the search within official statistics, we used the NORIS terminology, as described in the appendix of the document *Research activities and research needs in Statistics of the National Statistical Institutes of the EU and EFTA countries.* For example, we looked for references related to data collection, sampling, quality, variance estimation, risk, forecasting, multi-source environments,... before extending them to those related to common official applications: demography, government statistics, census, imputation,... The number of references reduced dramatically and are included in Appendix F.

1.4 Recent developments

Areas where Bayesian analysis, at large, has seen greater development[2] are the following:

1. *Bayesian computations (*)*, specially Markov chain Monte Carlo methods. Bayesian analysis requires for implementation powerful integration methods in multidimensional settings. Unavailability of such methods prevented their routine use in complex problems. Their recent availability has alleviated the problem and allowed widespread use of Bayesian methodology. Comprehensive references include Gilks, Spiegelhalter and Richardson (1995) and Gamermann (1997).

2. *Model choice (*).* An important problem in statistical analysis, from both frequentist and Bayesian perspectives, is the model determination or model choice. There has been enormous recent developments within this topic, both computational and theoretical. State of the art overviews may be seen in Racugno and Bertolino (1999). See also Draper (1995) and Kass and Raftery (1995) for important ideas.

3. *Robustness and sensitivity analysis methods.* Bayesian analysis has been questioned for its critical dependence on subjective inputs, specially the

[2]Here we only give a brief description of the developments. Some of them, marked (*), are more thoroughly described later

prior. Robust Bayesian analysis mitigates such issue. See Rios Insua and Ruggeri (2000) for a complete description.

4. *Nonparametrics.* This approach allows to mitigate dependence on a specific (parametric) model. Dey, Muller and Sinha (1999) provide an up-to-date description.

5. *Graphical models and influence diagrams* have provided means to qualitatively describe and communicate a model and efficiently compute, through local methods, probabilities of interest. See Lauritzen (1998) and Whitakker (1990) for good overviews.

6. *Bayesian Decision Analysis.* As we mentioned, a strong point of Bayesian Analysis is its coherent integration within decision making. This is specially interesting for Official Statistics, which will eventually integrate a decision making structure within public policy. Smith (1988) and French (1986) provide good introductions.

7. *Hierarchical modelling (*).* Though, formally, hierarchical models add little complication, they do provide further flexibility in modelling. On the other hand, they entail complicated computational problems, which have only been recently amenable to implementation. See Gelman et al (1995) and Carlin and Louis (1997) for excellent overviews.

8. *Many successful applications.* All the above developments have, in turn, opened up the possibility of more realistic and complicated modelling, leading to impressive applications in areas like medicine, artificial intelligence, image processing, econometrics, environmetrics, financial modelling, social statistics. There are excellent monographs devoted to Bayesian applications. Some of them are French and Smith (1997), Berry and Stangl (1996) and Dey, Ghosh and Mallick (1999). Note though that those for Official Statistics are not abundant.

From a more general point of view, most of these ideas are fully described in French and Rios Insua (2000). Another general introduction may be seen in Berger (1985). Bernardo and Smith (1994) is encyclopaedic in its coverage of Bayesian statistical theory. Less encyclopaedic and some 25 years old, DeGroot (1970) remains possibly the best introduction to Bayesian approaches to statistical decision theory. Other recent Bayesian views are provided by Robert (1994) and O'Hagan (1994). The history and development of the subject, can be charted through the following: Ramsay (1931), Lindley (1953), Savage (1954), Chernoff and Moses (1959), Schlaifer (1959), Raiffa and Schlaifer (1961), Ferguson (1967), DeGroot (1970), Berger (1980, 1985).

We finally describe some of the areas of special importance for Official Statistics which have seen some activity in the Bayesian field, as reflected in our bibliographic search. We group publications using the NORIS terminology, and give details in Appendix F.

- *Sampling (*).* Sampling methods are extremely important in Official Statistics activities. Though essentially frequentist, there has been some Bayesian activity. Ghosh and Meeden (1997) provide an excellent review. Of special importance is small area estimation. Ghosh and Rao (1994) provide a good overview.

- *Methods and tools for measurement and improvement of data quality*, including missing data imputation and outlier detection. Several issues of importance in official statistics relate to data quality, like imputation of missing data (for example, when data providers do not send data on time, or surveyees do not respond) and outlier detection. Gelman et al (1996, 1999) provides a description and examples. Related issues are those of variance estimation, as in Baskin (1992) in an application to the US Consumer Price Index.

- *Data analysis and knowledge extraction.* Though too a wide category, state-of-the-art ideas on Bayesian data analysis may be seen in Gelman et al (1996). As far as knowledge extraction is concerned, the Eurostat sponsored project BAKE, based on belief nets, should be mentioned.

- *Nascent techniques.* There has been some interest in Bayesian analysis of various nascent techniques. For example, see Neal (1996) or Muller and Rios Insua (1998) for Bayesian analysis of neural network models.

- *Time series analysis (*).* The best developed area in Bayesian forecasting is that related to dynamic linear models, see West and Harrison (1997) for an excellent review. As far as nonlinear models, see Menchero and Rios Insua (1999) for ideas concerning nonlinear autoregressions.

- *Meta-analysis and multi-source environments.* In some sense, as we have mentioned, one of the strengths of the Bayesian approach is its ability to integrate various sources of information, a problem often face by statistical offices when using both administrative and survey data. An interesting introduction to Bayesian meta-analysis may be seen in Morris and Normand (1992).

- *Data disclosure.* A recent issue of the Journal of Official Statistics (1999, 14, 4) contains a wealth of information in that problem from a Bayesian point of view.

Chapter 2

Why Official Statistics are not Bayesian?

2.1 Introduction

The previous chapter identified a state-of-the-art within Bayesian analysis in Official Statistics which may be briefly summarised as follows:*whereas Bayesian Statistics has had enormous methodological and computational developments, which has opened up the possibility of analysing very complex problems in many applied fields, its impact is yet to be seen in Official Statistics.*

Here we shall identify reasons which might justify this situation and counter them and provide strong points which could motivate the adoptuion of these methods within Official Statistics, mainly drawn from answers to our questionnaires.

Our support for Bayesian ideas is both conceptual and practical: the Bayesian approach provides a coherent framework which facilitates the analysis of decision making problems under uncertainty, see French and Rios Insua (2000) for a full development. See Berger (1985) and Robert (1994) for additional debate.

2.2 Three generic reasons against Bayesian analysis

We start by describing three traditional criticisms to the Bayesian approach which have stopped their adoption in practical fields. We call them generic, since they apply to any field. As we shall explain, they essentially consist of misconceptions which may be easily counterargued, though they still constitue common prejudices.

1. **Computations.** *Implementing the Bayesian framework leads to difficult computational problems. As a consequence, it is sometimes argued that it*

is necessary to limit attention to overly simplified models or to undertake a non-Bayesian analysis.

Although other goals are possible, most Bayesian computation is focused on calculation of posterior expectations $E^*[g(\theta)]$, where E^* represents expectation with respect to the posterior distribution and $g(\theta)$ is some function of interest. For instance, if $g(\theta) = \theta$, then $E^*[g(\theta)] = E^*[\theta] \equiv \mu$, the posterior mean; if $g(\theta) = (\theta - \mu)^2$, then $E^*[g(\theta)]$ is the posterior variance of θ; and, if $g(\theta)$ is 1 if $\theta > C$ and 0 otherwise, then $E^*[g(\theta)]$ is the posterior probability that θ is greater than C. All these problems essentially require the availability of powerful integration methods, specially in high-dimensional settings.

The recent development of Markov chain Monte Carlo methods, and other Bayesian computational machinery, has outdated this criticism, allowing for more realistic (and typically complex) modelling. Indeed, for complicated models, Bayesian analysis has arguably now become the simplest (and often the only possible) method of analysis. This will be the topic of section 3.2.4.

2. **Imprecision and sensitivity to the prior**. *It is often argued that the Bayesian framework demands excessive precision in the Decision Maker's judgements, particularly in regard to specification of the prior distribution.*

   This corresponds to a too narrow-minded perception of Bayesian ideas, and has led to alternative theories like fuzzy sets or Dempster and Shafer's theory. Note that a similar complain could apply to model or loss selection, but such phenomenon has not happened.

   We remind the reader that the Bayesian framework is normative. When applied, it serves as a guidance for action under uncertainty. However, in early stages of an analysis we may not be able to elicit precise information (prior, model, loss). To counter this fact, robust Bayesian methods have been developed. They essentially consist of undertaking a family of Bayesian analyses and basing conclusions on their common ground. If there are too many discrepancies, robust Bayesian tools suggest how to resolve them, guiding elicitation. We shall not pursue this important issue of robustness. The interested reader may follow the review by Rios Insua and Ruggeri (2000) and its discussion.

   As statistical inference is concerned, it should also be noted that there exists a well-developed version of Bayesian analysis which utilizes "default" or "noninformative" prior distributions, and hence which requires no more specifications than classical statistical methods. For reviews of this approach, see Berger (1985) and Kass and Wasserman (1995). We shall briefly dwell on this in Section 3.2.2.

3. **Descriptive**. *It is sometimes argued that actual Decision Makers do not conform to Bayesian postulates.*

Stemming from work by Allais (1953), many experimental studies have pointed out that some decision makers violate the Bayesian postulates in unaided tasks. This suggests weaknesses of the Bayesian approach as a descriptive theory. Some authors interpret this, as well, as threatening its normative status, although such an interpretation carries no logical force. In any case, many new theories which attempt to improve upon Bayesian analysis from a descriptive point of view have appeared, see Rios Insua (1994) for a review.

Whereas experience concerning the descriptive inadequacy of the Bayesian approach has accumulated, recent comparisons among the new theories have painted a somewhat different picture. Note first that by abandoning axioms for descriptive purposes, we have to abandon them for normative purposes: some theories have been proposed too lightly, since they violate principles like transitivity or stochastic dominance. Camerer (1992) observes that all alternative theories so far proposed run into one kind or another of descriptive problems; Bernasconi (1992) observes that performers of experiments seem to conform progressively to Bayesian postulates, after repetitions of experiments; finally, Hey and Orme (1994) observe that, for many subjects, Expected Utility fits equally well as other theories from a statistical point of view, whereas economic implications are not that important for those for which fit is not so good. To sum up, risky decisions do not fully conform to Bayesian postulates, but they approximate them very reasonably.

2.3 Specific criticisms

Whereas the previous criticisms were generic, in the sense of having somehow deferred the routine adoption of Bayesian methods in Statistics at large, here we describe some criticisms that, on the top of the above, might have prevented the use of Bayesian methods within Official Statistics. We mainly draw from responses to question 3 of our *academic questionnaire*, later compared with issues raised by official statisticians:

1. *The subjectivism of Bayesian Analysis.* There is a common perception that subjective input seems to be not well-accepted in Official Statistics, even having unconfortable connotations, many times being difficult to justify the adoption of prior information. There is a claim that official statistics must be objective.

   This is doubtful. On one hand, there is always subjectivism in any modelling activity performed (e.g., the choice of the likelihood is well acepetd by frequentists, but it is a subjective activity). On the other, in many official statistics problems there is plenty of information that may be used to the advantage of the nalysis. We believe that this information could be more profitably modelled proabilistically, through prior distributions. Finally, recall that there are being extensive developments in objective

Bayesian methods which require no more subjective inputs than frequentist ones. The interesting point is that we have a common framewrok for the cases in which we have or lack substantial prior information.

One point to stress, though, is that prior information is difficult to elicit. This problem actually remains as one of the most challenging issues within Bayesian analysis.

2. *The tradition in Official Statistics.* There is a established frequentist tradition within Official Statistics. This is reasonable in at least three respects:

   - From a practical point of view, the subject is relatively new, with the possibility of tackling complex problems only with the recent advent of BAyesian computational machinery and cheap computing power.
   - Changes take some time, specially if we recall that most statistical teaching, specially in Europe, is still mainly non-Bayesian.
   - There are few connections between academia and official statistics.

   The issue may be further enhanced because some National Statistical Institutes concentrate mainly in summarising data, hence not requiring very sophisticated modelling.

   Probably, the best way to turn over this situation is to proceed by example. Chapter 3 provides some of the and pointers to the literature.

3. *Lack of some methodological developments.* Some problems of interest within Official Statistics require good Bayesian answers. For example, the bulk of sampling theory is still frequentist,and although there have been many important Bayesian developments, see Chapter 3 for a review, there is a need to fully develop a Bayesian sampling theory. This will become more important with the increasing emphasis on model-based sampling. Other areas also require developments, for example Bayesian default tests, further comparison of frequentist and Bayesian methods,... Appendix E provides other ideas.

4. *Lack of Bayesian software.* Though the situation is changing, as our Appendix C shows, we should admit thatthere is a clear lack of software to implement Bayesian methods in a routine/automatic fashion. As we say, in some sense, 'the bulk of Statistics remains frequentist, because SAS is frequentist'. Projects like BUGS[1] are certainly helping, but much more work is still required. This is a very reasonable criticism: as Bayesian methods typically require more effort (being less automatic), the increase of effort imposed by the need to develop one's own software may make the approach really uncomfortable for many practicioners.

[1] BUGS is an acronym for Bayesian Statistics using Gibbs sampling, a software project from MRC described in Appendix C

2.4 The view from NSIs

Ten of the fifteen countries responded to the questionnaire submitted to NSIs. The names of experts contacted and effectively answering are shown in the following table [2]:

Country	Sent to	Passed to	Response
Austria	Biricz, Messmann	Haslinger	No reply
Belgium	Delanoy, Decuyper		No reply
Denmark	Andersen, Madsen		Reply
Finland	Laaksonen, Teikari, Notkola		Reply
France	Tranap, Desrosieres, Trognon		No reply
Germany	Ehling, Gertkemper	Speth	Reply
Greece	Stavropoulou, Rontos		No reply
Ireland	Jennings	Keogh	Reply
Italy	Cingolani, Barbieri, Scanu		Reply
Luxembourg	Weydert		Reply
Netherlands	Van de Laar, Betlehem	Zeelemberg	Reply
Norway	Saebo, Byfuglien		No reply
Portugal	Dias, Mateus		No reply
Spain	Cervera		Reply
Sweden	Blom, Lyberg	Lundstrom	Reply
United Kingdom	Jones, Pritchard, Holt	Charlton	Reply

In spite of the small number of respondents, a clear picture has emerged, specially if, because of the statistical tradition in the non respondent countries, we do not expect a much more favourable view, and some common features may be inferred about the perception and use of Bayesian methods in European NSIs.

- *There is little familiarity with Bayesian methods.*

  In most cases, the questionnaire changed hands, and, even, in this case, three of the respondents admitted no familiarity with Bayesian statistics and, moreover, they believed that it would be hard to find someone familiar at their NSIs. In positive cases, several times respondents mentioned that few people at their institutes would know those methods. However, in many cases we have identified key persons at NSIs with some familiarity, which may serve as contacts for further developments. Part of the reasons could be that, at least initially, we addressed senior staff at NSIs. However, as admitted by several of them, a main reason would be that their background in economics, econometrics or even statistics has not been permeated by Bayesian methods, with few Bayesian courses in European

[2] This may be useful for later contacts

universities, at least at the undergraduate level. It is interesting to note that in one country at least, the entrance examination includes a chapter on Bayesian methods for all the 'Grade A' statisticians.

- As such, it comes at no surprise that *there have been little applications of Bayesian methods in the production of statistics.*

  Many NSIs have tracked developments in areas like survey sampling, imputation and Bayesian networks and some have attempted comparisons for possible production of statistics, regarding, again, multiple imputation, neural networks and small area estimation. In some cases, the existing Bayesian software (SOLAS; BAYSEA) has been used or at least tested, with discouraging results. In very few cases (empirical Bayes methods in relation with small area estimation concerning unemployment; Bayesian time series methods in forecasting economic time series; Bayesian methods for imputation techniques), Bayesian methods have been used explicitly in the production of statistics.

  Some respondents have mentioned implicit use of Bayesian methods, possibly referring to explicit use of information; others declare that they would use them whenever necessary. In any case, most respondents showed an interest to receive further information or further experience with Bayesian methods.

- Several reasons were *repeatedly mentioned to justify the little use of Bayesian methods.*

  Many of them already emerged in the interview with the academic side, but we shall recall them, as expressed by NSIs officers, so as to emphasise important points:

  - Lack of software in some areas of interest in Official Statistics or unsuitability of the little software available, to the point of declaring Bayesian software discouraging or unusable.
  - Limited human resources, as expressed sometimes in the lack of methodology units, rendering impossible trying new ideas and methods.
  - Previous training of NSI staff has followed traditional courses of statistics, with little, if any, Bayesian material.
  - There is some misunderstanding that the majority of the statistical community does still not accept Bayesian methods.
  - There is a perception that Bayesian methods are still too new.
  - Some NSI staff suggests that there is not much emphasis in modelling within NSIs.
  - There is skepticism about modelling available information in prior distributions, in spite of prior information being used in other, non

probabilistic ways, implicitly in stratifying populations, selecting auxiliary information for estimation and calibration, allocating samples, etc...

  - Some, more technical, respondents refer to technical issues like the lack of reliable convergence detection methods in Markov chain Montecarlo algorithms.

- In many NSIs, however, *there is an organisational structure that would allow the spread of novel statistical approaches like Bayesian methods.*

  This may take the form of a methodological unit or regular seminars devoted to methodology.

As a conclusion, the previous responses clearly indicated that there is still much room for the innovation in the production of official statistical data with Bayesian methods.

The infrastructure needed for it indeed exists:

- Methodological units
- Facilities for training with technical lectures

but the superstructure is obviously lacking:

- little knowledge about Bayesian statistics
- skepticism about usability of Bayesian tools and methods for the regular production of official statistics.

2.5 Some strong points

Once described criticisms, we suggest several strong points of Bayesian Analtysis, which make it very powerful within Official Statistics. Most of them are generic advantages of Bayesian analysis; some of them are counterarguments of the previous criticisms. Again, we draw mainly from answers to our questionnaire.

1. *The possibility of (better) using prior information.* In most Official Statistics problems there is prior knowledge available (for example, we often have huge datasets on which prior distributions can be built up) and Bayesian techniques provide the suitable avenue for quantifying this prior knowledge. Moreover, the use of prior information is extremely important in measurement, design, estimation, prediction, testing, control and other problems. Bayesian techniques enable such prior information to be used along with sample information to get better solutions to many problems than those yielded by non-Bayesian methods.

2. *The Bayesian approach is better equipped, as far as communication is concerned.* The basic role of Official Statistics is to communicate information to the public and decision makers in government, eventually for public policy decision making. We believe that Bayesian output in the form of posterior densities and their summaries is far more intuitive and easier to grasp by non-experts, than the conclusions from classical statistical methods which can be almost impossible to understand for nonexperts, especially those describing the uncertainty in the data or conclusions, p values being notorious examples of misinterpretations.

   This is becoming more important as society faces more major environmental, medical, and other risk issues, where the ability of the Bayesian methodology to address risk assessment and management will make it much more attractive to statisticians working in government.

3. *The ability of Bayesian analysis to cope with large uncertainties, typical in Official Statistics data sets.* Bayesian methods are particularly relevant for situations with large uncertainties, with its ability to cope with complex hierarchical models needed for a proper analysis of many problems within that field. Increasingly, Official Statistics is going to be responsible for combining information from a large variety of different sources. Examples include imputation of missing data, models for nonresponse in surveys and prediction of future data. Similarly, Bayesian methods are better suited to investigate the hidden structure of large masses of data generated by Official Statistics, and the hierarchical nature of these data such as the presence of clusters, strata, etc.

4. *Improvement in census and survey related activities.* Bayesian methods may be very useful in various activities related with censuses and surveys, specially as modelling approaches are adopted for this activity. For example, Bayesian design could be useful to optimally replace non-controlled-error 'censuses' by controlled efficient surveys. Similarly, Bayesian small area estimation techniques could be very powerful: inference about small subpopulations will be very highly inefficient unless you combine information from other groups, which can be systematically done using the Bayesian approach of partial pooling, or shrinkage towards a regression model. As yet another example, in some problems, Official Statistics face huge difficulties to collect reliable data and would benefit from the correlated (or prior) information brought by other studies on similar populations. Meta-analysis and Spatial statistics have a lot to say in this respect. Again, the Bayesian solutions seem the most acceptable at the current time. At last, Bayesian computational techniques recently developed could be used in Official Statistics to devise more efficient surveys, even removing the exhaustive census.

Chapter 3

Some examples and developments

3.1 Introduction

In this chapter, we first give a quick overview of some important methodological developments in Bayesian analysis that might be relevant in Official Statistics: sampling (including small area estimation), validation, noninformative priors, computation and model selection. Some of the ideas are put into action in four small examples. We complete the chapter with a description of some other relevant applications.

3.2 Some methodological developments

3.2.1 Data collection: Bayesian sampling

One topic of great importance within Official Statistics is that of sampling. With the increasing importance of model-based sampling, Bayesian methods will attain great importance. In fact, it has raised some interest as shown in the bibliographic search. Most of this research is summarised in the recent monograph by Ghosh and Meeden (1997).

In finite population sampling, prior information is always used, either explicitly or implicitly. By making explicit the use of this information as a basic part of the model, Bayesian inference seems specially fit for this purpose. This is in contrast to design-based inference in survey sampling where prior knowledge is incorporated only as auxiliary information. Operationally, however, many estimators obtained by a frequentist methodology are also obtainable as Bayes estimators, typically through noninformative priors. The advantage of the Bayesian approach is that a variety of levels of prior information can be used in survey sampling. Situations considered range from a noninformative

Bayesian justification of standard frequentist methods when the only prior information available is the belief in the exchangeability of the units to a fully Bayesian model with a prior.

A noninformative Bayesian justification of the basic frequentist inferential procedures in finite population sampling may be given, the theoretical argument underlying this justification being based on the concept of admissibility. The proof of the admissibility of a variety of point estimators in finite population sampling follows by using a stepwise Bayes technique. This technique consists of defining a Bayes estimator by steps in such way that it is uniquely defined in all points of the sample space, and it is assured that the estimator is admissible as well. In such a way we may prove, for example, the admissibility of the sample mean for estimating the population mean, under squared error loss, by showing that it is a stepwise Bayes estimator.

An important instrument in the Bayesian approach to sampling is the Polya posterior which is a predictive distribution for the unobserved units in the population given the observations in the sample. It is appropriate when we believe that the observed sample is representative of the population as a whole, that is, in the judgment of the statistician the "seen" and "unseen" units of the population are roughly exchangeable. The Polya posterior does not depend on the design probabilities and can be used just like an ordinary posterior to make inferences even though it does not arise from any single prior. The Polya posterior has a stepwise Bayes justification, and can be used for both point and set estimation problems. There is also some relationship between the Polya posterior and the Dirichlet process priors and the Bayesian Bootstrap. The Polya posterior estimator of the population mean, under squared error loss, is the sample mean. However, this is the only estimator of interest, based on the Polya posterior, that can be found in closed form. In most cases the procedures cannot be found explicitly and must be approximated by simulation, by methods described below. For example, to estimate the median or interval estimates of the median or mean it is necessary to simulate the Polya posterior to find the estimators approximately.

The Polya posterior may be extended to allow the incorporation of different kinds of prior information into the analysis. For example, there could be a prior guess for the entire population or for each member of the population. Alternatively, there could be an auxiliary variable and the statistician's beliefs about the ratios of features of interest of the auxiliary variable are exchangeable, or we could have a stratified population but different levels of prior knowledge about stratification.

Another important case for which Bayesian solutions exist is that in which we must select not only an estimator but also a design from a feasible class of designs. The aforementioned stepwise Bayes technique can be used to identify uniformly admissible pairs of estimators and designs. Another problem is that of nonresponse; the Polya posterior along with an assumed model for the relationship between the respondents and nonrespondents leads to procedures similar in spirit to those arising from multiple imputation. Another case is that in which labels of units are such that members of the population whose labels

are close together are more alike than units whose labels are far apart. For such a population, the Polya posterior would not be appropriate since the necessary beliefs about exchangeability are not present. For such a case a sensible estimator of the population total is one that linearly interpolates between successive members of the sample. This estimator is shown to be admissible using the stepwise Bayes technique.

Empirical and hierarchical methods also apply in finite population sampling. These methods have become very popular recently, especially in the context of simultaneous estimation of several parameters. For example, they have been used to obtain estimates of income per capita, unemployment rates, crop yields and so forth for many state and local government areas. In such situations, estimates of certain area means, or simultaneous estimates of several area means, can be improved by incorporating information from similar neighboring areas. This type of estimation problems are usually referred as *small area estimation* problems.

The term *small area* is commonly used to denote a small geographical area, such as a county, municipality or a census division. It may also describe a small domain, that is, a small subpopulation or group of people within a large geographical area with a specified age, sex and race. For example, there may exist geographical subgroups within a given population that are far below the average in certain respects and need definite upgrading.

Bayesian methods are appropriate for small area estimation, including empirical and hierarchical Bayes methods. A situation where empirical Bayes procedure is adequate is one in which known relationships among the coordinates of a parameter vector allow use of the data to estimate some features of the distribution.

3.2.2 Noninformative priors

We have mentioned that one of the criticisms to the adoption of Bayesian methods in Official Statistics is the use of subjective prior information. In this section, we shall describe some ideas relative to noninformative (typically improper) priors which derive from certain formal rules to choose prior distributions, as an attempt to mitigate dependence from subjective inputs. Another possibility is the consideration of sensitivity analyses, which we shall not pursue here. See Rios Insua and Ruggeri (2000) for developments.

Though advances in prior elicitation are increasing the popularity of subjective methods in Bayesian Statistics, in practice, many Bayesian analyses are performed with noninformative priors. Such priors stem from formal rules which aim at, somehow, objectivise analyses, representing ignorance. They provide reference analyses. However, we believe that we should not use them indiscriminately, essentially ignoring relevant information we might have at hand.

The idea of choosing a prior distribution thorugh a rule, as a 'reference standard' goes back to Jeffreys. In fact, later efforts may be seen as modifications of his scheme. Jeffreys considers several scenarios to formulate his rules:

- For a discrete parameter space, he assigns equal probability to each parameter value.
- When Θ is a bounded interval of $\mathbf{R}$, or $(-\infty, \infty)$, he adopts a constant prior over the interval. In the second case, we have an example of an improper prior.
- When $\Theta = (0, \infty)$, typically associated with an unknown standard deviation θ, he suggests the prior distribution $p(\theta) = 1/\theta$, the reason being that this distribution is invariant with respect to power transformations.

In Jeffreys (1946) he proposes his general rule, consisting of choosing the prior distribution

$$p(\theta) \propto \det(\mathcal{I}(\theta))^{1/2},$$

where $\mathcal{I}(\theta)$ is Fisher's information matrix. With such rule, priors are invariant with respect to certain parameter transformations.

We briefly describe some other attempts to build reference prior distributions. Many of them lead to Jeffreys' priors or variations thereof.

- *The Insufficient Reason Principle,* (Laplace, 1820). This principle leads us to use a uniform prior, when the parameter space is finite. The problem with this rule stems from the *partition paradox* Shafer (1976). In practice, such paradox does not constitute a serious problem, since prior to applying Laplace's rule we must have undertaken a sufficiently fine partition of the parameter space. Its extension to a continuous parameter space leads to a flat prior, which is not invariant under parameter transformations.
- *Invariance.* The simplest case is that of invariance against permutations in a finite set. Should we have little information, it seems natural to demand that the prior is invariant under permutations. This leads to a uniform prior again. The ideas extend to more general cases, leading to Haar measures.
- *Maximum entropy.* The entropy of a distribution may be seen as an uncertainty measure. Maximum entropy priors are the less informative. The maximum entropy method chooses the prior maximizing the entropy. If we do not include constraints, the maximum entropy distribution is the uniform. In other cases, we need to solve the corresponding optimization problem. For a review see Zellner (1991).

There are other methods to determine reference priors, including Berger and Bernardo's (1992) reference prior, Jeffreys' *geometric* method (Jeffreys, 1961), Box and Tiao's *covering* method (Box y Tiao, 1973), *Rissanen*'s (1983),... A good overview may be seen in Kass and Wasserman (1994). A problem associated with reference distributions is that, in many cases, they are improper, potentially leading to problems of incoherence and inconsistency, inadmissibility, marginalisation paradoxes and improper posteriors.

3.2.3 Bayesian computation

Recent computational tools have allowed application of Bayesian methods to highly complex and nonstandard models. Indeed, for complicated models, Bayesian analysis has arguably now become the simplest (and often only possible) method of analysis. As we have argued, Bayesian analysis asks for powerful integration methods. We briefly review traditional methods and then concentrate on modern computational machinery. Later examples provide real illustrations.

Traditional numerical methods The 'traditional' numerical methods for computing $E^*[g(\theta)]$ are numerical integration, Laplace approximation, and Monte Carlo Importance Sampling. Brief introductions to these methods can be found in Berger (1985). Here we say only a few words, to place the methods in context and provide references.

A successful general approach to numerical integration in Bayesian problems, using adaptive quadrature methods, was developed in Naylor and Smith (1982). This was very effective in moderate (e.g., 10) dimensional problems. Also of interest are extensions of *Laplace approximation* method of analytically approximating $E^*[g(\theta)]$, leading to a reasonably accurate general technique, was carried out in Tierney et al. (1989). The chief limitations of the method are the need for analytic derivatives, the need to redo parts of the analysis for each different $g(\theta)$, and the lack of an estimate of the error of the approximation. For many problems, however, the technique is remarkably successful.

Monte Carlo importance sampling [see Geweke (1989) and Wolpert (1991) for discussion] has been the most commonly used traditional method of computing $E^*[g(\theta)]$. The method can work in very large dimensions, and carries with it a fairly reliable accuracy measure. Although one of the oldest computational devices, it is still one of the best, being nearly 'optimal' in many problems. It does require determination of a good 'importance function', however, and this can be a difficult task. Current research continues to address the problem of choosing a good importance function; for instance, Oh and Berger (1993) developed a method of selecting an importance function for a multimodal posterior.

Markov chain simulation techniques The newest techniques to be extensively utilized for numerical Bayesian computations are Markov chain simulation techniques, including the popular Gibbs Sampling. [Certain of these techniques are actually quite old — see, e.g., Hastings (1970); it is their application and adaption to Bayesian problems that is new.] A brief generic description of these methods is as follows:

Step 1.

Select a 'suitable' Markov chain on Θ, with $p(\cdot,\cdot)$ being the transition probability density (i.e., $p(\theta,\theta^*)$ gives the transition density for movement of the

chain from θ to θ^*). Here 'suitable' means primarily that the posterior distribution of θ given the data x, $p(\theta|x)$, is a stationary distribution of the Markov chain, which can be assured in a number of ways.

Step 2. Starting at a point $\theta^{(0)} \in \Theta$, generate a sequence of points $\theta^{(1)}, \theta^{(2)}, \ldots, \theta^{(m)}$ from the chain.

Step 3. Then, for large m, $\theta^{(m)}$ is (approximately) distributed as $\pi(\theta|x)$ and :

$$\frac{1}{m}\Sigma_{i=1}^{m} g(\theta^{(i)}) \cong E^*[g(\theta)]. \tag{3.1}$$

The main strengths of Markov chain methods are :

1. Many different g can simultaneously be handled via Step 3, once the posterior sample has been generated.

2. Programming tends to be comparatively simple.

3. Methods of assessing convergence and accuracy exist and/or are being developed.

The main weaknesses of Markov chain methods are :

1. They can be quite slow. It is not uncommon in complicated problems to need m to be in the hundreds of thousands, requiring millions of random variable generations if the parameter dimension is appreciable.

2. One can be misled into prematurely judging that convergence has obtained.

The more common Markov chain methods, corresponding to different choices of $p(\cdot,\cdot)$, will be discussed briefly. A recent general guide to these methods, and their use in practice, is Gelman et. al. (1995). See also Smith (1991) and Besag et. al. (1995).

Metropolis-Hastings algorithm : One generates a new θ^* based on a 'probing' distribution, and then moves to the new θ^* or stays at the old θ according to a certain 'accept-reject' probabilities, see Hastings (1970).

Gibbs sampling : The Markov chain moves from $\theta^{(i)}$ to $\theta^{(i+1)}$ one coordinate at a time (or one group of coordinates at a time), the transition density being the conditional posterior density of the coordinate(s) being moved given the other coordinates. This is a particularly attractive procedure in many Bayesian scenarios, such as analysis of hierarchical models, because the conditional posterior density of one parameter given the others is often relatively simple (or can be made so with the introduction of auxiliary variables). Because of its importance, here we give a simple illustrative example, with real cases provided later.

Example The following posterior density is a somewhat simplified version of posterior densities which occur commonly in Bayesian analysis, and which are particularly amenable to Gibbs sampling. Suppose the posterior density is :

$$\pi(\theta_1, \theta_2|\text{data}) = \frac{1}{\pi} \exp\{-\theta_1(1+\theta_2^2)\} \tag{3.2}$$

on the domain $\theta_1 > 0$, $-\infty < \theta_2 < \infty$. Many posterior expectations cannot be done in closed form. Gibbs sampling, however, can easily be applied to this distribution to compute all integrals of interest.

Note, first, that the conditional distribution of θ_2, given θ_1, is Normal with mean zero and variance $1/2\theta_1$; and, given θ_2, θ_1 has an exponential distribution with mean $1/(1+\theta_2^2)$. Hence the Gibbs sampling algorithm can be given as follows :

Step 0.	Choose an initial value for θ_2; for instance, the maximizer of the posterior, $\theta_2^{(0)} = 0$.
Step i(a).	Generate $\theta_1^{(i)} = \mathcal{E}/(1+[\theta_2^{(i-1)}]^2$, where $\mathcal{E}$ is a standard exponential random variable.
Step i(b).	Generate $\theta_2^{(i)} = Z/\sqrt{2\theta_1^{(i)}}$, where Z is a standard normal random variable.
Repeat	Steps i(a) and i(b) for $i = 1, 2, \ldots, m$.
Final Step.	Approximate the posterior expectation of $g(\theta_1, \theta_2)$ by :

$$\begin{aligned} E[g(\theta_1,\theta_2)] &= \int_{-\infty}^{\infty}\int_{1}^{\infty} g(\theta_1,\theta_2)\pi(\theta_1,\theta_2|\text{data})d\theta_1 d\theta_2 \\ &\cong \frac{1}{m}\Sigma_{i=1}^{m} g(\theta_1^{(i)}, \theta_2^{(i)}). \end{aligned} \tag{3.3}$$

For instance, the typical estimate of θ_1 would be its posterior mean, approximated by $\hat{\theta}_1 = (1/m)\Sigma_{i=1}^{m}\theta_1^{(i)}$. The table below presents the results of this computation for various values of m. Note that the true posterior mean here is 0.5.

Approximate values of posterior mean of θ_1 from Gibbs Sampling

m	100	500	1,000	10,000	50,000
$\hat{\theta}_1$	0.43761	0.53243	0.48690	0.49857	0.50002

There are other MCMC methods like *hit and run sampling* or *hybrid* methods.

3.2.4 Bayesian model selection

We assume here that the data, X, have arisen from one of several possible models $M_1, \ldots, M_m$. Under M_i, the density of X is $f_i(x|\theta_i)$, where θ_i is an unknown vector of parameters of f_i.

The Bayesian approach to model selection begins by assigning prior probabilities, $P(M_i)$, to each model; often, equal prior probabilities are used, i.e.

$P(M_i) = 1/m$. It is also necessary to choose prior distributions $\pi(\theta_i)$ for the unknown parameters of each model; sometimes these can also be chosen in a "default" manner, as will be illustrated later.

The analysis then proceeds by computing the posterior probabilities of each model,

$$P(M_i|x) = \frac{P(M_i)m_i(x)}{\Sigma_{j=1}^{m} P(M_j)m_j(x)}, \tag{3.4}$$

where $m_j(x) = \int f_j(x|\theta_j)\pi_j(\theta_j)d\theta_j$. Typically one selects the model (or models) with largest posterior probability.

Curiously, the Bayesian approach to model selection is less widely used than the Bayesian approach to estimation, even though the approach is arguably of even more value in model selection. The most obvious advantage of the Bayesian approach is the simplicity of interpretation of the answers; even those with limited statistical background can easily interpret the conclusion "the (posterior) probabilities that M_1 and M_2 are true are 0.93 and 0.07, respectively." This ease of interpretation is in stark contrast to the situation when classical measures such as P-values or Chi-square are used. Few understand what evidence is actually provided by such classical measures, and misinterpretation is the rule, rather than the exception (cf. Berger and Sellke, 1987; Berger and Delampady, 1987 and Delampady and Berger, 1990).

A second advantage of Bayesian model selection is that it is *consistent*, in the sense that, as one obtains more and more data, one is guaranteed to select the true model (or the model closest to the true model if none are true). Classical methods typically fail even this minimal criterion, usually by selecting models that are too complex when there is a large amount of data.

This is related to a third advantage of the Bayesian approach, namely that it acts as an automatic "Ockham's razor," selecting a simpler model over a more complex model if both are compatible with the data. Indeed, Bayesian analysis can be used to quantify Ockham's razor, making precise what has long been viewed as a fundamental, but heuristic, scientific principle. See Jefferys and Berger (1992) for discussion and illustration.

A fourth advantage of Bayesian model selection is that one can account for model uncertainty. Since each model will have a posterior probability, $P(M_i|x)$, one can maintain consideration of several models, with the input of each into the analysis weighted by the $P(M_i|x)$. Classical analyses, which select one model and base predictions upon this one model, are notorious for providing predicted precisions that are much too small. See Draper (1995) for general discussion.

A fifth advantage of Bayesian model selection is that it can be applied to comparison of multiple models, and applies very generally; the models need not be in standard families, and need not be nested.

There are, however, two difficulties in implementing Bayesian model selection (i) choosing the prior distributions $\pi_i(\theta_i)$, and (ii) computing the $m_i(x)$. A variety of strategies exist for carrying out the integrations necessary to compute the $m_i(x)$; see Kass and Raftery (1995) for discussion. Choosing the $\pi_i(\theta_i)$ is more of a problem.

It may well be the case that subjective knowledge about the θ_i is available, and can be incorporated into subjective proper priors for the θ_i. This is clearly desirable if it can be done. Often, however, the θ_i may be high dimensional and subjective elicitation of all the $\pi_i(\theta_i)$ may be impossible. There are then several possible "default" strategies one can follow.

The simplest default option is to use the approximation typically referred to as BIC (cf. Kass and Raftery, 1995). This is a quite accurate approximation if there is a substantial amount of data. Also, the approximation avoids the computational difficulty mentioned earlier.

For a moderate or small amount of data, BIC can be inaccurate. Sometimes (though not often!) it is possible to use "noninformative" or "objective" priors, $\pi_i^N(\theta_i)$. This can be done for some scenarios in which the dimensions of the vectors θ_i are the same for all models.

Unfortunately, it is somewhat rare for use of noninformative priors in model selection to be valid. Recently, however, two very general default Bayesian model selection tools have been developed, the "Intrinsic Bayes factor" approach of Berger and Pericchi (1996), and the "Fractional Bayes factor" approach of O'Hagan (1994, 1995). These approaches are too involved to describe here, but it is noteworthy that they apply to almost any model selection problems, and operate without the need for subjective proper priors. Although we cannot present the algorithms here, it is of value to look at an application, to see the nature of the conclusions that arise.

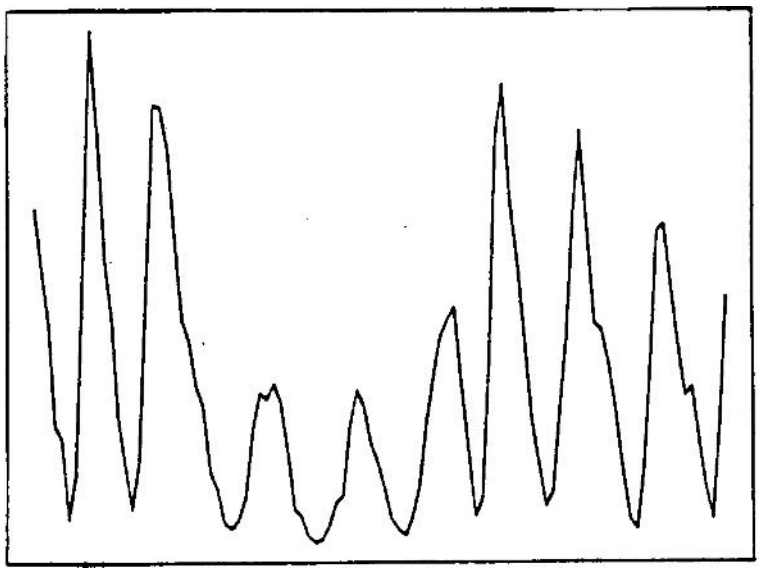

Figure 3.1: Time series data to be modelled

Example The Figure presents some time series data. It is decided to model this as a stationary autoregressive process with drift. For instance, the AR(1) model with a linear drift would be described as :

$$Y_t = \beta_1 t + \phi_1 (Y_{t-1} - \beta_1 (t-1)) + \epsilon_t, \tag{3.5}$$

where Y_t is the observation at time t, β_1 is the unknown linear coefficient, ϕ_1 is the unknown autocorrelation, and the ϵ_t are i.i.d. $\mathcal{N}(0, \sigma^2)$ errors, σ^2 also unknown.

It is decided to consider autoregressive models of order 1, 2, 3 and 4, and also to consider constant (C), linear (L), and quadratic (Q) drift. Thus the AR(j) model with drift of polynomial order k ($k = 0, 1, 2$) can be written :

$$Y_t = \Sigma_{\ell=0}^{k} \beta_\ell t^\ell + \Sigma_{r=1}^{j} \phi_r (Y_{t-r} - \Sigma_{\ell=0}^{k} \beta_\ell (t-r)^\ell) + \epsilon_t. \tag{3.6}$$

We are thus considering twelve models (any of the four AR models together with any of the three polynomial drifts).

The "intrinsic Bayes factor" approach applies directly to this problem. It utilizes only standard noninformative priors for the parameters (constant priors for the β_i and ϕ_i, and $1/\sigma^2$ for the variance, σ^2). Note that, because the models being considered are of differing dimensions, one cannot use these noninformative priors directly, but must use them through the "intrinsic Bayes factor" algorithm. There is also a computational complication: because of the stationarity assumption, $\phi = (\phi_1, \phi_2, \ldots, \phi_j)$ is restricted to the "stationarity region," and so the integration in computation of the $m_i(x)$ must be carried out over this region. Methods of doing this, as well as the relevant intrinsic Bayes factor algorithm, can be found in Varshavsky (1996). The results are summarized in the table.

Posterior probabilities of models assuming equal prior probabilities.

Model	$P(M_i\|x)$	Model	$P(M_i\|x)$
AR(1), C	~ 0	AR(3), C	0.740
AR(1), L	~ 0	AR(3), L	0.001
AR(1), Q	~ 0	AR(3), Q	0.001
AR(2), C	0.161	AR(4), C	0.076
AR(2), L	0.011	AR(4), L	0.006
AR(2), Q	0.001	AR(4), Q	0.001

There is clearly no support for a nonconstant drift. Among the models with constant drift, the AR(3) model is the clear winner, although the AR(2) model receives some support. It is of interest that classical model selection procedures choose substantially more complex models, such as the {AR(4), C} or even the {AR(4), Q} models. This is the "overfitting" of classical methods that was referred to earlier.

3.3 Some case studies

In this section, we illustrate with several case studies relevant for official statistics various important, novel ideas in Bayesian analysis, hopefully showing the potential of the Bayesian approach for the analysis of official data. In each case, we start describing the key methodological ideas and the application. The studies are sketched. Further details may be seen, respectively, in Marin (2000), Rios Insua et al (1999), Dellaportas et al (1999) and Moreno and Rios Insua (1999). We conclude by mentioning other cases found in the literature.

3.3.1 Unemployment forecasting: an application of Dynamic Linear Models

- *Key methodological ideas*
  - Introduce Dynamic Linear Models as a powerful framework for forecasting.
  - Show how we may robustify such models when departures from assumptions are identified.
  - Show a data-based assessement of prior.
- *Application*
  - Forecasting the unemployment and activity ratios in the State of Valencia.
- *Advantages of the Bayesian approach*
  - Move away from stationarity assumptions.
  - Ability to model standard features in official statistics time series, like trends, seasonality, regressions and autoregression.
  - Computational efficiency.
  - Open to external interventions.

Introduction

We describe a class of forecasting models that may be very useful for official statisticians, for example in the study of the evolution of unemployment rates and associated economic policies. Dynamic Linear Models (DLMs) stem from work by Harrison and Stevens (1976). However, during the late eighties and early nineties, numerous modeling and computational enhancements and the development of the user-friendly software BATS (Pole et al., 1994) have made them readily available for applications. Here, we shall describe the main ideas of DLMs and some extensions, and illustrate them with the forecast of unemployment rates in the State of Valencia, a NUTS 2 region in Spain. For a thorough review of Bayesian forecasting, see West and Harrison (1996).

DLMs allow for moving away from stationarity assumptions, since process parameters are time varying. Also, they are flexible enough to model usual behaviour of statistical time series used in the analysis of the economic and social conjoncture like seasonal patterns and trends, and permit the incorporation of covariates. They are also computationally fast, facilitating their use in real time decision making and large-scale simulations. Finally, they allow for the incorporation of all prior information, including that due to interventions, hence incorporating a principle of *management by exception*, fundamental in the Bayesian forecasting philosophy (West and Harrison, 1989): a set of models is routinely used for processing information, making inferences and forecasting, unless exceptional circumstances arise. In this case, the system is open to external intervention, typically by inclusion of additional subjective information. Forecasting is performed sequentially based on all available information.

Our problem is to forecast the next r values of a variable y_t, from instant T to instant $T+r$, given the available information D_T. For that we use DLMs, which in their simplest case have the following structure, for every instant of time t, $t = 1, 2, 3, ...$:

- *Observation equation :*

$$y_t = F_t\theta_t + v_t, \quad v_t \sim N(0, V_t) \tag{3.7}$$

  where y_t denotes the observed value, which depends linearly on the values of the state variables z_t, perturbed by a normal noise.

- *System evolution equation :*

$$\theta_t = G_t\theta_{t-1} + w_t, \quad w_t \sim N(0, W_t) \tag{3.8}$$

  describing the evolution of the state variables, linearly dependent on the variables in the previous state plus a random perturbation.

- *Initial information :*

$$\theta_0|D_0 \sim N(m_0, C_0) \tag{3.9}$$

  describing the prior beliefs of the forecaster.
  The error sequences v_t and w_t are independent, and mutually independent. Moreover, they are independent of $(\theta_0|D_0)$.

Updating procedures and the use of this model for forecasting are described in detail by West and Harrison (1997). Essentially, inferences about both parameters and forecasts, one or more steps ahead, are based on a normal model, with corresponding parameters computed recursively.

The model specification requires that $F_t, G_t, V_t, W_t, m_0, C_0$ are known. Modeling of these is fully described in West and Harrison (1997). Concerning F_t and G_t, the key idea is the *Superposition principle*, which states that linear combinations of independent DLM's lead to a DLM. This suggests a model building

strategy based on blocks, representing polynomial trends, seasonal patterns and dynamic regression, if covariates are available. Concerning W_t, the use of the *discount principle* allows for a semiautomatic modeling approach to that variance, based on the idea of information discounting. Finally, a typical strategy concerning V_t is to consider it constant, but unknown, and introduce a procedure for learning adaptively about it.

A key issue in Bayesian forecasting is that the output is the entire predictive distribution, not just summaries. Thus we can use this distribution for any purposes, taking expectations or simulating the future. As an aside, the standard method of: i) estimating model parameters; ii) plugging the estimates into the model; and iii) using the estimated model for prediction or simulation; typically greatly underestimates uncertainty in predictions, since the uncertainty in the model parameters is not taken into account.

Example

We show an application of a robust DLM to forecast the *activity ratio* Y_t and the *unemployment ratio* X_t in the State of Valencia (Spain). We have 23 pairs (y_t, x_t) of quarterly observations ranging from 1983 to 1988 available. These observations are given in Girón et al. (1989) and shown in the figure.

Activ.	**Unemp.**	**Activ.**	**Unemp.**
50.00	18.19	48.75	16.37
49.21	17.28	49.20	17.91
49.30	18.36	48.79	18.48
48.62	19.74	54.14	21.14
48.85	21.36	48.55	20.37
48.45	21.55	48.33	19.79
48.67	19.81	48.57	19.48
48.41	20.12	48.67	18.86
49.34	19.61	50.46	19.83
50.44	18.32	50.83	18.31
50.93	17.97	50.97	18.10
50.80	16.92		

We assume that a linear relation between activity ratio and unemployment ratio holds with the intercept and the slope varying through time. We take $F_t' = (1, x_t)$, hence including an independent term. We shall actually use a robust version of the standard DLM previously shown. In this version, we assume for the *initial vector* $(\theta_0, \nu_1, \omega_1, \ldots, \nu_n, \omega_n)'$ a multivariate exponential power distribution. This family includes many important multivariate distributions, like the normal, the double exponential and the uniform distributions. The exponential power distribution has a parameter β related with the multivariate kurtosis of the distribution. Then, we can consider distributions with higher or lower tails than the normal, and, hence, work with models in which normality is not acceptable. The exponential power linear dynamic model includes a

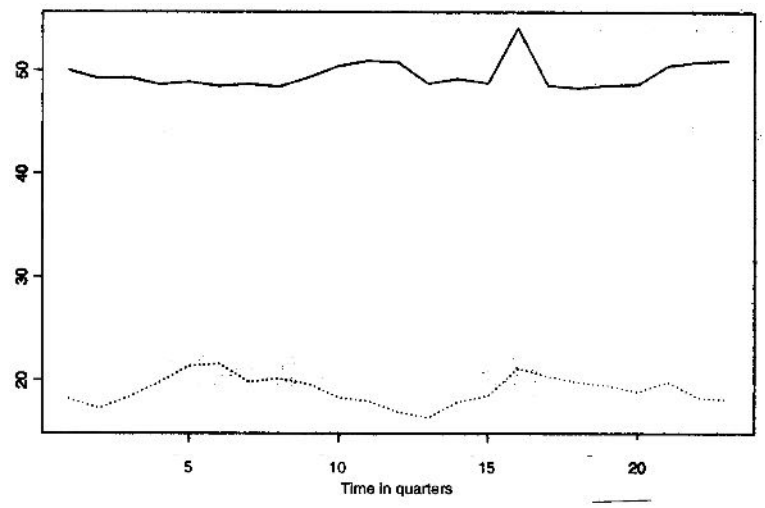

Figure 3.2: Quarterly activity and unemployment rates in Valencia. 1983-1988

recursive equation that determines the kurtosis of the distributions for each t. In such a way, we can identify the corresponding exponential power distribution that has this kurtosis.

We use the first seven pairs to adjust the model and the others to test the model. First, we make $G_t = I$, since there are no reasons to assume a systematic trend in the evolution of θ_t. We adjust the rest of the model by setting its initial parameters using the regression line between Y_t and x_t and the simulation of a sequence of values of θ_t from consecutive pairs (y_t, x_t), (y_{t+1}, x_{t+1}). This lead us to use a multivariate exponential power distribution with vector of means $m_0 = \begin{pmatrix} 50.78 \\ -0.10 \end{pmatrix}$, scale matrix $C_0 = \begin{pmatrix} 780.73 & -41.59 \\ -41.59 & 2.22 \end{pmatrix}$, and kurtosis parameter $\beta = 2.0$. We determine also $V_t = 0.26$ and $W_t = \begin{pmatrix} 0.35 & 0 \\ 0 & 1.19 \end{pmatrix}$.

Then, we calculate the updated distribution of θ_t and the updated characteristics of Y_t, as well as the values β of the exponential power distribution corresponding to the kurtosis of the updated distribution of θ_t for each t. The next table shows the mean vector, the covariance matrix of θ_t and the values of parameter β.

t	Mean	Covariance m.		β	t	Mean	Covariance m.		β
8	14.85	61.00	-2.89	1.03	16	19.23	81.05	-4.30	1.05
	1.86	-2.89	1.37			1.56	-4.19	0.23	
9	21.53	75.02	-3.51	1.04	17	19.67	82.09	-4.19	1.05
	1.28	-3.51	0.16			1.51	-4.19	0.21	
10	20.52	77.64	-3.81	1.05	18	19.82	83.14	-4.19	1.05
	1.38	-3.81	1.87			1.55	-4.19	0.21	
11	21.77	78.10	-3.62	1.04	19	18.86	82.99	-4.53	1.05
	1.24	-3.62	0.17			1.72	-4.53	0.25	
12	20.32	77.96	-3.94	1.04	20	18.31	84.44	-4.61	1.05
	1.42	-3.94	0.20			1.78	-4.61	0.25	
13	19.81	79.13	-3.99	1.05	21	18.01	85.55	-4.76	1.06
	1.46	-3.99	0.20			1.83	-4.76	0.27	
14	19.50	80.02	-4.11	1.05	22	18.10	86.79	-4.80	1.06
	1.49	-4.11	0.21			1.82	-4.80	0.27	
15	20.21	80.81	-4.02	1.05	23	17.36	86.95	-5.14	1.06
	1.40	-4.02	0.20			1.98	-5.14	0.30	

Updated Distribution of θ_t

Observe that the independent term is more important than the slope. Note also that the covariance between them is negative: when one increases, the other decreases. The parameter β is greater than 1, hence, the distributions of vector θ_t for each t are not normal, and there is a little trend towards more platykurtic distributions, that is, symmetrical distributions with lower tails than multivariate normal distribution.

The next table shows, the updated mean and standard deviation of Y_t, as well as the prediction error (the difference between the mean and the real observation).

t	Mean	S. D.	Error	S. E.	t	Mean	S. D.	Error	S. E.
8	48.75	1.46	-5.39	-3.70	16	46.65	0.61	2.03	3.33
9	54.53	0.22	-5.68	-25.76	17	49.84	0.43	-0.50	-1.15
10	47.59	0.46	0.96	2.10	18	49.67	0.26	0.79	3.01
11	50.17	0.56	-1.72	-3.06	19	48.13	0.73	2.31	3.16
12	46.27	0.76	2.06	2.73	20	50.42	0.24	0.41	1.69
13	48.36	0.23	0.31	1.35	21	50.23	0.30	0.70	2.38
14	48.19	0.28	0.38	1.38	22	51.17	0.26	-0.20	-0.77
15	49.53	0.38	-1.11	-2.94	23	48.83	0.65	1.97	3.02

Updated Characteristics of $\mathbf{Y}_t$

The errors have no trend and the standard errors are a little large; it may suggest that the value 0.26 chosen for V_t is too small. Nevertheless, the obtained results show that the exponential power model provides good predictions and that the distributions obtained in each time are a little more platykurtic than the normal, so the adjusted model is different to the standard normal one and allows considering outliers in predictions.

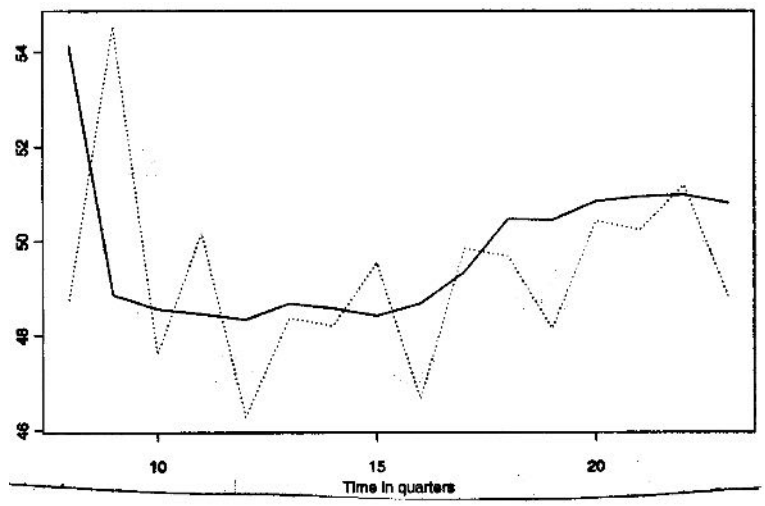

Figure 3.3: Predictions (dotted) and observations (solid) for successive periods of time.

There are many possible variants and generalizations of the standard normal model. Other possible ways, different than the exponential power model, consists of changing the assumption of normality, introducing, for instance, distributions like t-Student, elliptically contoured distributions and mixtures of normal distributions. Another possibility is the use of nonlinear models, including neural networks.

3.3.2 Industrial accident forecasting: an illustration in hierarchical modelling and model ellaboration

- *Key methodological ideas*
  - Hierarchical models as a flexible and powerful class of tools for modelling.
  - Progressive ellaboration of models based on robustness tools.
  - Assessment of priors from expert information.
  - Illustration of various computational strategies, depending on the complexity of the model.

- *Application*
  - Forecast accidentability of a certain industrial sector.
- *Advantages of the Bayesian approach*
  - Possibility of coherently combining all sources of information.
  - Possibility of coherently evolve the model when additional complexity is required.

Introduction

EU countries are paying increasing attention to safety at the workplace. Given the enormous socio-economic implications of labour accidents, evaluating accident proneness of companies should also help to identify where to allocate resources to improve safety. Indeed, this is recognised in several countries, including Spain, where, by law, companies need to spend part of their budgets in that task, typically through educational campaigns.

Answers to all these questions demand appropriate forecasting systems. Bayesian methods seem specially fit for the purpose, since we have access to several sources of information, including expert opinion and historical data, and we may consider all available knowledge in a coherent way. Expert information is specially relevant in presence of scarce historical data, or in a dynamic economic environment, where companies appear and change (grow, decay, merge) rapidly.

Our basic objective, hence, is to predict the number of accidents a given company or economic sector will have in a certain period, and provide information about the accident proneness of that company or sector. We describe three increasingly complex and realistic models. The first one is a Poisson-gamma model, standard in Bayesian reliability analysis. The number of accidents is assumed to follow a Poisson model, with rate dependent on the number of workers. Our main contributions here concern the assessment of prior distributions, typically the main criticism to Bayesian models.

The previous model may not be adequate if we try to evaluate accident proneness of companies in economic sectors with highly volatile numbers of workers. Hence, we couple our model with a forecasting model for the number of workers. For some of the issues we want to address, we are not able to provide analytic answers and we have to rely on simple simulation methods. This is a common case in Bayesian modelling: the more detailed the model, the more we have to rely on simulation methods to perform inference and prediction.

This is specially true for the third model, which allows for dynamic changes in the accident rate, i.e. for accident proneness growth (and decay). Inference and forecasting are done with a sampling-resampling simulation strategy. All the models are illustrated with real data.

A Poisson gamma model

We are interested in the following problem: given the accident history D_k of a company or economic sector (number of workers, number of accidents,...) prior to period k, we want to forecast the number of accidents X_k of the company in that period, whose duration is t_k. Initially, we assume the number n_k of workers in the period is known. This hypothesis is reasonable in economic sectors with stable labour force. We relax this assumption later. We shall assume that all period durations t_k are the same (or very similar), perhaps after an appropriate change of variable, and take it to be 1.

We are uncertain about X_k. We believe two quantities have influence on that number: n_k, here known, and λ, representing the accident rate of a worker. We assume independence of λ and n_k, which is reasonable in most contexts. This suggests that the joint distribution of λ and X_k is, taking into account dependence on D_k,

$$P(X_k, \lambda | n_k, D_k) = P(X_k | n_k, \lambda)\pi(\lambda | D_k),$$

where $\pi(\lambda|D_k)$ represents prior beliefs over λ at the beginning of period k. Our aim is to compute the posterior predictive distribution

$$P(X_k | n_k, D_k) = \int P(X_k | n_k, \lambda)\pi(\lambda | D_k) d\lambda,$$

and summarise it. Given the sequential nature of the problem, we shall be interested also in the posterior distribution of λ, after period k, which is

$$\pi(\lambda | D_{k+1}) \propto P(X_k = r_k | n_k, \lambda)\pi(\lambda | D_k).$$

We assume that $D_{k+1} = (D_k, (n_k, r_k))$, i.e. the only additional information are the numbers n_k of workers and r_k of accidents in period k. This distribution will be used as the prior for period $k+1$, in the absence of exceptions.

Assume that it is appropriate to use use Poisson models, i.e.,

$$X_k | \lambda, n_k \sim Po(n_k \lambda).$$

λ is now the only uncertain parameter. We have information concerning that rate available from experts. We need to model it, which amounts to assessing the prior distribution. For computational convenience and flexibility, we model expert beliefs by means of a gamma distribution with parameters a and p. We decided to elicit quantiles of the distribution of λ from the expert. For the assessment of quantiles, we use a computer protocol interactively convergent to the expert's quantiles, and based on fortune wheels, see e.g. French and Rios Insua (2000). We have chosen to assess the median l_1 and the 75 % quantile l_2 of λ. We then have

$$\begin{aligned} \Lambda(l_1, a, p) &= .5 \\ \Lambda(l_2, a, p) &= .75, \end{aligned}$$

where Λ is the distribution function of λ. Using an approximation procedure to solve that system of nonlinear equations we obtain an approximation $(\hat{a}, \hat{p})$ of the prior parameters. For consistency checking, we assess a third quantile (25 %) l_3: once assessed $\hat{a}$ and $\hat{p}$, we study whether

$$\Lambda(l_3, \hat{a}, \hat{p}) = .25.$$

If there is agreement, we use this as the prior for our model, with appropriate sensitivity checkings. If not, we ask the expert to reconsider his assessments and, eventually, correct them. If he does not want to modify them, we must adopt another form for the prior.

To sum up, we consider that

$$\begin{aligned} Pr(X_k = r_k | n_k, \lambda) &= (n_k\lambda)^{r_k} \exp(-n_k\lambda)/r_k! \\ \pi(\lambda|D_1) &\propto \exp(-a\lambda)\, \lambda^{p-1} I_{[0,\infty)}. \end{aligned}$$

Given the model structure, a simple application of Bayes formula shows that

$$\pi(\lambda|D_k) \propto \exp(-(a + \Sigma_{i=1}^{k-1} n_i)\lambda)\, \lambda^{\Sigma_{i=1}^{k-1} r_i + p - 1} I_{[0,\infty)},$$

corresponding to a gamma distribution with parameters $a + \Sigma_{i=1}^{k-1} n_i$ and $\Sigma_{i=1}^{k-1} r_i + p$. Simple computations provide the predictive distribution, from which predictive intervals follow easily. Two other convenient summaries of the predictive distribution are the predictive expected value E^p and variance V^p of the number of accidents, which are

$$E^P = n_k \times \frac{\Sigma_{i=1}^{k-1} r_i + p}{\Sigma_{i=1}^{k-1} n_i + a},$$

and

$$V^P = E^P \left(1 + \frac{n_k}{\Sigma_{i=1}^{k-1} n_i + a}\right).$$

Example In this particular case, the expert suggested the quantiles $l_1 = .58$, $l_2 = .98$ and $l_3 = .38$. As a consequence, we adopted a gamma distribution with parameters $\hat{p} = 1.59$ and $\hat{a} = 2.22$ as his prior. The third quantile suggested remarkable consistency. Note that this prior distribution is fairly vague.

We show results comparing predictions and number of accidents observed. $\sigma^P = \sqrt{V^P}$ is the predictive standard deviation. We also include a predictive interval of probability 0.9.

Forecasting with Poisson-gamma model

Year	E^P	σ^P	Acc	.9 Pr. Int.
1988	75.6	12.2	54	[55,94]
1989	62.2	9.6	68	[46,77]
1990	63.9	9.1	60	[48,78]

The predictive behaviour of the model seems good, improving each year. Note though, that some robustness checking is required, as done below. $\triangle$

A Bayesian analysis should always check sensitivity of conclusions (in our case, inferences and predictions) to the inputs (in our case, the prior). Here, we draw upon recent work in robust Bayesian analysis to illustrate some ideas. Robust methods have been developed as an attempt to overcome the traditional criticism to Bayesian analysis of demanding excessive precision in the the decision maker's (or expert) judgements. We may view these as a way of conducting a class of Bayesian analysis when, e.g., the prior distribution ranges in a class, and basing conclusions on the common grounds of those analyses. Should they essentially agree, we would have a robust model; otherwise, robustness would lack and we would either collect additional data or more precise prior information from the expert.

In our case, we could review critically the two hypotheses made to elicit the prior distribution:

1. From the quantiles, we have deduced the parameters of a gamma distribution.

2. We have done this with an approximation procedure.

For the first hypothesis, we consider that the prior distribution belongs to the class:

$$\Pi = \{\pi : \pi(0, a_1] = .25, \pi(a_1, a_2] = .25, \pi(a_2, a_3] = .25, \pi(a_3, \infty) = .25\}.$$

To check whether our hypothesis about a gamma distribution is critical, we undertake a global robustness analysis with respect to class Π: we are interested in determining whether the predictive distribution changes markedly when the prior π varies in the class Π. We start by computing the range of the predictive mean

$$[\underline{E}(X_k|n_k, D_k), \bar{E}(X_k|n_k, D_k)]$$

with

$$\bar{E}(X_k|n_k, D_k) = \sup_{\pi \in \Pi} E(X_k|n_k, D_k)$$

$$\underline{E}(X_k|n_k, D_k) = \inf_{\pi \in \Pi} E(X_k|n_k, D_k)$$

Numerical methods (see, e.g., Ruggeri, 1990) have been designed to solve these optimisation problems.

Example In our example, $a_1 = .38$, $a_2 = .58$, $a_3 = .98$. We solve the corresponding optimisation problems with the aid of a MATHEMATICA package, as reported in the table.

Bounds on scaled expected accidents

Year	$\underline{E}[X_k \mid n_k, D_k]/n_k$	$\bar{E}[X_k \mid n_k, D_k]/n_k$
1988	0.05	0.58
1989	0.05	0.58
1990	0.05	0.58

Note that additional data produce little variation in the accident range, perhaps because of the small annual variation of s/r. These results suggest that class Π is not robust in this example: expected forecasts may vary widely. $\triangle$

When, as in the example, robustness is lacking, we should be interested in reducing the imprecision, through elicitation of additional information from the expert. Fréchet derivatives (Ruggeri and Wasserman, 1993) and associated sensitivity measures, see Martin and Rios Insua (1997), help us to center additional elicitation efforts:

Example Using the mentioned tools, we detect great sensitivity mainly due to the first quartile (I_1). Hence, we elicit additional information about such quartile from the expert. We divide it in three subintervals, $I_{11} = [0, 0.15)$, $I_{12} = [0.15, 0.2)$ and $I_{13} = [0.2, 0.38)$ with probabilities 0.1, 0.05 and 0.1, respectively, so that we may keep the same gamma prior. Then, the sensitivty is much smaller. We compute again the upper and lower bound of the predictive expectation, with the new class:

Bounds on revised scaled expected accidents

Year	$\bar{E}[X_k \mid n_k, D_k]/n_k$	$\underline{E}[X_k \mid n_k, D_k]/n_k$
1988	0.15	0.24
1989	0.15	0.23
1990	0.15	0.23

Observe the reduction in ranges in both tables, suggesting much more robustness. $\triangle$

To sum up, we may consider that the analysis is fairly robust to the shape of the distribution, but not so much to the quantiles. Procedures suggesting where additional quantiles should be elicited and the impact of this additional information, may be provided. Obviously, the procedure would iterate until robustness is eventually achieved.

A Poisson-gamma model with covariates

We have suggested that the number of workers may change notoriously through different periods in some economic sectors. As a consequence, there may be great uncertainty about n_k, which should perhaps be modelled, specially when data is less aggregated. We first study under what conditions should we model that uncertainty. In such case, we propose a model and provide the corresponding forecasts.

Modelling uncertainty about the number of workers We may appeal to deterministic sensitivity analysis tools from decision analysis, to consider whether we should model uncertainty about the number of workers. Carrying on with the analysis, let $\phi_1(a, p, n_k)$ be the predictive expected value when n_k is the number of workers and a and p are the parameters of the gamma prior. Let $S = \{(a, p, n) : a \in [a_1, a_2], p \in [p_1, p_2], n \in [n_1, n_2], \}$ with $a_1, p_1, n_1 > 0$, be the range of parameters considered. Clearly,

$$\begin{aligned} \min_S \phi_1(a, p, n) &= \phi_1(a_2, p_1, n_1), \\ \max_S \phi_1(a, p, n) &= \phi_1(a_1, p_2, n_2). \end{aligned}$$

The maximum variability of ϕ_1 in S is given by

$$\Delta E_n^P = \frac{\min_S \phi_1(a, p, n)}{\max_S \phi_1(a, p, n)} = \frac{n_2}{n_1} \frac{\Sigma_{i=1}^{k-1} r_i + p_2}{\Sigma_{i=1}^{k-1} r_i + p_1}. \tag{3.10}$$

The quotient depends on n_2/n_1. Hence, if there is great uncertainty about n, predictions might change enormously.

Given the previous indication, it may be necessary to extend the model, taking into account uncertainty about n_k.

Continuing with our Poisson-gamma model, i.e. assume that, given D_k,

$$\begin{aligned} X_k|\lambda, n_k &\sim Po(\lambda n_k) \\ \lambda|D_k &\sim \gamma(\Sigma_{i=1}^{k-1} n_i + a, \Sigma_{i=1}^{k-1} r_i + p) \\ n_k|D_k &\sim f_k(n_k|D_k). \end{aligned}$$

Simple computations lead to

$$E(X_k|D_k) = \mu_k \times \frac{\Sigma_{i=1}^{k-1} r_i + p}{\Sigma_{i=1}^{k-1} n_i + a}.$$

Simple computations lead to the predictive variance:

$$V(X_k|D_k) = \frac{\Sigma_{i=1}^{k-1} r_i + p}{\Sigma_{i=1}^{k-1} n_i + a} \times \left(\mu_k + \frac{\mu_k^2}{\Sigma_{i=1}^{k-1} n_i + a} + \sigma_k^2 \frac{1 + \Sigma_{i=1}^{k-1} r_i + p}{\Sigma_{i=1}^{k-1} n_i + a} \right).$$

We could use these expressions to determine when to model uncertainty about n_k. Our suggestion is to do it when the predictive distribution is not too concentrated around the predictive mean.

Typically, we shall have to use numerical methods to undertake these computations in this model. Monte Carlo methods may be specially useful in this case, in spite of being one dimensional integrals, since we may reuse computations from r to $r+1$: we just need to generate once the n_k's and the integrand may be evaluated recursively.

The cases we have studied, in which the number of workers was fairly large, suggest modelling their evolution with dynamic linear models (DLM). With this choice, the simplest version of our model would be:

$$\begin{aligned} X_k|\lambda, n_k, D_k &\sim Po(\lambda n_k) \\ n_k = F_k\theta_k + v_k & \quad v_k \sim N(0, V_k) \\ \theta_k = G_k\theta_{k-1} + w_k & \quad w_k \sim N(0, W_k) \\ \lambda|D_1 &\sim \gamma(a, p) \\ \theta_0|D_1 &\sim N(m_0, C_0) \end{aligned}$$

with $F_k, G_k, V_k, W_k, m_0, C_0$ known. We introduce state parameters θ_k, which evolve linearly (plus a random perturbation w_k). The observable variable n_k depends linearly on θ_k (plus a random perturbation v_k).

Finally, the expressions for predictive distributions and moments of X_k, take into account the expressions from predictive moments and densities in DLM's mentioned above. The example illustrates these ideas.

Example We reanalyse now the monthly data, modelling uncertainty in the number of workers. The data suggests a second order polynomial model, given its (changing) linear pattern. Also, we introduce a low coefficient, first order autoregressive term to improve short-term forecasts. Specifically, the model we adopt for the number of workers is

$$\begin{aligned} n_k &= \theta_k^1 + \theta_k^3 + v_k \\ \theta_k^1 &= \theta_{k-1}^1 + \theta_{k-1}^2 + w_k^1 \\ \theta_k^2 &= \theta_{k-1}^2 + w_k^2 \\ \theta_k^3 &= r\theta_{k-1}^3 + w_k^3 \end{aligned}$$

θ_k^1 and θ_k^2 designate, respectively, the (dynamic) intercept and slope in the linear trend. θ_k^3 designates the autoregressive term. Hence, $F_k = (1, 0, 1)$ and

$$G_k = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & r \end{pmatrix},$$

where r is the autoregressive coefficient. Also, $v_k \sim N(0, v)$, i.e. the observation variance is constant but unknown. We allow for learning about this variance by modelling a prior

$$\begin{aligned} (\theta_0^1, \theta_0^2, \theta_0^3)|\phi &\sim N(\mathbf{m_0}, vC^*) \\ \phi &\sim \gamma(n_0/2, d_0/2) \end{aligned}$$

where $\phi = v^{-1}$. Finally, the system error $\mathbf{w_k} = (w_k^1, w_k^2, w_k^3)$ is distributed according to $\mathbf{w_k} \sim N\left(\mathbf{0}, \begin{pmatrix} vW_k^* & 0 \\ 0 & \sigma^2 \end{pmatrix}\right)$, where σ^2 is the autoregressive variance. W_k^* is defined by a discount principle (West and Harrison, 1996), with discount factor .9.

Note that with this model in mind, we would err our forecasts in boom and crisis periods, when the slope trend changes sign, though our model would adapt after a while. This kind of information can be actually incorporated by intervention. We have introduced fairly mild ones, by modifying the model prior slope to 0 and increasing its prior variance, to allow for faster learning.

The unspecified prior parameters were assessed judgementally with standard techniques; for example, for θ_0^1, we assessed m_0^1 by asking the expert the median number of workers, which coincides with the mean in the normal case. Their sensitivity was thoroughly checked, by tring alternative prior specifications within reasonable neigbourhoods of the initial assessments. The next two figures provide forecasts of the number of workers and number of accidents with our model, respectively, showing its effectiveness.

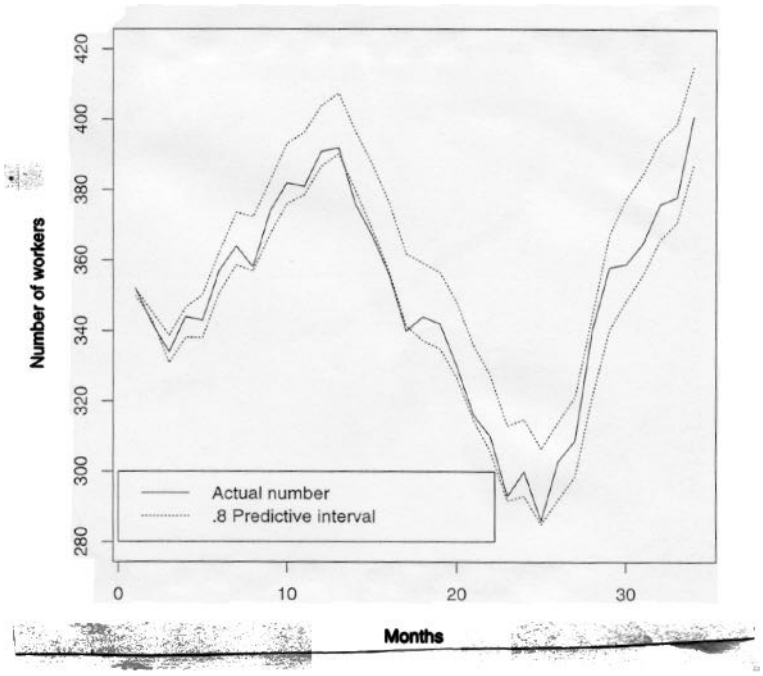

Figure 3.4: Performance of forecasting model

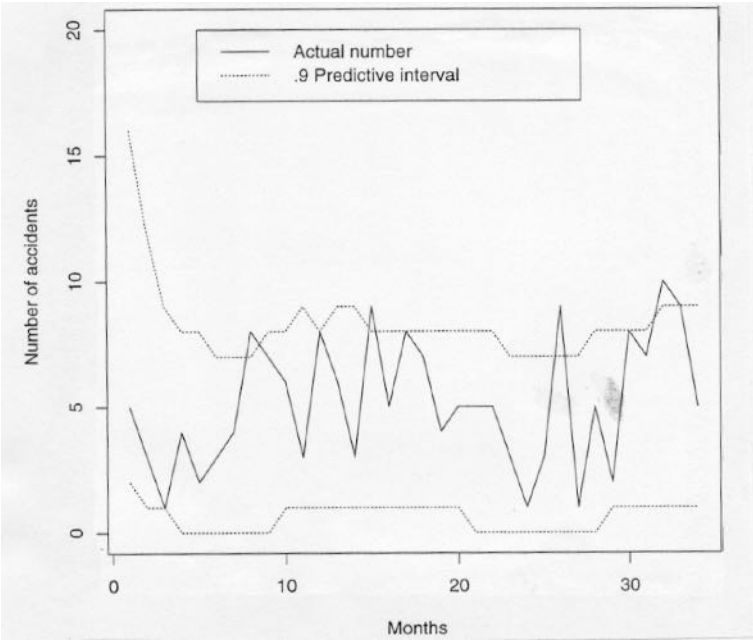

Figure 3.5: Performance of forecasting model

Poisson gamma model with reliability decay

The previous models suggest that the accident rate λ remains the same through the process, though our beliefs about it change. In a more realistic model, we would expect it to evolve. For example, under normal circumstances, we would expect worse performance of companies, due to progressive forgetting of safety rules, until some educational campaign or a nasty accident takes place. This is designated *reliability decay* in the reliability literature. Analogously, we may talk about reliability growth. We shall provide a model for reliability decay in Poisson-gamma models, which may be justified as above. Our final discussion suggests other models, including reliability growth.

The main feature now is an evolution term for the rate $\lambda_k = \lambda_{k-1} + \beta_k$, where β_k is a random variable independent of λ_{k-1}. We assume that all β_k's are i.i.d according to an exponential distribution with parameter β. Evolution will be slow when β is big. Since the β_k's are positive, a priori we have $E(\lambda_k) = E(\lambda_{k-1}) + E(\beta_k) \geq E(\lambda_{k-1})$. Hence, a priori, we expect the rate to increase and reliability to decay. Note, though that, a posteriori, once we have received the data, it may happen that $E(\lambda_k|D_{k+1}) < E(\lambda_{k-1}|D_{k+1}) < E(\lambda_k|D_k)$: a posteriori reliability may grow. Hence, we are modelling a pattern of reliability decay, though the model is flexible enough to allow for reliability growth.

We consider then the following model, which assumes evolution in both

the number of workers and the accident rate, though they are conditionally independent:

$$
\begin{aligned}
X_k|\lambda_k, n_k, D_k &\sim Po(\lambda_k n_k) \\
n_k = F_k\theta_k + v_k & \quad v_k \sim N(0, V_k) \\
\theta_k = G_k\theta_{k-1} + w_k & \quad w_k \sim N(0, W_k) \\
\lambda_k &= \lambda_{k-1} + \beta_k \\
\lambda_0|D_1 &\sim \gamma(a, p) \\
\beta_k &\sim \gamma(\beta, 1) \\
\theta_0|D_1 &\sim N(m_0, C_0)
\end{aligned}
$$

with given $F_k, G_k, V_k, W_k, m_0, C_0, a, p$ and β.

Predictions with this model necessarily requires the use of simulation methods. Specifically, we use a weighted bootstrap resampling strategy described in Smith and Gelfand (1992).

Example We apply the above model to the data. Exploratory analyses on the rate suggested a slow evolution, and we chose $\beta = 1000$. The rest of model is as before. The figures show forecasts for the number of accidents and the evolution of the expected value of the rate.

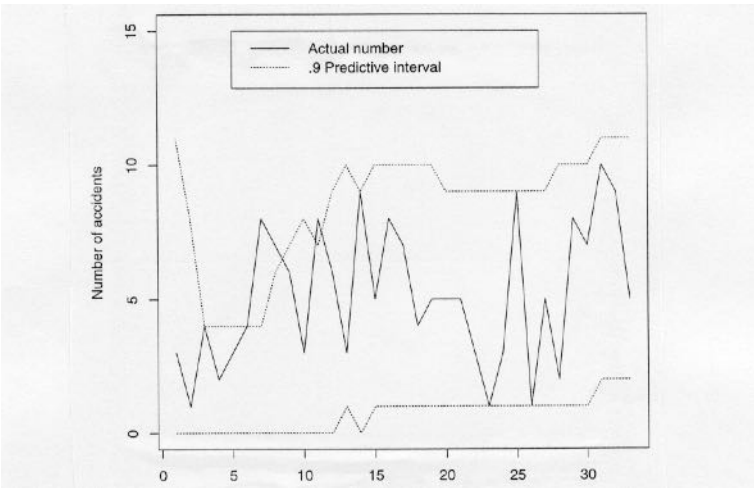

Figure 3.6: Forecasting number of accidents

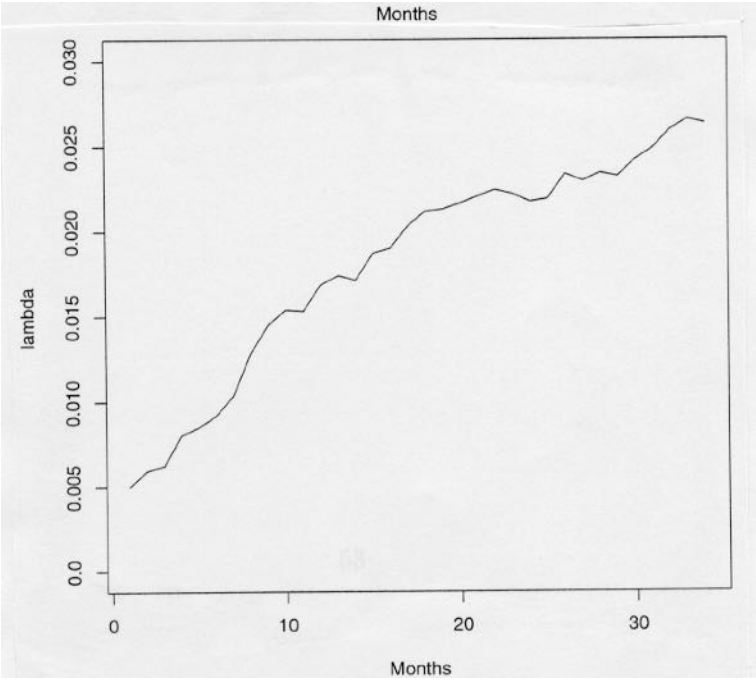

Figure 3.7: Expected value of rate

3.3.3 Mortality forecasting: nonlinear models with incomplete data

- *Key methodological ideas*
  - Illustrate modern computational machinery in a complex model.
  - Illustrate a method for imputing incomplete data.
  - Illustrate informative prior assessment.
- *Application*
  - Forecasting mortality among English and Welsh females.
- *Advantages of the Bayesian approach*
  - Avoid instability of weighted least squares estimates.
  - Mitigate problems of overparameterisation.
  - Mitigate problems of data incompleteness.

Introduction

Representation of mortality data via a parametric model has attracted the attention of actuaries, demographers and statisticians for over a century. The most famous such model is that of Gompertz (1825), still used. This approach of summarising mortality patterns presents many advantages over other ways of describing such data, particularly because it facilitates comparisons over time and space; for example, between cohorts, periods and regions.

A recent attempt to represent mortality across the entire age range has been the eight-parameter non-linear model of Heligman and Pollard (1980). Their suggested model has been used in the past for a wide range of mortality data resulting in satisfactory representations of a variety of patterns. However, estimation of the parameters is problematic due to the overparameterisation of the model, and, also, numerical instabilities in the (commonly used) weighted least squares approach can only be removed by fixing two parameters to be constant. Furthermore, such numerical difficulties create large fluctuations in parameter estimation over time or space, resulting, due to similar inconsistencies in the dispersion matrices, in lack of substantive interpretability.

Here we show a Bayesian inference approach which has the following advantages over currently used methods. First, because the parameters of the model have a straightforward interpretation, use of informative prior distributions resolves the problem of overparameterisation. Secondly, the non-normality of the likelihood surface in the parameterisation usually adopted means that the least square estimates are inadequate. Thirdly, application to an incomplete life table can be routinely made using simulation-based Bayesian computation methodology. Fourthly, posterior densities of other quantities of interest such as the joint lifetime of a couple or the median lifetime of a person can be derived.

Inference for complete life tables

Heligman and Pollard (1980) suggested a model which represents the underlying life-table probabilities π_x of dying between exact ages x and $x+1$, for $x = 0, 1, 2..., n$, via the curve

$$\frac{\pi_x}{1-\pi_x} = A^{(x+B)^C} + D\exp\left(-E(\log(x/F))^2\right) + GH^x. \tag{3.11}$$

Note that the idea behind the above model is to decompose the odds that an individual of age x will die before he or she attains age $x+1$ into three parts: a child mortality curve, an accident hump in early adult life, and an adult mortality curve. As an example, in the figure we present $\log(\pi_x)$ and its three parts evaluated from (3.11) for certain values of A, B, C, D, E, F, G, H.

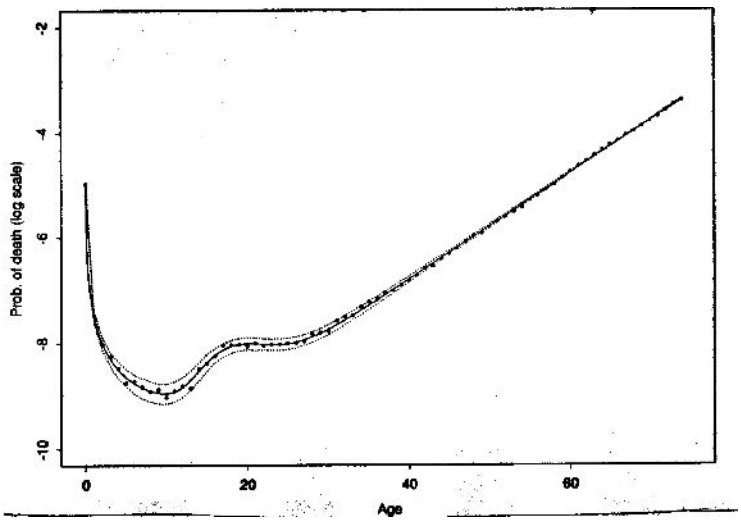

Figure 3.8: Probablity of death (log scale)

The interpretation of the parameters in the above model is straightforward. A, taking values in the interval $(0,1)$, represents the infant mortality rate. B represents the one-year-olds mortality rate, taking values within the interval $(0,1)$. C takes values in $(0,1)$ and is closely associated with the rate of mortality decline, or, the rate at which an individual adapts to his environment. The three parameters of the mid-life mortality component, D, E and F, reflect what is often referred to in the demographic literature as the accident hump. In particular, D indicates the severity of the accident hump and takes values in $(0,1)$; E is defined in $(0,\infty)$ and is related to the spread, with large values indicating a concentrated accident hump; and F is indicative of the location with domain the interval $(15, 110)$. Finally, G and H have as domains the intervals $(0,1)$ and $(0,\infty)$ and indicate the base level of later adult mortality and the rate of increase in mortality at the later adult ages, respectively.

A lot of authors have noticed that the estimation of the parameters $\boldsymbol{\theta} = (A, B, C, D, E, F, G, H)$ by least squares methods is problematic. In fact, Heligman and Pollard (1980) suggested the use of weighted least squares with weights equal to $w_x = 1/{q_x}^2$, where q_x is the observed raw frequency estimate of π_x. In an experimental study, Kostaki (1992) notes that other forms of weights w_x cannot provide any form of convergence. Rogers (1986) and Congdon (1993) fix two parameters to avoid numerical instabilities in the non-linear least squares minimisation procedure. Note that in general, with the exception of the work of Congdon (1993), all authors who have used the above model, do not report estimates of the dispersion matrix, a task which poses further computational problems.

A non-linear logistic model Following the Bayesian framework, we treat all model parameters as unknowns and we specify prior information via probability density functions. The demographic interpretability of the parameters will result in informative priors for nearly all the parameters. Let n_x be the population at risk having age in the interval $[x, x+1)$ and d_x the number of people who die at that age. Treating counts of dead persons at different ages as independent random variables, each of them follows a binomial distribution

$$p(d_x) = \binom{n_x}{d_x} \pi_x^{d_x} (1 - \pi_x)^{n_x - d_x}$$

and so if $\boldsymbol{d}$ is the vector with elements $d_x,\ x = 0, 1, \ldots, n$ we obtain

$$p(\boldsymbol{d}) = \prod_{x=0}^{n} \binom{n_x}{d_x} \pi_x^{d_x} (1 - \pi_x)^{n_x - d_x}.$$

If we denote the right hand side of (1) by $K(x)$ it follows that

$$p(\boldsymbol{d}) = \prod_{x=0}^{n} \binom{n_x}{d_x} [K(x)]^{d_x} [1 + K(x)]^{-n_x}. \tag{3.12}$$

Note that this is a problem of estimating the parameters of a generalised non-linear model.

Denoting by $p(\boldsymbol{\theta})$ the prior distribution for $\boldsymbol{\theta} = (A, B, C, D, E, F, G, H)$, it follows that for data $\boldsymbol{d} = d_x, x = 0, 1, \ldots, n$ the resulting posterior is of the form

$$p(\boldsymbol{\theta} \mid \boldsymbol{d}) \propto p(\boldsymbol{\theta}) p(\boldsymbol{d}) = p(\boldsymbol{\theta}) \prod_{x=0}^{n} \binom{n_x}{d_x} [K(x)]^{d_x} [1 + K(x)]^{-n_x}.$$

Analytic integration of this 8−dimensional nonnormalised posterior joint distribution is not feasible so we adopt a simulation based approach to obtaining the desired posterior summaries. In particular, we suggest using a Markov chain Monte Carlo algorithm at each iteration of which we update all elements of $\boldsymbol{\theta}$ simultaneously, through Metropolis steps.

An illustrative example We illustrate the methodology of the previous subsections using the 1988-1992 mortality data of English and Welsh females. Estimated resident populations on the 30th of June of each year and the yearly counts of deaths were summed up across the five years to avoid having the observed mortality rates influenced by extreme random fluctuations.

To each component parameter, we assign, independently, a log-normal distribution $\log\theta_i \sim N(\mu_i, \sigma_i^2)$, where μ_i, σ_i are implicitly defined by specifying z_{i1}, z_{i2} to be suitable lower and upper percentiles of the normal distribution (in this application, we obtained the 1% and 99% percentile values by discussion with a colleague). The 1% values are set at

$$(10^{-4}, 10^{-4}, 10^{-2}, 5 \times 10^{-5}, 0, 15, 10^{-7}, 1)$$

and the 99% values at

$$(2 \times 10^{-2}, 15 \times 10^{-2}, 3 \times 10^{-1}, 10^{-2}, 20, 110, 10^{-3}, 12 \times 10^{-1}).$$

These represent vague, but still informative, prior beliefs. Given experience with a series of such problems, more informative forms of prior beliefs would, of course, evolve, as would some knowledge of parameter correlations.

We are now in a position to perform the Markov chain Monte Carlo analysis algorithm. For our example, the chosen sampling strategy was to simulate a single chain. After a number of "burn-in" iterations (100000) required for convergence we stored the outcome of every 50th iteration until a sample of 2500 realisations was formed. The figure shows the results. The dots represent the observed probabilities of death. For each model, curve (3.11) was evaluated at the posterior mean of $\boldsymbol{\theta}$ which was estimated by the corresponding sample mean.

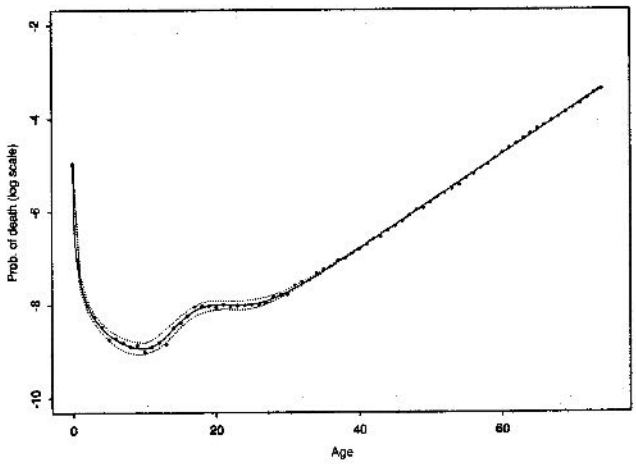

Figure 3.9: Probability of death (log scale)

The figure also shows 95% predictive intervals for the logarithm of the observed probabilities of death across age. They were calculated as follows. For the logistic non-linear model and for each integer age we have, via the stored output sample of $\boldsymbol{\theta}$, 2500 estimates of the true probability of dying at that age, $q_x(i)$ say, $i = 1, \ldots, 2500$. For each $q_x(i)$ we sample one value $d_x(i)$ from the corresponding binomial distribution $Bin(b_x, q_x(i))$ and thus we obtain a posterior predictive sample $d_x(i)/n_x(i)$ for the observed probability of dying. The empirical 2.5% and 97.5% quantiles of that sample form the above mentioned intervals. We see that the estimated curve provides a good fit for the observed frequencies.

When plotting the marginals, we may see that these are markedly skew to the right, underlying the danger of posterior or likelihood normal approximations in the original parameterisation.

Inferences from incomplete life tables

Mortality data are often collected using five-year age groups rather than individual years of life, except for the first five years, which are presented in two intervals $[0, 1)$ and $[1, 5)$. Thus, the data available - so-called incomplete or abridged life tables- are of the form d_0, D_1, D_x, $x = 5, 10, 15, \ldots, n'$, where D_x denotes the number of people who died at age in the interval starting at x, $[x, x+1, \ldots, x+k]$ say, and n' is the starting age of the last group. Such data are collected, for example, from the World Health Organisation. This is a common method of describing mortality patterns for reasons of convenience, especially in countries with incomplete and unstable documentation of vital statistics.

The problem of extending an incomplete to a full life table has been stud-

ied by Mode and Busby (1982), Pollard (1989) and Kostaki (1991). ¿From a Bayesian perspective, the incomplete life table problem can be seen as an incomplete data problem, or, as we shall show below, as a constrained parameter problem. We can then use a general approach to such situations using a Markov chain Monte Carlo strategy.

We assume, as in the case of complete data, that the true probability π_x of a person dying at age x is given by HP formula. We again denote by $\boldsymbol{\theta}$ the parameter vector, and by d_x the number of people dying at age x which is, as before, binomially distributed with parameters n_x and q_x. The only known non-random quantities are the grouped population counts N_x and the population with age 0 at last birthday and the only observed data are the grouped death counts D_x and the number of dead under one year olds. All the other quantities, i.e. $\boldsymbol{\theta}$, d_x's and n_x's, will be considered, in a Bayesian perspective, as unknowns. Let $\boldsymbol{N}$, $\boldsymbol{D}$, $\boldsymbol{n}$ and $\boldsymbol{d}$ be the vectors of the corresponding quantities. Note that $\boldsymbol{n}$ and $\boldsymbol{d}$ do not contain the quantities for $x = 0$.

In this case the full model for the data and the unknowns given everything that is known will be

$$p(\boldsymbol{D}, \boldsymbol{d}, d_0, \boldsymbol{n}, \boldsymbol{\theta}|\boldsymbol{N}, n_0) = p(\boldsymbol{D}|\boldsymbol{d}, \boldsymbol{n}, \boldsymbol{\theta}, \boldsymbol{N}, d_0, n_0)p(\boldsymbol{d}|\boldsymbol{n}, \boldsymbol{\theta}, \boldsymbol{N}, d_0, n_0)\times$$

$$p(d_0|\boldsymbol{n}, \boldsymbol{\theta}, \boldsymbol{N}, n_0)p(\boldsymbol{n}, \boldsymbol{\theta}|\boldsymbol{N}, n_0)$$

We now examine each of the above conditional densities in some detail. First, we note that $\boldsymbol{D}$ depends only on $\boldsymbol{d}$ since

$$D_x = \Sigma_{i=x}^{x+k} d_i$$

and therefore

$$p(\boldsymbol{D}|\boldsymbol{d}, \boldsymbol{n}, \boldsymbol{\theta}, \boldsymbol{N}, d_0, n_0) = \prod p(D_x|d_x, \ldots, d_{x+k}),$$

a product of indicator functions, the product ranging over $x = 1, 5, 10, \ldots, n'$.

When $\boldsymbol{n}$ and $\boldsymbol{\theta}$ are known, $\boldsymbol{d}$ depends only on these quantities, and, as in the previous section, its components are independent random quantities binomially distributed:

$$p(\boldsymbol{d}|\boldsymbol{n}, \boldsymbol{\theta}, \boldsymbol{N}, d_0, n_0) = p(\boldsymbol{d}|\boldsymbol{n}, \boldsymbol{\theta}) = \prod_{x=1}^{n} p(d_x|n_x, \pi_x(\boldsymbol{\theta})).$$

where by $\pi_x(\boldsymbol{\theta})$ we denote the probability of dying as a function of the parameters of the Heligman-Pollard formula.

Similarly,

$$p(d_0|\boldsymbol{n}, \boldsymbol{\theta}, \boldsymbol{N}, n_0) = p(d_0|n_0, \pi_0(\boldsymbol{\theta})).$$

Finally, we can consider that, *a priori*, $\boldsymbol{\theta}$ is independent of anything else and that $\boldsymbol{n}$ given $\boldsymbol{N}$ is independent of $\boldsymbol{\theta}$ so that

$$p(\boldsymbol{n}, \boldsymbol{\theta}|\boldsymbol{N}, n_0) = p(\boldsymbol{\theta})p(\boldsymbol{n}|\boldsymbol{N}, n_0)$$

where $p(\boldsymbol{\theta})$, is the same prior of $\boldsymbol{\theta}$ as in the complete data case, while $p(\boldsymbol{n}|\boldsymbol{N}, n_0)$ is, likewise, the expression of our prior beliefs about population counts per year of age in the light of the observed counts alone. The prior we choose is of the form

$$p(\boldsymbol{n}|\boldsymbol{N}, n_0) = \prod p(n_x, \ldots, n_{x+k}|N_x).$$

The product is, again, over $x = 1, 5, 10, \ldots, n'$. We assume that counts from different age groups are independent given the grouped counts and that within each group the distribution of the individual counts is uniform over the set of all combinations of positive values that sum up to the grouped count. We do this because *a priori* we have no information about $\boldsymbol{n}$.

Simulation specifics The quantity of interest is $\boldsymbol{\theta}$, but since we have missing data we have to also carry these along during the simulation stage. Therefore, we seek to sample from $p(\boldsymbol{\theta}, \boldsymbol{d}, \boldsymbol{n}|\boldsymbol{D}, \boldsymbol{N}, d_0, n_0)$. This is proportional to the full model mentioned in the previous section. A convenient simulation method is the Gibbs sampler.

We update each vector separately because of their different characteristics. The vectors have complicated conditional posterior distributions, and therefore for each vector we use the Metropolis-Hastings algorithm which can handle distributions of arbitrary complexity.

Based on the results of the previous section it is seen that the conditional posterior distribution of $\boldsymbol{\theta}$,given everything else, is

$$p(\boldsymbol{\theta}|\cdot) \propto p(\boldsymbol{d}|\boldsymbol{n}, \boldsymbol{\theta}) p(d_0|n_0, \pi_0(\boldsymbol{\theta})) p(\boldsymbol{\theta})$$

the same as in the full data case. The conditional posterior distribution of $\boldsymbol{d}$ is

$$p(\boldsymbol{d}|\cdot) \propto p(\boldsymbol{D}|\boldsymbol{d}) p(\boldsymbol{d}|\boldsymbol{n}, \boldsymbol{\theta})$$

and that of $\boldsymbol{n}$ is easily seen to be

$$p(\boldsymbol{n}|\cdot) \propto p(\boldsymbol{d}|\boldsymbol{n}, \boldsymbol{\theta}) p(\boldsymbol{n}|\boldsymbol{N}, n_0).$$

An example For the purpose of illustration -and to aid direct comparison with our earlier analysis of the English/Welsh data- we have grouped the data into the above-mentioned age intervals. We have used exactly the same prior distributions for $\boldsymbol{\theta}$ as in the complete data case and a similar simulation strategy. The figure show the fitted curve and the 95% predictive intervals, which are slightly wider than those of the full data case.

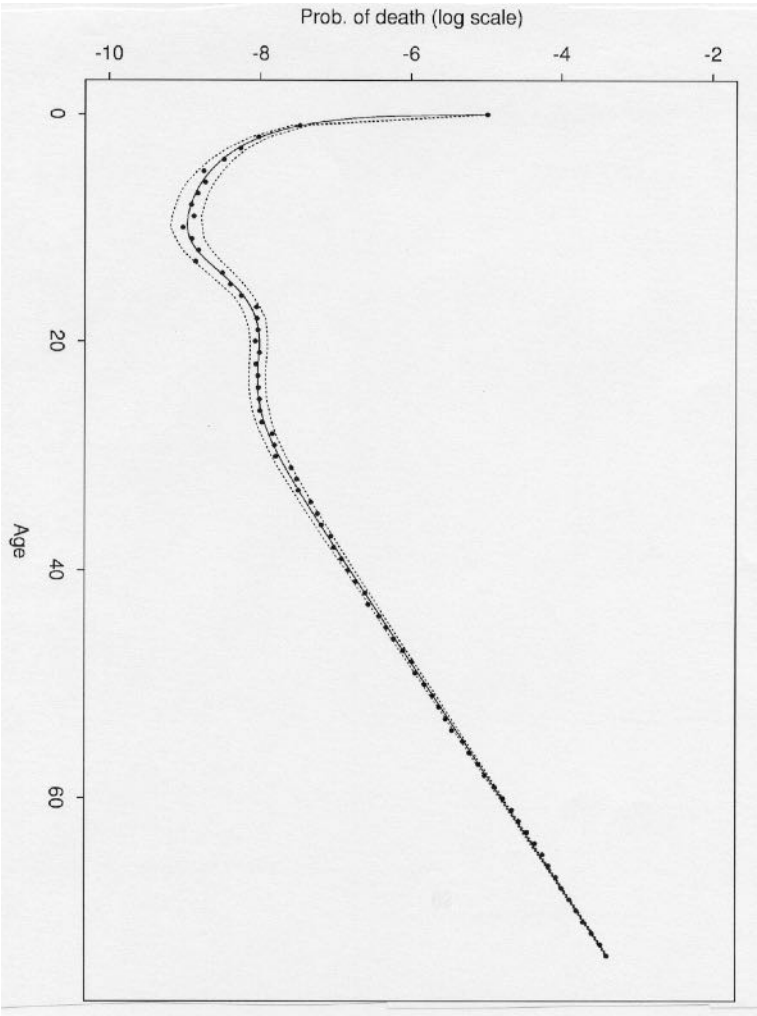

Figure 3.10: Probability of death (log scale)

Many other quantities derived from the underlying model parameters may be of interest to official statisticians and demographers. These include:

1. The survival probability, defined as the probability of surviving from age x to age $x + t$.
2. The median lifetime.

Least squares approaches are not well suited to deriving inferences for non-linear transformations of basic parameters. However, the simulation-based Bayesian framework provides an easy way to calculate such quantities.

3.3.4 Institutional evaluation: An application of Bayesian Ordinal regression

- *Key methodological ideas*
  - Describe a hierarchical ordinal regression model as an example of a Bayesian generalised linear model.
  - Illustrate modern computational methods.
  - Latent variables.
- *Application*
  - Evaluate performance of an institution, identifying key factors affecting such evaluation.
- *Advantages of the Bayesian approach*
  - More naturally and efficiently deal with ordinal data.
  - Avoid numerical unstability of maximum likelihood estimates.
  - Avoid relying on large sample properties on mle's, unsuitable in this complex problem.

Introduction

Institutions are paying increasing attention to service quality as a way to take the lead over competitors. For example, the Spanish government has recently issued a law to evaluate their services, and various autonomous governments have quality of service units to perform such evaluations through citizen surveys. This will become more and more important as some economies focus on services, as is happening in Spain with the enormous importance of tourism. This creates the need for much more work on service quality modelling, an underdeveloped research opportunity in its quantitative side, at least by comparison with industrial quality control. Rust and Metters (1996) provide an excellent introductory review.

Modelling issues refer mainly to service quality measurement (what and how to measure), modelling (how to use measurements to predict customer

behaviour, quality improvement,...) and decision analysis (how to motivate service suppliers, how much to invest in service improvement programs,...).

As far as measurement is concerned, there is considerable discussion in the literature distinguishing concepts like customer satisfaction, perceived quality, disconfirmation,... Various ways have been used to measure satisfaction including postal and telephone customer surveys, internal satisfaction surveys,... SERVQUAL (Parasuraman et al. 1988, 1991) has somehow emerged as a measurement standard, but see various criticisms in Parasuraman et al. (1991). Once with measurements, emphasis has been placed on building linear indicators relating service quality or customer satisfaction with explanatory variables, see e.g. Peña (1996), yet, for reasons outlined in Johnson (1996), these might not be entirely appropriate, since we shall be typically using ordinal scales. Based on these regression models, we could predict the impact of various variables on satisfaction; classify customers according to satisfaction; compare branches, or other subdivisions, of the organisation,...

Here, we discuss several issues concerning service quality models. We suggest statistical models to analyse the involved issues. The general theme is that of multivariate hierarchical generalized linear models and ordinal regression models. The framework proposed in modelling ordinal regression is based on the so-called category boundaries or threshold approach.

Data and Problem Description

Our initial problem is that of evaluating consumer satisfaction within an organisation, e.g. a government office, and relate it with influencing factors, with the aim of improving satisfaction. Assume the organisation is divided into suborganisations, e.g. branches. A customer satisfaction survey is undertaken: n_j customers, $j = 1, ..., J$, are questioned at each of J branches. In it we demand:

- Satisfaction with the branch and with the organisation as a whole, which we designate respectively y_{ij}^{11} and y_{ij}^{12}, $i = 1, ..., n_j$, $j = 1, ..., J$.
- Quality of various branch specific aspects related to satisfaction, as suggested by the organisation which, in, e.g., the case of government offices, might include staff efficiency, staff kindness, accuracy of written information, queueing times,..., which we code as y_{ij}^{2h}, with $i = 1, ..., n_j$, $j = 1, ..., J$ and $h = 1, ..., H$, H being the number of aspects demanded.

These aspects will vary from one organisation to another. These variables will be jointly designated as y^1 and y^2. Assume all its components are coded in an ordinal scale, with K levels and labels going from, say, *totally disagree* to *totally agree*, and higher levels indicating greater satisfaction. Since scales are ordinal, standard statistical techniques are not directly applicable. We also have access to a customer database which provides us with covariates for each individual, say sex, years as customer, income level,..., which we designate x_{ij}^s, for individual i at branch j, with $s = 1, ..., S$, and S the number of covariates, jointly designated as x. Our initial aim is to relate the key variables y^1 with explanatory variables

y^2, conditionally or unconditionally on x. Such model would allow us to, among other things:

- Identify which of the y^2 variables are more influential on the y^1 variables, allowing us, therefore, to find out weak and strong points of the organisation and/or branch.
- Rank branches within an organisation according to service quality, allowing us to identify branches requiring remedial action.

Eventually, these models would drive important organisational decisions like where to place money in educational campaigns, which employees to promote,...

Model formulation

In this section, we describe a three stage model for our basic problem. The key assumption is that of an underlying continuous process which is later discretised to provide responses y^1 and y^2. In addition to this latent trait, a hierarchical model structure incorporates parameters useful to compare branch performance and facilitate inference.

Relating y^1 with y^2 and x We first relate y^1 with y^2 and x. We shall assume that there are underlying latent variables z_{ij}^{11} and z_{ij}^{12}, for each individual i within branch j, that describe the underlying evaluation process for the branch and the organisation. z_{ij}^{11} will be governed by covariates y^2 and x, and z_{ij}^{12} by z_{ij}^{11}, specifically

$$z_{ij}^{11} = \beta_j^1(1, y_{ij}^2, x_{ij}) + \epsilon_{ij}^1,\ \epsilon_{ij}^1 \sim N(0, \sigma_{1j}^2)$$
$$z_{ij}^{12} = z_{ij}^{11} + \epsilon_{ij}^2,\ \epsilon_{ij}^2 \sim N(0, \sigma_{2j}^2)$$

We therefore assume a linear relation between z^{11} and (y^2, x), with weights depending on the branch, whereas z^{12} is a perturbation of z^{11}. We then assume that

$$y_{ij}^{11} = m \iff \theta_{m-1}^{1j} \leq z_{ij}^{11} < \theta_m^{1j}$$
$$y_{ij}^{11} = k \iff \theta_{k-1}^{2j} \leq z_{ij}^{12} < \theta_k^{2j}$$

where $\theta_0^{ij} = -0.5 < \theta_1^{ij} < ... < \theta_K^{ij} = K - 0.5, i = 1, 2$, are category thresholds for branch j, $j = 1, ..., J$. Parameters in this stage include regression weights β^1, variances σ_1^2, σ_2^2 and category cutoffs θ^1 and θ^2. Implicit in this model is the assumption that there is certain evaluation homogeneity within each branch (same variances, cutoffs and regression weights). Conditional exchangeability is assumed for y_{ij}^2, given covariates, and for y_{ij}^1, given y_{ij}^2 and the covariates.

As priors are concerned, for the regression weights we shall assume that

$$\beta_j^1 \sim N(\beta_0, \Sigma_0).$$

The rest of the prior model follows Johnson (1996). For variances, we assume that

$$\sigma_{ij}^2 \sim IG(\alpha_i, \lambda_i) \propto (\sigma_{ij}^2)^{-(\alpha_i - 1)} \exp(-\lambda_i / \sigma_{ij}^2), i = 1, 2$$

To simplify the discussion, we shall assume a flat prior over $\lambda_i, i = 1, 2$, and fix $\alpha_i, i = 1, 2$ at appropriate values. With two parameters, we obtain a flexible range of inverse-gamma models, even if we fix one of the parameters. Specifically, we shall do $\alpha_i = 2$, to allow for certain vagueness. Cutoffs θ^{ij} will follow a uniform prior, over the appropriate region.

Relating y^2 with x Similarly, we assume an underlying latent variable $z_{ij}^{2h}, h = 1, ..., H$, for each explanatory aspect, so that

$$z_{ij}^{2h} = \beta_j^{2h}(1, x_{ij}) + \epsilon_{ijh}^3, \ \epsilon_{ijh}^3 \sim N(0, \sigma_{3hj}^2)$$

and

$$y_{ij}^{2h} = l \Longleftrightarrow \theta_{l-1}^{3hj} < z_{ij}^{2h} \leq \theta_l^{3hj}$$

with similar notation as above. We assume independence of all H responses for each customer, given the explanatory variables. This hypothesis may be reasonable for appropriately designed studies, e.g. following SERVQUAL dimensions. Extensions to non-independent cases are simple.

Parameters include the regression weights β^{2h}, variances σ_{3h}^2, and category cutoffs θ^{3h}. Prior formulation is similar to that above, with independent priors for different classes of parameters.

Computations The joint model, conditional on x and including latent variables, may be described as

$$f(y^1 g z^1, y^2, x, \beta_1, \theta^1, \theta^2, \sigma_1, \sigma_2) f(y^2 g z^2, x, \beta_2, \theta^3, \sigma_3).$$

The general form of the likelihood will be

$$l(z^1, \beta_1, \theta^1, \theta^2, \sigma_1, \sigma_2) l(z^2, \beta_2, \theta^3, \sigma_3).$$

Then, taking into account the independence of priors, we may actually perform inference in two stages for the y^1 and y^2 models. We describe in detail a posterior simulation scheme for y^1. The second one is similar.

We may see that the joint posterior density is proportional to:

$$\prod_{j=1}^{J} [\prod_{i=1}^{n_j} f(z_{ij}^{12} | z_{ij}^{11}, \sigma_{2j}^2) f(z_{ij}^{11} | \beta_j^1, (y_{ij}^2, x_{ij}), \sigma_{1j}^2) I(\theta_{y_{ij}^{12}-1}^{2j} \leq z_{ij}^{12} < \theta_{y_{ij}^{12}}^{2j})$$

$$I(\theta_{y_{ij}^{11}-1}^{1j} \leq z_{ij}^{11} < \theta_{y_{ij}^{11}}^{1j})] \ p(\sigma_{1j}^2 | \lambda_1, \alpha_1) p(\sigma_{2j}^2 | \lambda_2, \sigma_2) p(\beta_j^1 | \beta_0, \Sigma_0)$$

in the appropriate region.

Given the previous expression, all conditional posteriors are proportional to known density functions. Hence an MCMC scheme can be implemented in our model with two underlying latent variables. The steps may be summarized as follows:

1. Initialize parameters. Fix initial values for θ, from $\theta_0 = -0.5$ to $\theta_K = K - 0.5$, with other values equally spaced. For each observed y_{ij}^{11}, sample a value for z_{ij}^{11} from a uniform distribution in $(\theta_{y_{ij}^{11}-1}, \theta_{y_{ij}^{11}})$. Proceed similarly for z_{ij}^{12}. σ_{1j}^2 and σ_{2j}^2 can be initialized as the variance of the previous values. For β_j^1 take the classical least squares estimator $(X_j^T X_j)^{-1} X_j^T Z_j$ as initial value, where Z_j is an n_j vector containing the previously obtained values z_{ij}^{11}, and X_j is the $n_j \times (1 + H + S)$ design matrix whose rows are $(1, y_{ij}^2, x_{ij})$, for $i = 1, ..., n_j$. Set $\lambda_1 = \lambda_2 = 2$. (We are initially assuming relatively vague priors on each variance parameters).

   Until convergence is detected, iterate through

2. For each branch j and customer i, sample z_{ij}^{11} from a normal distribution with mean
$$\frac{z_{ij}^{12}/\sigma_{2j}^2 + \beta_j^1(1, y_{ij}^2, x_{ij})/\sigma_{1j}^2}{1/\sigma_{1j}^2 + 1/\sigma_{2j}^2}$$
   and variance $(1/\sigma_{1j}^2 + 1/\sigma_{2j}^2)^{-1}$ truncated at $(\theta_{y_{ij}^{11}-1}^{1j}, \theta_{y_{ij}^{11}}^{1j})$.

3. For each branch j and customer i, sample z_{ij}^{12} from a normal distribution with mean z_{ij}^{11} and variance σ_{2j}^2 truncated at $(\theta_{y_{ij}^{12}-1}^{2j}, \theta_{y_{ij}^{12}}^{2j})$.

4. For each branch j, sample from a $(1 + H + S)$-normal distribution a value for β_j^1 with mean vector
$$\beta_j^* = (\Sigma_0^{-1} + X_j^T X_j)^{-1}(\Sigma_0^{-1}\beta_0 + X_j^T Z_j)$$
   and variance-covariance matrix
$$(\Sigma_0^{-1} + X_j^T X_j)^{-1}$$
   with Z_j a vector with entries z_{ij}^{11}.

5. For each branch j, sample σ_{1j}^2 from an inverse gamma distribution with, respectively, shape and scale parameters
$$n_j/2 + \alpha_1 \qquad \text{and} \qquad \lambda_1 + \Sigma_{i=1}^{n_j}(z_{ij}^{11} - \beta_j^1(1, y_{ij}^2, x_{ij}))^2/2$$

6. For each branch j, sample σ_{2j}^2 from an inverse gamma distribution with, respectively, shape and scale parameters
$$n_j/2 + \alpha_2 \qquad \text{and} \quad \lambda_2 + \Sigma_{i=1}^{n_j}(z_{ij}^{12} - z_{ij}^{11})^2/2$$

7. For each branch j, and level k, $k = 1, ..., K-1$, sample θ_{1k} from a uniform distribution on the interval
$$\left(\max_{\substack{i \\ y_{ij}^{1j}=k}} z_{ij}^{11}, \min_{\substack{i \\ y_{ij}^{1j}=k+1}} z_{ij}^{11} \right)$$

and θ_{2k} from a uniform distribution on the interval

$$\left(\max_{\substack{i \\ y_{ij}^{2j}=k}} z_{ij}^{12}, \min_{\substack{i \\ y_{ij}^{2j}=k+1}} z_{ij}^{12} \right)$$

assuming the intervals are well-defined. Otherwise modify them as in Albert and Chib (1993).

8. Sample λ_1 from a gamma density with parameters $J\alpha_1 + 1$ and $\Sigma_{j=1}^{J} 1/\sigma_{1j}^2$

9. Sample λ_2 from a gamma density with parameters $J\alpha_2 + 1$ and $\Sigma_{j=1}^{J} 1/\sigma_{2j}^2$

Conceptual answers to the posed problems

Once with a scheme for posterior simulation, we describe answers to the problems posed. An additional difficulty in this setup is the provision of a meaningful way to communicate results to users, which are not very technical.

Describing how variables affect customer satisfaction. The basic issue is how do we convey the meaning of our regression model and how improvements in explanatory aspects might improve satisfaction. The procedure we suggest is as follows. Start assuming that x is fixed. Based on the predictive distribution of $y^2|x$, we may define a typical customer, e.g., through its medians $(y_x^{21}, ..., y_x^{2H})$. To see the effect of an improvement on each of the explanatory aspects, we may consider the improved customer $y_x^{*2h} = (y_x^{21}, ..., y_x^{2h}+1, ..., y_x^{2H})$ and base inference on the predictive distribution of $y^1|y_x^{*2h}, x$ through plots or moments. We have found also useful to communicate the meaning of the model considering the worsened customer $y_{*x}^{2h} = (y_x^{21}, ..., y_x^{2h} - 1, ..., y_x^{2H})$ and provide similar inferences based on $f(y^1|y_{*x}^{2x}, x)$.

Comparing branches. Implicitly, our model considers branch differences through parameters β_j^1, σ_{1j}^2 and σ_{2j}^2. Adopting an approach similar to that in Goldstein and Spiegelhalter (1996), after each iteration of the sampler we may aggregate the z_{ij}^{11} and z_{ij}^{12}, that is, consider

$$z_{.j}^{11} = \Sigma_{i=1}^{n_j} z_{ij}^{11}/n_j \ \ and \ \ z_{.j}^{12} = \Sigma_{i=1}^{n_j} z_{ij}^{12}/n_j.$$

We may, then, plot 95% intervals for $z_{.j}^{11}$ and $z_{.j}^{12}$, which would give an intuitive idea of differences in customer satisfaction between branches, and customer satisfaction between branch and organisation.

Example We describe a simplified case study undertaken for a regional organisation in Spain. Fifty branches of its network were surveyed, with around fifty customers per branch. Six explanatory aspects were included (effectiveness, competent advice, friendliness, written reports, privacy and waiting time). A 1-7 scale was used. An additional open question demanded suggestions for

possible improvements. The survey was conducted by phone. 5000 iterations were allowed for burn-in, and 4000 observations were later collected. The figure shows boxplots for customer satisfaction (y^{11}) at eight branches and satisfaction with organisation, branch and service aspects at one of the branches. They suggest, in general, high satisfaction.

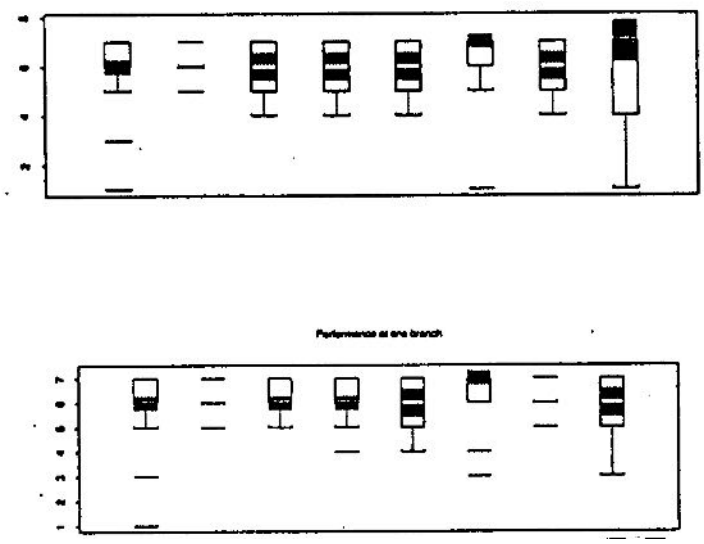

Figure 3.11: Performance at one branch and customer satisfaction at eight branches.

The next figure shows 95% credible intervals for $z_{\cdot}^{11}$ and $z_{\cdot}^{12}$ at those eight branches. We may appreciate, e.g, greater satisfaction within branch 6, both as far as the organisation and the branch are concerned. Evaluations of branch and organisation are fairly similar for branches 2 to 7; branch 1 evaluates branch better than organisation; branch 8 evaluates organisation better than branch.

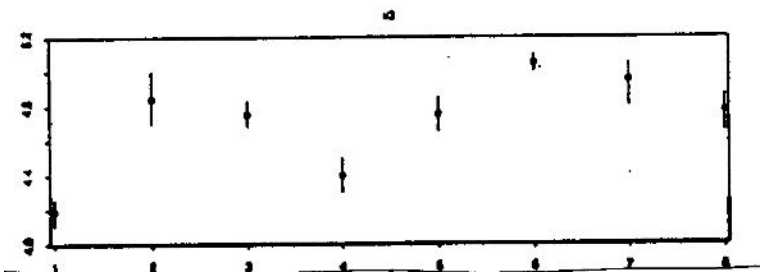

Figure 3.12: z_{11} and z_{12} at eight branches.

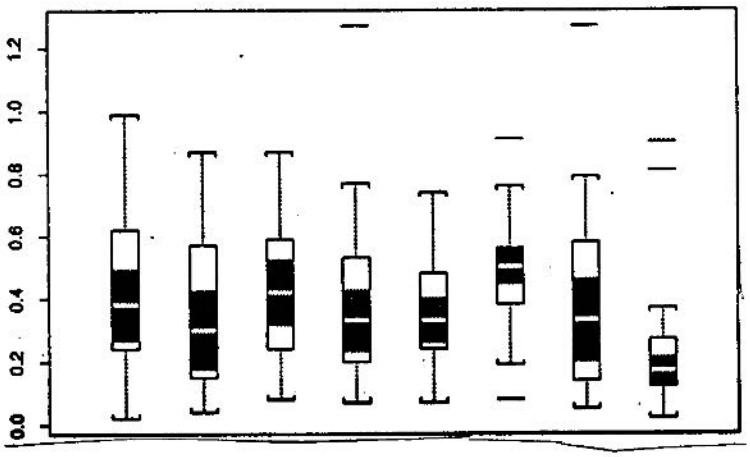

Figure 3.13: Posterior distribution of β_3.

We also show boxplots for the posterior distribution of β_3, coefficient for competent advice, at the eight branches. Branch 6 gives slightly more weight than other branches to competent advice, whereas branch 8 gives slightly less weight than the others.

The table shows expected predictive probabilities for customer satisfaction of 6 or 7 for the improved customer. The baseline value, obtained for the typical customer based on the medians, was 0.4. Note the improvements in satisfaction, and specially the bigger improvement when advice is demanded to be more competent (aspect 2).

Effectiveness	0.499
Compet.Adv.	0.522
Friendliness.	0.499
Written Info.	0.500
Privacy	0.501
Waiting time	0.504

Expected probabilities for improved customer

3.3.5 Other Case studies

We conclude with a quick overview of case studies and applications available in the literature or mentioned by our surveyees.

As we have mentioned, small area estimation techniques are to a great extent Bayesian, see Ghosh and Rao (1994) for an appraisal. Examples include the Federal State Cooperative Program initiated by the US Bureau of the Census, to provide high-quality consistent series of county (and subcounty), population estimates, the estimation of per capita income, for the same organisation and

the Erickssen-Kadane proposal for census adjustment. See Dick (1995) for a related problem, solved with an empirical Bayes approach, to the Canadian census adjustment. Other examples presented by Ghosh and Rao refer to estimation of areas under corn and soybean and the estimation of mean wages and salaries. Fay, Nelson and Litow (1993) further illustrate small area estimation and empirical Bayes methods to estimate median income for 4-person families. Ghosh et al (1996) treat a dynamic version of the problem with an autoregressive model.

Ghosh and Meeden (1997) provide several examples in their monograph in Bayesian smapling. For example, they explain how to deal with nonresponse in the Current Population Survey, a large monthly survey by the Census Bureau to provide data on yearly income. An example of cluster sampling, hierarchical models and empirical Bayes is given by Malec et al (1997) for the National Health Interview Survey, where binary variables are considered.

Work by Baskin and collaborators (1992, 1993) describe variance estimation methods for the US consumer price index. Zaslavsky (1993) illustrates the combination of various data sources to estimate population shares, mitigating undercounts.

There is some work as well in Bayesian demography. Apart from our example taken from Dellaportas et al (1999), we should mention the work by Kadane et al (1997) on projections of the Kurd population.

Bernardo (1990, 1996) has discussed various models for election forecasting.

Some organisations are, in fact, implementing on a large scale Bayesian methods. One example is the Devices Division of the US Food and Drug Administration. Simon French is undertaking several risk analysis consulting problems with the British Department of Health. Charles Whiteman (Iowa) has consulted with the Governor of Iowa to implement Bayesian methods to forecast tax receipts and manage state budgets.

Many other issues in relation with Bayesian analysis within Official Statistics may be seen at the presentations of the ISBA 2000 world meeting at http: www.ntua.gr/ISBA2000.

Chapter 4

A strategy to foster Bayesian methods in Official Statistics

4.1 Introduction

In this chapter, we summarise some findings of this project, and suggest a strategy to promote the adoption of Bayesian methods by the community of official statisticians.

4.2 Main conclusions

The basic conclusions of our survey, as already outlined are:

1. There has been an enormous growth in Bayesian research, facilitated mainly by recent computational developments and availability of cheap computational power. Specific areas with important developments include Bayesian computation, Model selection, Robustness and sensitivity analysis, Nonparametrics, Graphical models, Decision analysis and Hierarchical models. These, in turn, have opened up the possibility of undertaking more comprehensive Bayesian approaches and models in many application areas.

2. However, these new applications have permeated little the field of Official Statistics. Reasons include the perceived subjectivism of Bayesian analysis, the relative novelty of these methods, lack of specific methodological developments in the field, lack of Bayesian software in areas of interest to official statisticians, limited human resources, and the tradition in statistical education within European universities.

3. In many NSIs, however, there is an organisational structure that would allow the spread of novel statistical approaches like Bayesian methods, in the form of a methodological unit or regular seminars devoted to methodology.

We have suggested various attractive points of Bayesian analysis for official statisticians and illustrated through small examples their applicability and the previous responses clearly indicated that there is still much room for the innovation in the production of official statistical data with Bayesian methods. The infrastructure needed for it indeed exists:

- Methodological units
- Facilities for training with technical lectures

but the superstructure is obviously lacking:

- little knowledge about Bayesian statistics
- skepticism about usability of Bayesian tools and methods for the regular production of official statistics.

Given that *we are convinced of the potentiality of Bayesian methods within official statistics* and that networking of official and academic statisticians has proven to be an effective tool to transmit new approaches and methodologies into official statistics production, we emailed a new message with the first stage report, the questionnaire and an invitation to take part in an eventual network of official and academic statisticians interested in Bayesian methods.

The specific message was

> Dear colleague:
>
> The EUROSTAT sponsored research project 'BayOff: A feasibility study on the Application of Bayesian Methods of Inference to the Treatment and Analysis of Official Statistics Data' has completed its first stage on the identification of the state-of-art of Bayesian techniques and their possible application in NSIs.
>
> I am pleased, as a member of the research team, to send you a copy of the draft report, which is being examined by Eurostat. We will be very grateful to have your reactions on it.
>
> We have sent a questionnaire to NSIs (some of you have already answered) to know about the awareness and willingness to apply Bayesian methods.
>
> For the second stage, we would like to identify NSIs willing to be part of a future pilot project of application of Bayesian techniques to the following fields:
>
> - missing data imputation
>
> - small area estimation

- time series forecasting

These techniques have been already tested in some Institutes.

I look forward to your news and reactions. With warmest regards,

The following NSIs or individual experts in NSIs have responded positively (at least until they receive further details of the network) with topics and potential contact persons:

- United Kingdom: Tim HOLT, John CHARLTON
- Finland: Seppo LAAKSONEN (Data imputation and small area estimation)
- Denmark: Birger MADSEN (Data imputation)
- Netherlands: Bert KROESE (Data imputation and small area estimation)
- Italy: Mauro SCANU (Data imputation)
- Spain: José L. CERVERA, Javier MUÑOZ

The Federal Statistical Office of Germany declined to participate, because of current lack of human resources, in spite of their interest in the project report.

This clearly shows *the willingness of European NSIs to find out more about Bayesian methods.* This will be taken into account in the design of a future strategy to foster the application of Bayesian methods in Official Statistics.

4.3 A strategy to foster Bayesian methods in Official Statistics

4.3.1 The context: the 5th Framework Programme of R&D

BayOff has revealed that although there has been an enormous advancement in Bayesian methods, they are little used in the production and analysis of Official data, in spite of, we believe, their enormous potential. Little awareness of those methods and techniques and the traditional reluctance of official statisticians to methods based on the formalisation of prior information are partly the causes of that little impact, *a situation that should surely change, to the benefit of the European Statistical system.*

The 5th Framework Programme of Research and Development (5FP) of the European Union provides an excellent context for examining in depth the way Bayesian methods and techniques can solve important problems in Official Statistics. The 5FP highlights the need for research, at a European level, on new statistical methods, indicators and applications. Indeed, Cross-Programme action number 4 (CPA4) of the Information Society Technologies research programme (IST) explicitly states the need to develop and demonstrate new statistical tools and methods, to use statistical tools and methods in applications, and to develop indicators of the new economy.

Following this recommendation, statistical offices of the EU, together with EUROSTAT, have identified their Research and Development needs, developing the *European Plan for Research in Official Statistics (EPROS)*. In EPROS, a mention is made to nascent techniques as neuro-fuzzy systems, including Bayesian network learning using Markov Chain Monte Carlo methods (Research Priority 4.2 of EPROS). Consequently, this opportunity should be used to put into practice a strategy for fostering the use of Bayesian Methods in the field of Official Statistics.

4.3.2 Subject areas and Bayesian techniques

In the drafting of such a strategy, areas in Official Statistics should be matched with techniques that could be used. However, statistical operations carried out by NSIs share common phases (such as sample design, data editing, estimation, and data analysis), independently of the fact that they deal with diverse subjects(social statistics, economic statistics) or are based on information from different elementary units (persons, households, enterprises). It is then advisable *to focus on methods and techniques rather than on Official Statistics topics.*

Taking into account the huge amount of Bayesian theory and applications described in the Bibliography and the case studies presented, and also of the contacts with official statisticians, the *most promising methods* and techniques seem to be, as far as Official Statistics are concerned, are:

- Sampling techniques and complex designs
- Improvement of data quality: imputation of missing data, detection of outliers
- Knowledge extraction and automated data analysis
- Small area estimation
- Time series analysis
- Integration of data from different sources (meta-analysis)
- Ordinal data analysis
- Variance measurement
- Data disclosure

In Annex E, some concrete applications are reviewed in more detail, presenting current research suggested by the members of the BayOff Team and external experts. Suggestions are given about their possible applications in Official Statistics.

4.3.3 The design of a strategy

In the next section, proposals are made for a *strategy* of the European Statistical System to foster the use of Bayesian techniques in the different phases of production of Official Statistics.

With respect to the *methodology* used to design a strategy of a programme for the use of Bayesian techniques in the European Statistical System, it has to be said that BayOff is just an initial picture. Even if the programme is somewhat structured in *Specific objectives* and *Activities* and *Means*, it is clear that systematic approaches like a *SWOT* (Strengths, Weaknesses, Threats and Opportunities) analysis or the *Logical Framework* (mainly used to put into place development co-operation programmes) would be clearly beneficious.

The *assessment of external conditions* to develop such a strategy has to include hypotheses on the context of the European Statistical System. In particular, the design of a strategy should take into account the *workload foreseen* that should be fully integrated in the Community Statistical Programme, the opportunity given by the 5FP and the commitment of the actors (where Eurostat should play a leading role).

The full benefits of the 5th Framework Programme for R&D, and in particular the EPROS (and related programmes) should be taken into account, from a financial and organisational point of view.

4.3.4 Proposals for a strategy

General Objective of the Strategy. To foster the application and use of Bayesian methods and techniques in the production and analysis of Official Statistics

Specific objectives. Two specific objectives can be aimed at, according to what has been identified:

- To increase the awareness and trust of official statisticians on Bayesian methods
- To provide them with the necessary tools to apply them in their daily work

Activities and means. Some activities are suggested below, with a brief detail of their content. More detailed is the proposal for a **Bayesian network of Official and Academic Statisticians**, that was worked out after personal contacts with some experts[1] and circulated in statistical meetings (similar initiatives are currently circulating, such as AMRADS).

[1] In particular, Timo ALANKO (Eurostat), Andrea SALTELLI (JRC ISPRA) and the previously mentioned experts.

Activities

- Training in Bayesian methods
  Means
  - Manuals on Bayesian statistics with case studies in Official Statistics
  - Training courses on Bayesian statistics
    * Selection of topics
    * Selection of lecturers
    * Preparation of materials
    * Dissemination (BayOff Web site, TES Working Group, etc.)
- Providing Bayesian tools for statistical practice
  Means
  - Maintaining the BayOff Web site with more Bayesian resources
    * Bibliography
    * Software
  - Design of Bayesian modules for the most common statistical packages used in NSIs (SPSS, SAS)
  - Adaptation of existing Bayesian software to the needs of NSIs
  - Methodological projects concerning new Bayesian developments in Official Statistics, possibly to be funded through Framework 5 Program.
- Increasing the interest of official statisticians in Bayesian statistics
  Means
  - Bayesian Network in Official Statistics
  - Mailing list of OS on Bayesian methods Organisation of a specialised workshop on Bayesian methods and participation in Official Statistics conferences such as ISI/IAOS, NTTS,
  - Promoting sessions on Official Statistics in specialised Bayesian meetings (ISBA, Valencia meetings)

According to the identification of the most promising techniques and the daily practice of Official Statisticians, the topics on which training courses could be designed and organised could include:

- Introduction to Bayesian Statistics
- Bayesian forecasting techniques for Time Series
- Bayesian methods of missing data imputation
- Small area estimation with Bayesian techniques
- Integration of different data sources with Bayesian techniques

The courses could be prepared on CD-ROMS or a web site to facilitate dissemination.

4.3.5 A network of Bayesian Official and Academic Statisticians

It is proposed to set up a network of Official and Academic Statisticians interested in the applications of Bayesian techniques to Official Statistics. A draft was already circulated by a member of the BayOff team in the DGINS meeting of Oporto (may 2000).

General objective To increase the impact of Bayesian analysis in the production of Official Statistics in the European Statistical System, increasing the value of data produced for the wider community of users.

Specific objective To set up a network of academic and official statisticians interested in exploring the application of Bayesian analysis to a selected number of open problems in official statistics, and to disseminate the results of such exploration to foster its use.

Expected results

- a selection of important problems with real data produced by NSIs.
- an increased awareness of the possibility of using Bayesian methods in Official Statistics
- a set of case studies ready to communicate and share by means of workshops and publications.

As far as organisation is concerned, every interested country would constitute a national node including an official statistician (in the NSI or other statistical office) and a principal researcher or a team of researchers in universities and research centres.

The role of the official statistician would be to suggest an open important problem in Official Statistic and prepare and provide the necessary data.

The role of the academic team would be to offer a Bayesian solution to the problem and draft a self-contained report on the methods and techniques applied, the results obtained and the difficulties faced.

A secretariat of the network would co-ordinate the works in order to present the results in a final workshop and publication in leading journals in Official Statistics.

Funding of the activities of the Network should be found through the Accompanying measures programme of the 5FP, as a thematic network through IST, perhaps in connection with the AMRADS initiative (network of Academic and Official Statistics).

So far, the following institutions have manifested their interest in constituting that network:

- SPAIN: National Statistical Institute (INE)
  Universidad Rey Juan Carlos of Madrid

- GREECE:Athens University of Economic and Business
  National Statistical Service of Greece
- ITALY:Italian Statistical Institute (ISTAT)
  Università di Pavia
  Consiglio Nazionale delle Ricerche, IAMI
  Università idi Roma – La Sapienza
- United Kingdom:University of Southampton
- European Commission:
  EUROSTAT (preliminary contacts with Mssrs Nanopoulos and Alanko)
  Joint Research Centre (ISPRA) (contact with Mr Saltelli)

4.3.6 Other ideas

There are other ideas that might help in promoting Bayesian Methods within NSIs:

- The creation of chairs on Official Statistics in universities.
- Sabbatical leaves in Official Statistics research and development centers.
- Easier access to Official data.
- Handling the management of some Official Statistics datasets by consulting groups in universities.

4.4 Conclusions

BayOff suggests an impressive amount of methodological developments in Bayesian analysis. There is also an increasing catalogue of Bayesian software, and interesting applications carried out by academic statisticians. Yet simultaneously, there is general concern that Bayesian methods are not permeating the daily work of official statisticians at National Statistical Institutes.

We have identified the willingness of several NSIs to consider Bayesian methods, possible fields of application, and a strategy to go ahead. We hope that this strategy will be implemented in the near future in spite of the usual NSI environment characterised by a lack of human resources and the pressure of society (governments, media, citizens,...) that permanently demands more and better information about the economic and social situation.

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Appendix A

Questionnaire on Bayesian Analysis and Official Statistics. Academic

A.1 The questionnaire

To further explore the reasons of the status depicted above, we produced a questionnaire trying to ascertain important recent developments in Bayesian Analysis, their actual and potential impacts on Official Statistics and possible ways of changing the situation found. The heading and questionnaire prepared were the following:

We have been asked by EUROSTAT (the European Statistical Office) to undertake a prospective study on the feasibility and potentiality of Bayesian methods in the field of Official Statistics (BayOff project). The study might have great impact in at least two directions: the adoption by European Union Statistical Offices of Bayesian methods (currently almost nil) and the promotion of Bayesian research in the European Union.

As part of the study, we have to interview some twelve-fourteen leading figures in the field on prospects and we would like you to be included within that group. If, as we hope, you are willing to take part, we shall be grateful if you could answer the enclosed questionnaire, if possible, within two weeks. Should you not be in a position of doing so, we would appreciate if you could suggest alternative leading figures that might take part in this survey.

With best wishes

QUESTIONNAIRE

1. Which are the most relevant developments within the field of Bayesian Statistics over the last ten years?

2. Which are, according to you, important open problems that Bayesian Statistics still needs to face?

3. Bayesian Analysis has had little impact within Official Statistics. Could you give reasons why?

4. Could you mention any important work concerning Bayesian Analysis within Official Statistics?

5. Could you describe why Bayesian Analysis could be relevant within Official Statistics?

6. Could you suggest ways in which Bayesian Analysis might permeate the field of Official Statistics?

7. May we include your name together with your answers, or do you prefer to keep them anonymous?

The questionnaire was sent to twenty leading (academic, but many of them with some relations with Official Statistics) figures in the field of Bayesian Statistics by email. We have summarised responses all throughout the main text. Below we provide the detailed answers.

A.2 The detailed answers

Jose M. Bernardo (U. Valencia)

1. MCMC methods; complex applications (hierarchical models, dynamic forecasting,..); developments and appraisal of objective methods in bayesian inference; model choice

2. Large set of non-subjective results with commonly accepted 'conventional' priors

3. Subjective input is not usually acceptable in such area: see 2 above.

4. Not aware of any

5.
   - Bayesian output in the form of posterior densities is far more intuitive and easier to grasp by non-experts.
   - Bayesian analysis is able to cope with complex hierarchical models needed for a proper analysis of many problems within that field.
   - Bayesian design could be useful to optimally replace subject to non-controlled-error 'censuses' by controlled efficient surveys

6. Produce interesting case-studies.

James O. Berger (Duke University)

1. Markov Chain Monte Carlo Bayesian Robustness Objective Bayesian Model Selection Bayesian Nonparametrics Graphical Models and Influence Diagrams

2. (i) Developing suitable software, allowing routine implementation of Bayesian methods in the same manner as classical methods are implemented. This involves creating default Bayesian procedures for the large variety of situations where classical procedures now exist. This could be done today, i.e., there are no serious technical hurdles. The rest of the problems I mention are more technically challenging. (ii) Finding suitable objective priors for Bayesian nonparametrics (iii) Interactive implementation of robustness/elicitation (iv) Development of multiscale Bayesian methods capable of dealing with huge amounts of data (i.e., methods whose complexity varies with the amount of data being processed and the local need for high level methods) (v) Deciding upon optimal methods of model selection and prediction; this is currently in a high state of flux. (vi) Clarifying the role of frequentism in the Bayesian statistics world of the future.

3. (i) Methods used in official statistics are, almost of necessity, codified and somewhat slow to change. Bayesian analysis became easy to implement in practice only relatively recently, and so its penetration into official statistics is probably just now beginning. (ii) Rather misleadingly, Bayesians in the last half of this century tended to portray their field as that of subjective statistics, which carries uncomfortable connotations for many official statisticians. This was misleading because Bayesian statistics, for nearly 240 years, has primarily been an objective approach to statistics, and the great bulk of applications will always remain so.

4. (i) The Devices Division of the U.S. Food and Drug Administration is aggressively implementing a Bayesian approach to its statistical analysis. (ii) Many small area estimation techniques in survey sampling are effectively Bayesian.

5. (i) Probably the overriding reason is that of communication. The job of official statistics is to communicate information to the public and to decision makers in government, in ways that can be understood and properly utilized. The conclusions from classical statistical methods can be almost impossible to understand for nonexperts, especially those describing the uncertainty in the data or conclusions. (P-values are but one notorious example.) In contrast, Bayesian analysis gives easily interpretable conclusions. (ii) Increasingly, official statistics is going to be responsible for combining information from a large variety of different sources. Bayesian analysis is essentially the only way that this can be intelligently done, in general.

Stephen E. Fienberg (Carnegie Mellon)

1. MCMC; Bayesian work on graphical models; multiple imputation

2. Bayesians are not especially good at large data problems, involving large numbers of parameters, and especially settings where the dimensionality is also large.

3. As in other areas the practitioners in the field of Official statistics are slow to change and they are suspicious of "subjective" techniques for Public data that are supposed to be objective.

4. The most important application I knoewof are in the development of models For disclosure limitation and the use of Bayesian smoothing models for small area estimation, especially in the US.

5.

6. Bayesian statistics will permeate the field of Official Statistics only when Bayesian take Official Statistics problems seriously and treat them as of interest in their own right rather as simple another application of know techniques and problems.

Simon French (U. Manchester)

1. In the context of displaying official statistics both to politicians and the general public, then surely the development of graphical methods within Bayesian statistics, and I would also include Bayesian decision analysis, is clearly an important advance. I think one of the strengths of the Bayesian Approach is that it provides a much easier Framework within which to communicate with all sections of the community.

2. I guess there are two problems, intimately related. We need better more straightforward ways of eliciting opinions, both from experts and from non-experts. Secondly, we still need to make the use of subjective opinion much more acceptable. Our strength is that we deal with subjective opinion explicitly; but that is also one about greater weaknesses.

3. Well, speaking from pure prejudice, I guess one of the reasons is that we do not pretend to be objective, and official statistics should clearly be objective. Then, of course, most statistical education is non-Bayesian, so most of the recruitment into the statistical civil service draws upon non-Bayesian statisticians. I suppose that one might also claim that within science the issues facing inference are much more difficult and need the more powerful, more coherent techniques Bayesian statistics can bring. In many ways, official statistics are more straightforward and the older, more conventional non-Bayesian methods still work, at least passibly.

4. Sorry. Although I am usually prepared to give answers based upon pure prejudice, this one really does require some factual basis! So let me omit this question.

5. I think that as society becomes much more risk where and we face many major environmental, medical, and other risk issues, then the ability of the Bayesian methodology to address risk assessment and management will make it much more attractive to statisticians working in government. I have become particular aware of this in working with the British Department of Health. Classical statistics, with its abject fear of rejecting a true hypothesis claimed throughout the BSE crisis that there was no evidence of a risk to the public; and, of course, that was totally correct. There was no evidence of a risk to the public. But equally there was no evidence of NO risk the public. classical statistical methods could not, and in fact still cannot, answer the real questions of interest to the politicians and the public. On the other hand, a Bayesian Approach with its close fit with decision analysis could have addressed the questions in a much more useful fashion.

6. Possibly through the medium of supporting risk analyses and the process of risk communication with the public. As I noted above, Bayesian methods of formal suitable for risk analyses than are conventional classical statistical methods.

Dani Gamerman (U. F. Rio Janeiro)

1. Certainly the development of new computational tools to undertake the calculations required in complex problems.

   Main keywords: stochastic simulation, MCMC, quadrature approximation

2. Implement multi-purpose software to tackle standard statistical problems, currently being tackled by frequentist statitical techniques

3. I think part of it is the long-established frequentist tradition in the field of sampling.

4. I think there has been some early work in the 80's by people like O'Hagan and Pfefferman that dealt with sampling from (linear at least) Bayesian perspective.

5. With the recent trend in Official Statistics towards model-based approaches, the Bayesian approach will naturally play a key role.

6. As suggested in the previous answer, once established that explicit incorporation of a model into the inferential procedures, Bayesian inference will be very important. Areas such as (multi-level) hierarchical models will become vital in the correct description and estimation of aspects of sampling procedures.

Andrew Gelman (Columbia)

1. Routine use of Bayesian methods in applications (for example, hierarchical random and mixed effects models in biostatistics; shrinkage methods in small area estimation for public health and public policy data)simulation-based computation methods, incorporation of graphical and exploratory methods into Bayesian modeling, inference, and model checking

2. models for multivariate outcomes routine computation and software for models that we currently know how to fit, but not in an automatic way (for example, hiearchical generalized linear models)

3. Association of Bayesian methods with "subjectivity" (I do not believe that this association is appropriate; the probability distributions used in Bayesian inference can be as "objective" as likelihoods or other statistical procedures; for example, classical confidence intervals are subjective in the choice of estimator or class of estimators to use. More generally, the True "subjectivity" in a statistical procedure comes not from a probability distribution but from the choice of what information to include in the model. Bayesian methods have the potential for being much more objective than classical methods because the Bayesian probabilistic framework is general enough to allow one to combine information from many sources–for example, to use a predictor variable that is partially missing).

   Beyond this, I am not familiar enough with the history of official Statistics to say more.

4. Multiple imputation–dealing with uncertainty about missing data in public data sets modeling nonresponse in surveys and censuses small-area estimation: getting inference for subsets of the population (for example, small geographical areas, incidence of rare diseases, etc.) for which data are sparse and so simple, non-model-based estimates do not give good answers

5. Bayesian methods are particularly relevant for situations with large uncertainties. Exampes include impuattion of missing data (even if you know a lot about a general population, you probably will not know much about missing data on a particular individual; thus Bayesian methods can be useful in combining the available information to make a reasonable probabilistic assessment of the uncertainty),small-area estimation (once again, inference about small subpopulations will be highly inefficient unless you combine information from other groups ,which can be systematically done using the Bayesian approach of partial pooling, or shrinkage toward a regression model), models for nonresponse in surveys (which includes methods for poststratification weighting), prediction of future data (mathematically, this is similar to the problem of imputing for unseen units)

6. In general, the simplest thing is the idea of expressing estimates as probability statements. The next thing would be the general use of mixed-effects regressions in situations where classical linear and generalized linear models are currently being used.The advantage of mixed-effects regressions is that thay allow for separate estimates for subsets of the population (for example, if the random effects in such a regression represent different regions of a country), and also because these models are more complex, they allow more realism in fitting public data, which of course do not really follow exact additive relationships as assumed by regression models. Finally, Bayesian methods can be used in combining information from different data sources. A nice recent example of this is the work of Brad Carlin and collaborators on spatial models for combining information from U.S. Census and U.S. public health records to get higher-resolution disease maps.

Anonymous

1. Doubtless, the greatest contribution in the last decade has been the development of Bayesian Computation, specially the field of Markov chain MonteCarlo Methods.

2. One of the most challenging problems not yet satisfactorily solved is the problem of model selection.Another important problem is the developement of objective or reference priors for may statistical models, icluding the most versatile hierarchical and non-linear Models.

3. The order of the following reasons does not mean neither decreasing nor increasing order of importance. Historical reasons, which are hopefully evolving towards a more sympathetic view of the Bayesian methodology. The way in wchich the teaching of Statistics is carried over in many Faculties,overemphasizing classical Statistics methods and classical Data Analysis. Up to recently almost all Statistical Software was based on classical models and classical statistics, so that most practicioners are unaware of the potenciality of the, say, MCMC methods.

   The few number of people, above all in Europe, working or teaching in Bayesian Statistics, as compared with the whole number of statisticians. Difficulties in eliciting priors in complex problems deter many practicioners from applying Bayesian methods.

4. The analysis of missing data in large data base sets are those used in Official Statistics by using multiple imputation. Analysis of longitudinal data. Bayesian Data Mining. Detection of outliers and robust methods appropiate in large data base sets. Prediction of short and medium term Socio-Economic indicators.

5. One important reason is that Bayesian methods are better suited to investigate the hidden structure of large masses of data generated by Official

Statistics, and the hierarchicla nature of these data such as the presence of clusters, strata, etc.

6. By making presentations of Bayesian methods in action, that is for example, applications of MCMC methods to complex real data problems which can be easily and fully analyzed (performing sensitivity analysis, variable selection, model cheking, etc.) By showing off the usefulness and suitability of Bayesian modelling to the important problems of prediction -an important task in Official Statistics- for which Bayesian Methods are specially well suited, such as dynamic models and the most recent hierarchical dynamic models.

Jay Kadane (Carnegie Mellon)

1. I think that unquestionably the vast developments in computing are the most important. Many of these are special cases or relatives of Markov Chain Monte Carlo methods.

2. There is still important work to do in computing, especially to learn when chains are not necessary. I also think problems of several decision makers, including experimental design, are important. Then there are the usual problems of modeling and elicitation.

3.
   - Bayesian statistics is still somewhat new. Official statistics is very conservative methodologically.
   - Bayesian statistics requires informed judgment to be used explicitly. While it is my impression that official statisticians use judgment often, it is rarely made manifest so that others can see what judgments were made and how they affected the analysis.

4. I am not very current on this, but my own efforts are now in Bayesian demography. My colleagues and I published an article on this subject in JASA, and we are now waiting to hear whether we shall have visitors from the government of South Africa this summer to help them implement our methods in their circumstances.

5. Mainly because it correctly reflects uncertainty, thus helping policy-makers develop flexible responses to current problems.

6. I think we have to do excellent work on specific problems, and impress by example.

Daniel Peña (U. Carlos III, Madrid)

1. Gibbs Sampling and MCMC, advances in hierarchical modelling, Computing.

2. Robustness problems, meta-analysis, multivariate analysis.

3. Offical Statistics must be objective and prior information is difficult to justify

4. No

5. For meta analysis and in general for combining information from different sources

6. .

Luis Raul Pericchi (U. Simon Bolivar, Venezuela)

1. Developments of Hierarchical Modelling; Monte Carlo Markov Chain Methods for computation Laplace asymptotic expansions Default Bayes Factors; Robust Bayes

2. Strategies to reach practice; Applications to Official Statistics and Censuses; A default Bayes Factor in its greatest generality; A general theory to construct default priors for model selection; Test to whether an MCMC output makes any sense;

3.
   - The community is too official, meaning there are specified rules to follow, rather than experiencing research; You have to follow say United Nations regulations, and everybody is suspcting that you are cheating, so follow the rules
   - Lack of understanding of complex hierarchical modelling which has potentially enourmus impact;
   - Lack of computer packages dedicated to the specific objectives of Official Statistics.
   - Lack of the theory of Bayesian Default tests;
   - The curious missunderstanding that Bayesians do not want to ramdomize
   - Lack of good and simple test (and theory) of Finite Populations under a Bayesian view

4. I would say any theoretical developments in finite populations and also in hierarchical modelling: Come to mind the book by Ghosh and Meeden. Also all the developments in missing data by imputation etc

5. i)Hierarchical Modelling; ii) Missing Data; iii) Model Selection through posterior probabilities;

6. Find ways so that accept that research advances in the last 40 years are needed by Official Statistics as well as the rest of practice....For example what about making a case to the International Statistical Institute?

Christian Robert (U. Rouen)

1. - Bayesian Nonparametrics on a large scale - Bayesian Model Choice for complex models - Spread of Bayesian techniques outside Mathematics (Signal Processing, Artificial Intelligence, Econometrics, Ecology, to name a few...) - Upsurge in modelling abilities (incl. Spatial Statistics, graphical models) - Considerable increase in computational abilities and correlated disappearance of academic models and priors, incl. linear approximations - Meta-analysis

2. - Improved assessments of prior and modelling sensitivity, incl. use of infinite dimension models as in reversible jump techniques - More generic derivation of reference priors - Reinforcement of theoretical basis of tests and model choice - Complete abandon of the ghetto mentality, still too present among some Bayesians

3. Judging from the French point of view, e.g. from the situation at INSEE, (1) Bayesian Statistics have not been taught in Statistics programmes for many years. While it is now at ENSAE and ENSAI, the two French schools for Official Statistics, it is still not taught in most universities, either at the undergraduate or graduate levels. (2) Moreover, there is a certain lag between the development of new techniques and their use by Official Statisticians. For instance, the use of simulation tools like SIR, which dates back to the 80's is only starting at INSEE. This can be due to the fact that people higher up in the hierarchy are less in touch with the latest developments. (3) But this is also due to the few connections between Academia and the world of Official Statistics. For instance, it is difficult if not impossible to validate a Ph.D. thesis on survey sampling in a French university as a Statistics thesis. Meetings where people from Official Statistics attend, like the ISI, are often shunned off by academic statisticians and the reverse also applies.

4. Capture recapture studies for the estimation of difficult parts of the population (homeless people, &tc.) Multiple imputation techniques for missing data in surveys Small area estimation

5. From one side, Official Statistics often enjoys huge datasets on which (a) prior distributions can be built up, (b) refined processing can be led, most often in a hierarchical manner or, equivalently, at a local level, a feature for which the Bayesian Analysis is most convenient. In the most complex cases, it actually is the only solution. When the dataset gets really huge (e.g., an INSEE survey which involves hundred of thousands of companies and millions of workers), the current Bayesian techniques do not necessarily work either, but they can provide more satisfactory approximations. On the other side, some parts of Official Statistics are facing huge difficulties to collect reliable data and would benefit from the correlated (or prior) information brought by other studies on similar populations. This is the realm of Meta-analysis, but also of Spatial Statistics. Again, the

Bayesian solutions seem the most accessible, if underused, solution at the current time. At last, the computational techniques recently developped by Bayesians could be used in Official Statistics for the devising of more efficient surveys for instance, incl. the removal of the exhaustive census!

6. - More easily accessible datasets - Wider bridges between Academia and Official Statistics by the creation of chairs of Official Statistics in universities and a wider possibility of sabatical leaves in Official Statistics research and development centers (as this is the case in France with the research center of INSEE) - Management of some Official Statistics datasets by consulting groups in universities

Arnold Zellner (U. Chicago)

1. Many SUCCESSFUL APPLICATIONS of BAYESIAN STATISTICS in solving inference, decision, forecasting and other important private and public problems.

2. Improved MODEL FORMULATION procedures and more general, improved INFORMATION-PROCESSING. RULES that will permit us to get better solutions to old problems and new solutions to as yet unsolved problems using Bayesian methods.

3. Most government statisticians have not received good training in Bayesian Statistics. More efforts are needed to educate government statisticians to the benefits of applying Bayesian analysis to their problems. E.G. Prof. Charles Whiteman of the U. of Iowa, has consulted with the Governor of Iowa and his staff and implemented Bayesian procedures for forecasting tax receipts and managing state budgets using asymmetric loss functions, Bayesian predictive densities, Bayesian probability statements about future outcomes, etc. and reports that the Governor and his aides understand and appreciate his Bayesian efforts and results. And many years ago, H. AKAIKE and his colleagues at the Institute for Statistical Mathematics in Tokyo created a BAYESIAN COMPUTER PROGRAM FOR SEASONAL ADJUSTMENT AND ANALYSIS (BAYSEA) that can perhaps serve as a model for more modern Bayesian seasonal analysis and adjustment programs.

4. In addition to the two examples mentioned in the previous question, the Federal Reserve Bank of Minneapolis operated a Bayesian vector autoregressive forecasting model, due to Robert Litterman, for many years. Also, Malay Ghosh, S. James Press and others have worked with the U.S. Bureau of the Census on Bayesian shrinkage estimation, spatial models, etc.

5. Prior information is extremely important in measurement, design, estimation, prediction, testing, control and other problems. Use of Bayesian techniques enables such prior information to be used well along with sample

information to get better solutions to many problems than those yielded by non-Bayesian methods.

6. For each important official problem, be it a design, estimation,prediction, or control problem, compare current non-Bayesian solutions with Bayesian solutions theoretically and in practice. The results will, in my opinion, be very illuminating and promote more widespread use of Bayesian methods in the field of Official Statistics. If current workers do not have the time or knowledge to perform such comparative studies, this work can be done under contract with university and industry researchers. The users of official government statistics should also be involved in these studies.

Please indicate that these responses were kept short on purpose and hopefully are understandable.

Adrian Smith (Queen Mary and Westfield College, London)

1. Hierarchical models, graphical models, Markov chain Monte Carlo
2. Novel methods for presenting analyses in the form of many prior to posterior mappings (ie sensitivity analyses)
3. Lack of education caution about use of priors (whose?) a lot of official statistics takes form of summarising rather than analysing data
4. Not really aware of any at present.
5. Ideal for examining a range of scenarios (models) should be ideal for data fusion (combining survey and administrative data sources)

Malay Ghosh (U. Florida)

1. The phenomenal development in Bayesian computing, making Bayesian methods readily available to practitioners. One consequence of this is that Bayesian methods now find wider applications in many real life problems.
2. Development of more default priors so that Bayesian methods can be more routinely used. More important is the development of Bayesian tools for model diagnostic and model checking.
3. Lack of Bayesian software which can be routinely used. BUGS has made significant preliminary inroads, but more remains to be done.
4. Bayesian small area estimation techniques. The original surveys are targeted towards a certain population. the direct survey estimates are usually unreliable for lower level of aggregation. Bayesian methods have been very useful for 'borrowing strength' to produce more reliable local area estimates.
5. There is always prior knowledge available in most Official Statistics problems. Bayesian techniques provide the suitable avenue for quantifying this prior knowledge.

6. More dialogue between classical and Bayesian statisticians; organization of seminars, workshops and conferences.

Alicia Carriquiry (Iowa State)

1. Computational methods, which now permit fitting realistic models to complex data. Key words: MCMC, complex model fitting.
2. Because Bayesian analysis is heavily model-dependent (except the new non-parametric or semi-parametric approaches), anything related to model choice and validation is of high priority in my view. In particular:
   - choice of prior distributions in realistic problems,
   - model validation and diagnostic measures, particularly in complex problems,
   - diagnostics for convergence in numerical approaches,
   - development of robust (to model choice) approaches.
3. Bayesian methods have not been adapted in Official Statistics because of:
   - Practicality: carrying out Bayesian analyses, even with the new computational approaches, is still very hard. Setting up reasonable models is a difficult task for the non-academic statistician.
   - Lack of user-friendly software.
   - Type of problems in Official Statistics: a large proportion of what statisticians do in government is create tables and tables and tables of summary statistics. In general, little inferential work is done, exceptions being forecasting and a few others. The Bayesian approach does not present any obvious advantage over the frequentist approach when computing a mean, with reasonable sample sizes. Furthermore, users of official statistics are often not interested in anything *but* means.
   - Mission of government agencies: agencies produce statistics for multiple users, each with their own expectations, biases, and requirements. The Bayesian approach is typically heavily model-based, and a given model may not be acceptable to all users. Governments tend to cover their backs and minimize the potential for controversy, so an approach that depends on subjective model choices is likely to be resisted.
   - Governments move at a slower pace, because of what I mentioned in the previous point: caution. Given that the Bayesian approach has moved from the realm of theory into the realm of applications only recently I am not surprised that the approach has not found its way yet into Official Statistics. One exception: missing data and imputation, where at least in the US, several public-use nationwide surveys are now being "completed" using multiple imputation a la Rubin.
4. The only one I am familiar with is imputation for the National Survey on Health Statistics conducted by the Department of Health and Human Services of the US. The US Census Bureau and the Bureau of Labor Statistics

have adopted Bayesian methods for small area estimation in some surveys, but I do not know the details.

5. Bayesian analysis can be relevant in *some* areas of Official Statistics. The most obvious ones are:
   - Forecasting and prediction: updated filtering approaches that allow for non-Gaussian, non-linear models are often easier to use from within a Bayesian framework.
   - Any other prediction problem, such as election results.
   - Missing data and imputation: the Bayesian approach allows the analyst to "carry" the uncertainty about imputed values into the rest of the analyses.
   - Small area estimation: typically done using some version of a BLUP. Mixed effects models as those used for small area estimation can be fitted with fewer assumptions and in a more efficient way from a Bayesian perspective.
   - Whenever data are collected sequentially.
   - Whenever data are scarce but information about the problem can be obtained reliably.
   - Whenever inferences are based on multi-layer models.
6. Making Bayesian analyses the approach of choice will not be easy and it will only happen when:
   - Bayesian statisticians demonstrate, in *real* official statistics problems, that the Bayesian approach offers a significant advantage (from a PRACTICAL, not philosophical) point of view.
   - Software becomes available. Software needs to be easy to use (like SAS) and require little statistical knowledge.
   - Users are "educated" in the sense of expecting more than just means and maybe a variance from an analysis. I believe that Bayesian methods will become common place in Official Statistics only when users of OS demand the additional information (such as posterior (distribution) that is now not available to them.
   - Bayesian methods that are not so model dependent, or that are robust to poor model choices become widely available.

Appendix B

Questionnaire on Bayesian Analysis and Official Statistics. Professional

B.1 The questionnaire

Background information The BayOff project was undertaken in the frame of the SUP.COM programme of research financed by the European Commission (Eurostat). It aims to carry out a prospective study on the feasibility of applying Bayesian methods and techniques to the problems encountered in Official Statistics.
The first stage of BayOff included a follow up of recent references in journals and proceedings, contacts with leading academic statisticians, identification of existing software and of interesting applications in the field of practical data analysis. A report was produced, which will be sent later to you, for further information.
The main conclusion drawn from the 'academic side' can be summarised as follows: although Bayesian Analysis has experienced enormous growth in the last few years, it has permeated very little the field of Official Statistics

Second stage of the BayOff project The second stage of BayOff envisages more directly the potential application of Bayesian techniques in Official Statistics. It includes the following actions:
- Contacts with National Statistical Institutes (NSIs) and Eurostat
- Identification of possible applications in NSIs and Eurostat
- Assessment of the willingness of the NSIs and Eurostat to apply

Bayesian techniques
As a result of stages I and II, a future strategy to apply this work to Official Statistics will be planned.

Contacting the National Statistical Institutes In order to contact NSIs, Eurostat provided the research team with the addresses of contact persons, selected by their participation in the DOSIS/EPROS Working Group on Research in Official Statistics. As other Working Groups of the European Statistical System, it is composed by representatives of the 15 EU Member States' National Statistical Institutes. Even if the profile of the representative is quite general, they can act as 'entry points' to the NSIs[1]. Generally, they are aware of research and development activities in their respective NSI, as well as of the current Eurostat's policy of fostering R&D activities in the field of Official Statistics.
The contact persons are thus receiving a copy of the BayOff stage I report and a questionnaire that will be used to get a picture of Official Statisticians' view on the feasibility of applying Bayesian techniques in Official Statistics (the report will be distributed later, to avoid response biases).
The BayOff team thanks the willingness of these contact persons in answering the questionnaire or forwarding it to the right respondent. Their responses will be kept anonymous if it is so wished. Some of the questions were already answered by the leading figures of Bayesian Statistics on the 'academic side'.

The questions In order to get a better understanding of the little impact of Bayesian methods in Official Statistics, we need your valuable comments to a small set of questions. Please feel free to add any comment or reference. These are the questions:

Question 1: Are you familiar with the Bayesian methodology for Statistical Inference? If not, please forward this questionnaire to a colleague familiar with it.
Question 2: Do you know whether your organisation has ever applied Bayesian methods in the production of statistics? If yes, in which areas and which methods?
Question 3: In the case that you could not provide examples of use of Bayesian techniques in the production of official statistics, could you mention some possible reasons for it?
Question 4: Are there in your organisation regular presentations of statistical methodology?

[1] One of the members of the BayOff research project has been national delegate in the DOSIS/EPROS Working Group.

Question 5: Is there any methodological or research unit that evaluates new techniques and methods to improve statistical estimations? Otherwise, how are new methods tested?
Question 6: May we include your name together with your answers, or do you prefer to keep them anonymous?

B.2 The detailed answers

The detailed individual responses to the questionnaire are following, since they are interesting enough and reflect the personal views of perhaps the most convenient experts in the NSIs. We include the answers of Seppo LAAKSONEN (Finland). Kees ZEELEMBERG (the Netherlands), Nico WEYDERT (Luxembourg), Birgen MADSEN (Denmark), Sixten LUNDSTROM (Sweden), Gerhard KEOGH (Ireland), and three anonymous experts.

Seppo Laaksonen, Finland.

Question 1: I am not very familiar but we have not any other persons being essentially more familiar.
Question 2: We have not applied really but we have followed achievements of Bayesian techniques in some areas such as estimation of sample survey data, multiple imputation (Rubin's method we know quite well and its Bayesian background but real applications we have not done to the production of statistics but some comparisons for research purposes), and Bayesian networks (we participated in the EU/FP4/Keso project 1996-99 which developed a method in this area but it was not running very well). On the other hand, we have implicitly used Bayesian techniques in our thinking.
Question 3: As mentioned, implicitly we have done something but the explicit utilisation is not easy for example due to not-good software's in this area, and because the method is fairly marginal in statistics production. We maybe want to see whether Gibbs sampler, Mice or Schafer's/Rubin's software's would help something. Solas 2.0 we have tested and it may help in some marginal tasks.
Question 4: I do not understand what means 'presentation'?
Question 5: We have this R&D department or Statistical Methodology Department with non-reasonable resources (approx. 10 permanent and 5-15 temporary). We focus on sampling design, variance estimation, reweighting (adjustments for nonresponse and frame errors), editing and imputation (this is now much preferred), small-area/domain estimation and complex survey/econometric analysis. As far as we have time, we look forward to whatever new techniques (recently such as statistical disclosure control methods, data mining, symbolic data analysis, multi-level modelling, data linking, enterprise demography) ,

Kees Zeelenberg, Statistics Netherlands, kzlg@cbs.nl
Question 1: Yes.
Question 2: Empirical Bayes methods have been used to produce small-area statistics of unemployment. Bayesian methods are used in the analysis of imputation techniques.
Question 3:
Question 4: Yes: once a week there is a presentation on methodology (broadly defined, including not only mathematical methods but also presentations on pilot projects for new statistics).
Question 5: Yes: there is a Department of Statistical Methods, whose task it is to improve the quality of statistics by using methods from mathematical statistics and applied economics.

Nico Weydert, Luxemburg

Question 1: I must admit that I'm not familiar with Bayesian methodology and worse, even though we are a very small office I don't know any colleague familiar with it. It won't be difficult to draw right now some conclusion ...
Question 2: As far as I know we have never applied Bayesian methods.
Question 3: A very easy reason is that we don't even have a small unit dealing with methodology. We shall try to have one in the future, but up till know our human resources are eaten up by everyday work. Almost all our senior staff is trained economists, with some knowledge in econometrics, statistics as well, but none with thorough knowledge in statistics.
Question 4: Alas, not.
Question 5: The answer has already been given before.

Birger Madsen, Statistical Mehods Unit, Statistics Denmark Unit, bms@dst.dk

Question 1. Yes.
Question 2. We are experimenting with SOLAS for multiple imputation, which can be viewed as a Bayseian approach.
Question 3. Otherwise we have not used Bayseian techniques, for no particular reason.
Question 4. Yes.
Question 5. Yes.

Sixten Lundstrom, Statistics Sweden, Working at the R&D unit, Statistics Sweden, with methods in sampling design, estimation and analysis.

Question 1. I do not think that any person at Statistics Sweden can answer "yes" on this question. For many years ago my boss Dr. Lars

Lyberg studied if it should be appropriate to use Bayesian methodology at Statistics Sweden. The result of his work was discouraging. My knowledge is just from a course at university; I have never used Bayesian methodology at Statistics Sweden.
Question 2. See answer for question 1.
Question 3. I think that the number of teachers familiar with Bayesian methodology at Swedish universities is small, which of course affects the situation.
In my opinion Bayesian methodology is impractical in our typical surveys, that is, there you have many study variables, many doimains and many types of parameters. I suppose that you must have an apriori distribution for each quantity? In survey sampling we actually use apriori information when we stratify the population, when we allocate the sample, when we select auxiliary information in the (calibration) estimation, etc. However, this information seems to be much easier to use than in the Bayesian approach.
Question 4 Yes.
Question 5 Yes, I am working in such a unit.

Gerard Keogh, Statistical Methods & Development, Central Statistics Office, Ireland, Gerard.Keogh@CSO.IE

Question 1. Yes, I am familiar with Bayesian Inference
Question 2. To my knowledge, this statistical office has NOT used Bayesian methods in production statistics. Some time ago we examined "Solas Multiple Imputation Software" based on Rubin's (Bayesian) Multiple Impuation methodology. On the basis of our tests we found the software unsuitable for general use. We have also kept abrest of research on Neural Networks in imputation. As yet we feel the jury is very much out on this approach to imputation.
Question 3. The main reason for lack of use of Bayesian Methods is partly due to their lack of acceptance in the general statistical community. However, those more familiar with Bayesian methods would have difficulty identifying suitable 'prior' distributions and also with issues associated with Gibbs sampling convergence etc.
Question 4. Yes, there are presentations of statistical methodology relating to improving the quality and methods for the production of official statistics.
Question 5. Yes, the Statisitical Methods Development unit evaluates new statistical methods for possible wider use.

John Charlton, Office of National Statistics, United Kingdom John.Charlton@ons.gov.uk

Question 1: Yes.
Question 2: I have consulted with several colleagues: it does not appear

to have applied Bayesian methods, except insofar as estimates resulting from Multilevel modelling using MLWin software can provide empirical Bayes (shrinkage) estimates, and some have been produced as part of a small area estimates project.
Question 3: Bayesian methods have not been used because without easy-to-use software they are somewhat intractable because of the integration required, but also because the emphasis of the NSI has not been on modelling, but on producing design-based estimates. This is changing, and modelling is becoming more important. Also the need to combine data from many different sources suggests Bayes as a useful approach. We should after all use all the available information. The development of WinBugs may also change the situation.
Question 4: Yes, we have regular methodology lunchtime seminars, including one recently on Bayes methods (the Eurostat-funded BAKE project - which was unfortunately less well attended than the usual ones)
Question 5: Yes, we have a Methods and Quality Division which does just this.

Anonymous 1

Question 1: Yes.
Question 2: To my knowledge, Bayesian methodologies have never been applied for the production of statistics in this Institute.
Question 3: Since the application of the Bayesian point of view in Official Statistics is relatively new, my feeling is that researchers and management are still not acquainted with Bayesian methodologies and aware of its advantages. At the same time, it does not seem that Bayesian methodologies are consolidated: in particular, controversies on the reasonability of the subjective definition of probability still make difficult its application in the context of Official Statistics.
Question 4: Yes, there are, although these presentations are seldom given.
Question 5: Yes, there is a methodological unit. Some researchers are currently adopting also a Bayesian point of view in two research areas: statistical disclosure control and record linkage.

Anonymous 2

Question 1. Yes, there are some colleagues which are familiar with the Bayesian methodology.
Question 2: The Institute has used the method BAYSEA by Akaike, H. and Ishiguro, M. (1980) for forecasting economic time series.
Question 3: We will use Bayesian methods whenever necessary.
Question 4: Yes.
Question 5: Yes.

Anonymous 3

Question 1. Yes, I have been working with those techniques.
Question 2: For the moment, no applications have been made. Many years ago, a proposal was made for a methodology of updating National Accounts with partial information using Bayesian inference methods, but never applied. There is however some interest in testing Bayesian methods for small area estimations.
Question 3: The staff is not familiar with those techniques. Routine analyses are carried out, with little innovation. Moreover, the use of standard software (SAS) does not allow to apply Bayesian techniques easily.
Question 4: Occasional presentations took place in the last years, organised by the professional association of senior statisticians. Currently the experience has not been repeated.
Question 5: A Directorate for Statistical Methodology has been recently created, including the Unit for Sampling Design, in which estimation methods are developed. There is not any R&D Unit that centralises research activities.

Appendix C

Bayesian Statistical Software

C.1 Introduction

Availability of Bayesian user-friendly software would push the usage of Bayesian methods. To a certain extent, Statistics remains frequentist because SAS is frequentist.
Two of the main advantages of the Bayesian approach create certain difficulties in the development of generic software:

(a) The extreme flexibility of Bayesian analysis as, virtually, any model built may be analysed. Classical packages provide access only to a few models, with well-defined scenarios, for which a classical procedure is available.

(b) The possibility of using extensive subjective prior information. For that, we believe that the software should include some kind of ellaborate expert system for the assessment of prior distributions. This is difficult, in part, because there remains much to be done to empirically determine optimal ways of assessing prior distributions. Note that, strictly speaking such system would not be necessary, since we might base the software on non-informative priors.

We describe three possible approaches to the development of Bayesian software. The first one uses general packages. The second one is the development of generic Bayesian software. Finally, it is also possible to develop context specific software. The authors of this report, in their usual practice, tend to develop generic programs in a general purpose language (FORTRAN, C or Java) and the graphical and exploratory data analysis utilities of Splus.

C.1.1 Inference with generic packages

Recently, various approaches, mainly oriented to teaching, have been proposed to Bayesian Statistics based on standard packages, such as Minitab (Albert, 1993; Marriot and Naylor, 1993) or Splus (Gasparini, 1994). The ideas may be translated, in part, to other environments like Statgraphics, SAS or SPSS. Albert (1996) provides an excellent introduction.
The main idea is that based on the importance of simulation methods, taking advantage of high- level commands, e.g. of graphical type, on one hand and, on the other, of the possibility of programming at low level, we may provide support in many cases to the statistical computation needs. In that sense, we should mention the many public domain functions exisiting, e.g. in Statlib, which facilitate the implementation of Bayesian ideas.

C.1.2 General Bayesian packages

There have been, as well, several attempts to develop general Bayesian packages.
One attempt has been BAYES-FOUR, a library of FORTRAN routines, which implements gaussian quadrature methods developed by Naylor and Smith. It is not very user- friendly, requiring the writing of FORTRAN routines for the likelihood function, the prior distibution and the integrand.
Specific languages have been also developed. One of them is Lisp-Stat (Tierney, 1990). As its name indicates, it derives from Lisp. In spite of its advantages, like its ellegance and dynamic graphics, its syntax is, at least initially, somewhat hostile. Its computational capability is somewhat limited, but this may change with developments provided by their users, see e.g. Doss and Narasiman (1996).
Another relatively recent proposal is BUGS (Thomas et al 1992) a program to analyse general hierarchical models based on Gibbs sampler. The suggested general structure is based on a graphical model. In the current version, model nodes have to be univariate, only log-concave densities are allowed and matrix and vector indexing is limited to three levels. The modelling language is close to S, hence resulting a bit obscure.

C.1.3 Bayesian special purpose packages

There are also specific packages for certain types of models or applications.
One of them is BATS (Pole, West, Harrison, 1994) for the Bayesian analysis of time series based on dynamic linear models. The environment is powerful and user-friendly, allowing for the modelling of

many time series following a decomposition based on trends, regression terms, seasonal part and autoregression terms.
Among the second ones, there is BayRes (Rios Insua et al, 1996) a package for reservoir management, following the methodology described in Rios Insua et al (1996). It essentially consists of time series analysis module, a preference modelling module and an optimising module.
We describe now, as an example, a possible implementation of a system to forecast industrial accidents, as described in Chapter 3. The models there described are clearly useful, however, its application is far from simple. Hence, it seems convenient to automate it, implementing it in a user-friendly program.
The system should ask the user to specify his beliefs by means of quantiles, by using some "non-technical" methods for assessing them, based e.g. on a betting scheme or the one described in French (1986), based on fortune wheels. Simple tests and prior knowledge (e.g. on accident proneness growth, would help the user to choose models. By means of graphs, the user is helped in interpreting predictions and refining his assessments, avoiding him all those checkings that the system could perform by itself, e.g. the local sensitivity analysis. For that we could use the following scheme:

- The *data base* contains accident histories of companies, including their description, accidents, numbers of workers, causes and consequences, and the model used. It is updated with subsequent accidents and forecasts, and other information.
- The *model base* includes models, like those described here, and tools to facilitate model selection, both to forecast the number of accidents and workers and model evolution rates. The model base is incremented once the available models do not seem adequate and another one needs to be devised.
- The *knowledge base* sums up the information available by the experts and includes a system for assessing and combining priors. Companies are classified according to sectors and sizes, to simplify prior elicitation.
- *Bayes* includes algorithms to compute predictive and posterior distributions for the above models. The module performs sensitivity analysis, with the possibility of requiring new information from the expert.
- *Predictor* is a module to display forecasts through graphics and moments.

Recall that this program would be part of a forecasting system, which would include users that could intervene when required by introducing additional subjective information. Moreover, this forecasting system could be part of other upper level systems, such as a system for claim forecasting or a decision analytic based decision support system for

aiding in some or all of the decisions suggested.

C.1.4 Bayesian expert systems

There is, as well, an abundant production of software base on Bayesian networks and influence diagrams, including BAIES, HUGIN, India or DIANA.

C.2 A software catalog

Smith (1988) provides an interesting dicussion on Bayesian software. Goel (1987) and Press (1988) provide initial catalogs, which are somewhat dated already. A more recent one is in Carlin and Louis (1996). Ours may be seen as an updated version of theirs. Before moving to it, we should mention that big software companies like Microsoft or SAS are already investing important efforts in developing Bayesian software.
What follows is a collection of software references. A short description is given and whenever possible: author, operating system or running environment and a link to a www or ftp address.

C.2.1 Bayesian packages and languages

XLISP-STAT .
Author: Tierney
OS/RE: MSWindows(16bit/32bit); Unix running X11; Macintosh
Link: http://www.stat.umn.edu/ luke/xls/xlsinfo/node1.html

A version of Lisp-Stat based on the XLISP language. Lisp-Stat is an extensible statistical computing environment for data analysis, statistical instruction and research, with an emphasis on providing a framework for exploring the use of dynamic graphical methods. Extensibility is achieved by basing Lisp-Stat on the Lisp language, in particular on a subset of Common Lisp. Lisp-Stat extends standard Lisp arithmetic operations to perform element-wise operations on lists and vectors, and adds a variety of basic statistical and linear algebra functions. A portable window system interface forms the basis of a dynamic graphics system that is designed to work identically in a number of different graphical user interface environments, such as the Macintosh operating system, the X window system, and Microsoft Windows. A prototype-based object-oriented programming system is used to implement the graphics system and to allow it to be customised and adapted. The object-oriented programming system is also used as the basis for statistical model representations, such as linear and non-linear regression models and generalised linear models. The S language motivated

many aspects of the system design. Free, well supported and well documented. Comes with some tools for approximate Bayesian analysis.

SBAYES .
Author: Tierney
OS/RE:
Link: http://lib.stat.cmu.edu/S/sbayes

S functions for approximate Bayesian analyses. Comprises similar Bayesian tools to those in LISP-STAT, for use in S. Computes approximate posterior moments and univariate marginal posterior densities via Laplace method. First order (normal) approximation and importance sampling are also available as computational methods. MCMC methods are not available. This limits its applicability to problems of moderate dimension. But it is applicable to any model for which the user can write down the log-likelihood and log-prior.

BART (Bayesian Approaches to Randomized Trials) .
Author: Spiegelhalter, Freedman Parmar
OS/RE:
Link: e-mail to author: davids@mrc.nsu.cam.ac.uk

S code for analysing approximately Normal likelihoods with some standard priors. Used for implementing a Bayesian approach to clinical trial monitoring and analysis via an indifference zone.

BATS (Bayesian Analysis of Time Series) .
Author: West, Harrison Pole
OS/RE: MSDOS; Windows3.1; Windows95
Link: ftp.stat.duke.edu/pub/bats/

Menu driven system for time series analysis and forecasting using Bayesian Dynamic modelling. It focuses on the set of dynamic linear and non-linear models, including Kalman filter models. It was originally developed in APL at the University of Warwick and described in the paper by West, Harrison and Pole (1987). Today, a substantially enhanced version of it forms the computing basis for the popular Bayesian forecasting and time series book by Pole, West and Harrison (1994).

[B/D] (Beliefs adjusted by Data) .
Author: Goldstein, Wooff
OS/RE: MSWindows; Unix running X11
Link: http://fourier.dur.ac.uk:8000/stats/bd/

[B/D] is an interactive programming language which allows complete a priori and diagnostic analyses of Bayes linear statistical problems. Bayes linear problems are subjective statistical analyses based on expectation and covariance structures, rather than on distributional assumptions. It provides an environment in which we quickly and easily specify beliefs about quantities of interest, attach data to some or all of those quantities, and carry out the general process of Bayes linear adjustment. There is a great deal of information available about the adjustments, and diagnostics for conflicts between the data and your prior beliefs. It is also possible to produce interactive Bayes linear influence diagrams for the adjustments, providing simple graphical summaries of the adjustments and accompanying diagnostics. Contains both programming and some statistical package elements. It is primarily a tool for organising beliefs, rather than a statistical package for doing standard sort of analyses.

BUGS (Bayesian inference Using Gibbs Sampling) .
Author: Spiegelhalter, Thomas Gilks
OS/RE: MSDOS; MSWindows; Unix
Link: http://www.mrc-bsu.cam.ac.uk/bugs/Welcome.html

BUGS is a piece of computer software for the Bayesian analysis of complex statistical models using Markov chain Monte Carlo (MCMC) methods. The Classic BUGS program uses text-based model description and a command-line interface, and versions are available for major computer platforms. A prototype Windows version, WinBUGS, has an option of a graphical user interface. The programs are reasonably easy to use and come with a range of examples. BUGS lets you write Splus like code to define complex hierarchical models. The program then determines the full conditional distributions necessary for the Gibbs algorithm. The only input required is the model specification, the Gibbs sampler implementation is automatic, but it should be used with caution.

CODA (Convergence Diagnosis and Output Analysis) .
Author: Nicky Best et al
OS/RE: S-Plus; R
Link: http://www.mrc-bsu.cam.ac.uk/bugs/Welcome.html

This set of S-plus functions contains a wide range of diagnostics and plots for presenting the results of an MCMC analysis in general, and BUGS output in particular, and checking convergence of the simulation. The program enables various summary statistics, sample traces and density plots, sample autocorrelations (within a chain for one parameter) and crosscorrelations (across one or more chains for two pa-

rameters) and other popular statistics and plot convergence diagnosis (Gelman and Rubin, 1992; Geweke, 1992; Raftery and Lewis, 1992)

BAYES FOUR .
Author: Naylor, Shaw
OS/RE: Fortran-77; C/C++ in Unix
Link: e-mail to author: j.naylor@maths.ntu.ac.uk

Library of Fortran-77 functions for numerical integration and importance sampling. User supplies Fortran code for a likelihood and optionally specifies a prior, parameter transformations, desired posterior expectations, etc. The program may compute posterior moments as well as expectations of parameters of interest. One- and two- dimensional marginal posterior and predictive densities are also available numerically and as graphical displays. The program uses numerical quadrature for problems of dimension up to 8, depending on the regularity of the posterior surface and noninterative Monte Carlo methods for dimensions beyond that (typically up to about 20). A user written C/Fortran function which evaluates the posterior density pointwise is compiled with the Bayes-Four library.

BINGO (Bayesian Inference: Numerical, Graphical Other) .
Author: Shaw
OS/RE: APL
Link: http://www.warwick.ac.uk/statsdept/Staff/JEHS/

This program is an extended version of BAYES FOUR. BINGO features a multi-kernel version of the Naylor and Smith(1982) algorithm, highly efficient numerical integration rules, device-independent graphics and facilities for other approaches, including built-in reparametrizations, Monte Carlo importance sampling and the Metropolis-Hastings algorithm.

FIRST BAYES .
Author: O'Hagan
OS/RE: MSWindows
Link: http://www.maths.nott.ac.uk/personal/aoh/1b.html

First Bayes is a teaching package for elementary Bayesian Statistics. Particular features include: plotting and summarization of thirteen standard distributional families and mixtures of these families; analysis of one-way ANOVA and simple linear regression models; automatic updating of posterior and predictive summaries following user-made changes to the data or prior distributions; sequential analysis.

MCSim .
Author: Bois et al.
OS/RE: Macintosh; Unix
Link: ftp://sparky.berkeley.edu/pub/mcsim

MCSim is a simulation package that allows you to design your own statistical or simulation models (eventually dynamic, via ODEs) and perform Bayesian inference through Markov Chain Monte Carlo simulations. The only fully implemented likelihood function is the log-normal, but other choices may be employed via a user supplied C subroutine.

XBAIES (Bayesian Analysis In Expert Systems) .
Author: Cowell
OS/RE: PC/Alpha/HP/Dec/Sparc
Link: ftp://assun3.city.ac.uk/pub/xbaies1.3/

XBAIES is an expert system shell. It includes several advanced modelling features, such as chain graphs utilities and different algorithms for learning about parameters, Monte Carlo propagation, but lacks defining belief functions.

BELIEF / GRAPHICAL-BELIEF .
Author: Almond
OS/RE: Common Lisp, CLtL 2
Link: http://www.stat.washington.edu/almond/belief.htmlBELIEF

BELIEF allows defining belief functions and parameter uncertainty. Its commercial version GRAPHICAL- BELIEF also allows knowledge based models and has a graphical user interface. Almond mantains a comprehensive list of resources for belief networks at
http://www.stat.washington.edu/almond/belief.html

MBSN (Windows95 NT) (Microsoft) .
Author: Microsoft
OS/RE: MSWindows
Link: http://www.research.microsoft.com/research/dtg/msbn/

MSBN allows the creation, assessment and evaluation of Bayesian belief networks.

HUGIN .
Author:
OS/RE: MSWindows; SunOS; Solaris

Link: http://hugin.dk

The HUGIN System is a tool enabling you to construct model based decision support systems in domains characterized by inherent uncertainty. The models supported are Bayesian belief networks and their extension influence diagrams. The HUGIN System allows you to define both discrete nodes and to some extent continuous nodes in your models.

Bayesian Knowledge Discoverer .
Author: Marco Ramoni
OS/RE: MS Windows 95, NT
Link: http://kmi.open.ac.uk/ marco

A program to learn Bayesian Networks from incomplete databases. Related to the BAKE project.

C.2.2 Function repositories

There is a main resource for functions to perform Bayesian inference written in Splus, XLisp-Stat, ... (http://lib.stat.cmu.edu/). Some examples of Splus functions are:

- BICREG. Bayesian model selection and accounting for model uncertainty in linear regression models, using the BIC approximation.
- BIC.LOGIT. Logistic regression with Bayesian model selection and accounting for model uncertainty, using the BIC approximation.
- GLIB. Bayesian generalized linear modelling, model comparison and accounting for model uncertainty.

Many of them may be found in http://lib.stat.cmu.edu/S/. XLisp-Stat specific functions can be found in http://lib.stat.cmu.edu/xlispstat/. Several functions in Fortran and Splus can also be found in http://www.epi.mcgill.ca/ web2/Joseph/software.html

Appendix D

Presentation of BayOff to the International Society for Bayesian Analysis.

The ISBA meeting Web page
http://www.ntua.gr/ISBA2000/new/Fin_sessions_1.html
http://www.ntua.gr/ISBA2000/new/Fin_sessions_1.html
includes the following information about the BayOff Project:
Bayesian Analysis in Official Statistics:
The EUROSTAT "Bayoff" Project
Dellaportas, P.[1], Rios-Insua, D.[2], and Cervera, J. L.[2]*
[1]Department of Statistics, Athens University of Economics and Business, Greece and
[2] Department of Statistics, University Rey Juan Carlos, Spain INE, Spain
BayOff is an exploratory project supported by EUROSTAT to study the feasibility of the application of Bayesian methods to the analysis of official statistics. We describe the development and main findings of the project, providing a strategy for the adoption of Bayesian methods within NSIS
Keywords: Official Statistics, Bayesian Methods.
The slides used are herewith enclosed.

A feasibility study on the application of Bayesian methods to Official Statistics

Rey Juan Carlos University, Spain
National Statistical Institute (INE), Spain
Athens University of Economics and Business, Greece

BAYOFF

The BayOff project

- EU Financed (Eurostat) under SUP.COM
- Research team:
  - Academic Statisticians (URJC, AUEB)
  - Official Statisticians (INE–Spain)
- Objective:
  - ***a prospective study on the feasibility of applying Bayesian methods and techniques to Official Statistics***

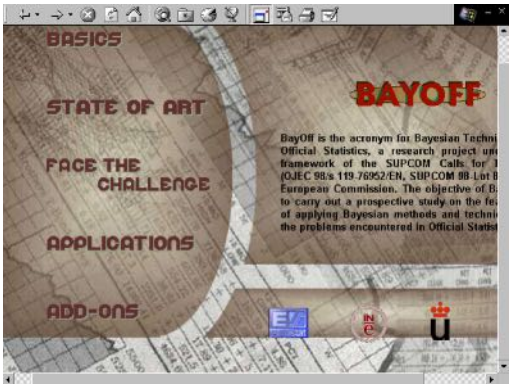
Project planning: Stage 1

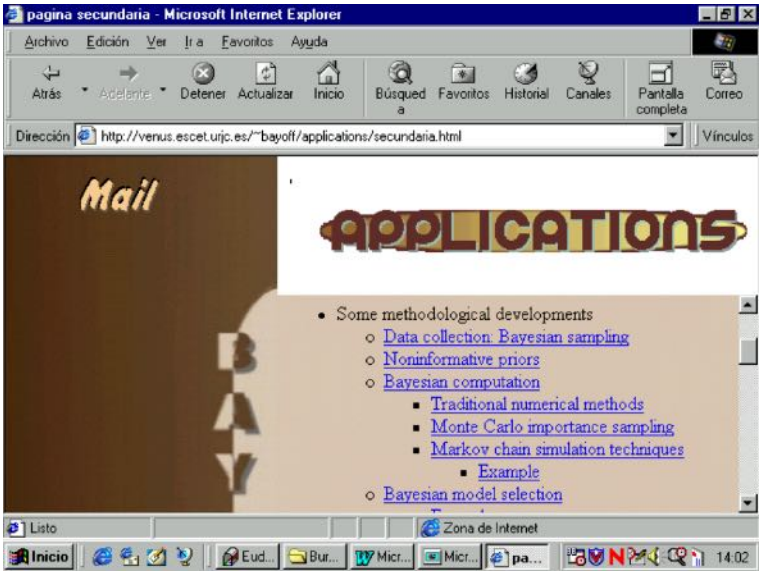
- Research on recent references in journals and proceedings
- Contacts with leading Bayesian statisticians
- Identification of interesting applications in the field of practical data analysis
- Identification of existing software
- Set up of a Web site on the project

Project planning: Stage 2:

- Contacts with NSIs and Eurostat
- Identification of possible applications in NSIs
- Assessment of willingness of NSIs and Eurostat to apply Bayesian techniques
- Planning a future strategy to apply this work to Official statistics

Web page:
http://venus.escet.urjc.es/~bayoff/





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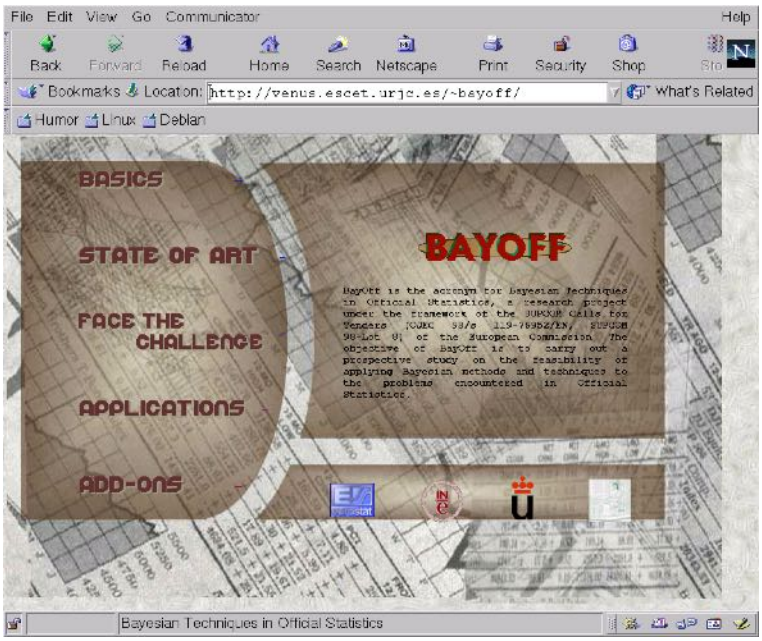
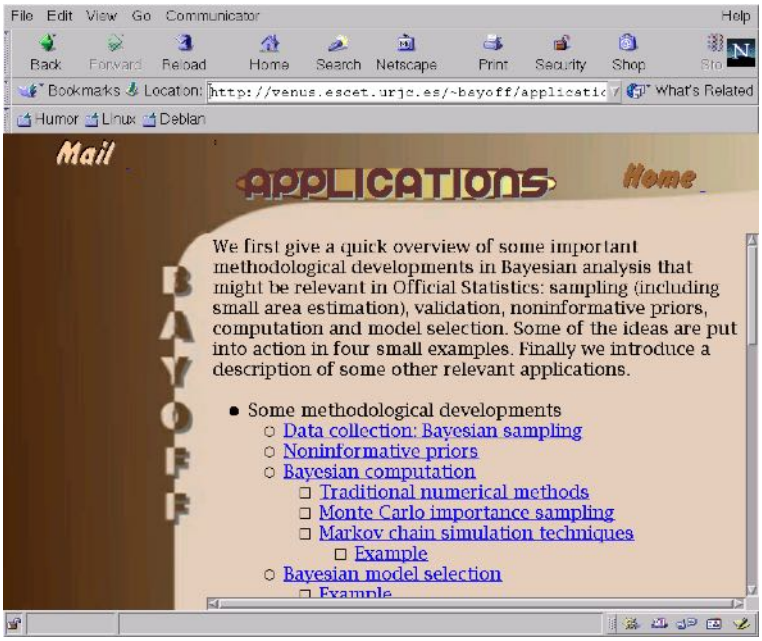
Vínculos
Mail
ADD-ONS
system or running environment and a link to a www or ftp address.
Bayesian packages and languages
XLISP-STAT
SBAYES
BART (Bayesian Approaches to Randomized Trials)
BATS (Bayesian Analysis of Time Series)
Zona de Internet
Inicio
14:04

Appendix E

Web site of the BayOff project

The URL address of the BayOff project is
http://venus.escet.urjc.es/~bayoff
http://venus.escet.urjc.es/~bayoff
Selected pages are included.




File Edit View Go Communicator Help
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Bookmarks Location: http://venus.escet.urjc.es/~bayoff/applicatio What's Related
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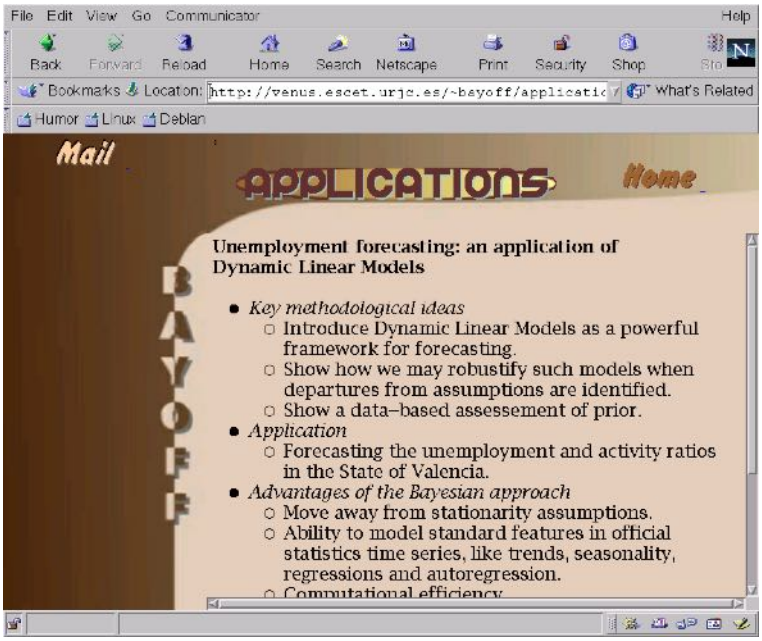
BAYOFF
We first give a quick overview of some important methodological developments in Bayesian analysis that might be relevant in Official Statistics: sampling (including small area estimation), validation, noninformative priors, computation and model selection. Some of the ideas are put into action in four small examples. Finally we introduce a description of some other relevant applications.
Some methodological developments
Data collection: Bayesian sampling
Noninformative priors
Bayesian computation
Traditional numerical methods
Monte Carlo importance sampling
Markov chain simulation techniques
Example
Bayesian model selection


File Edit View Go Communicator Help
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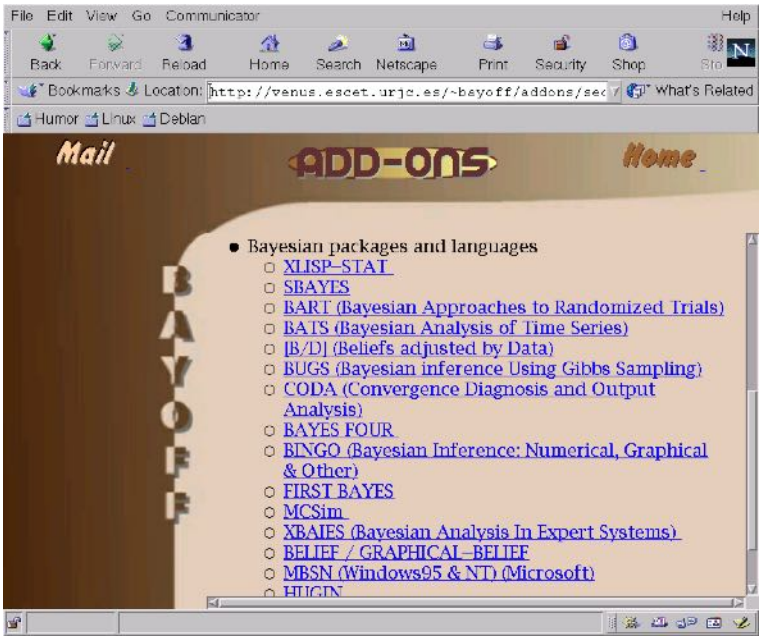
Bookmarks Location: http://venus.escet.urjc.es/~bayoff/applicatio What's Related
Humor Linux Debian
Mail
APPLICATIONS
Home
BAYOFF
Unemployment forecasting: an application of Dynamic Linear Models
Key methodological ideas
Introduce Dynamic Linear Models as a powerful framework for forecasting.
Show how we may robustify such models when departures from assumptions are identified.
Show a data–based assessement of prior.
Application
Forecasting the unemployment and activity ratios in the State of Valencia.
Advantages of the Bayesian approach
Move away from stationarity assumptions.
Ability to model standard features in official statistics time series, like trends, seasonality, regressions and autoregression.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What's Related
Humor Linux Debian
Mail
ADD-ONS
Home
BAYOFF
Bayesian packages and languages
XLISP-STAT
SBAYES
BART (Bayesian Approaches to Randomized Trials)
BATS (Bayesian Analysis of Time Series)
[B/D] (Beliefs adjusted by Data)
BUGS (Bayesian inference Using Gibbs Sampling)
CODA (Convergence Diagnosis and Output Analysis)
BAYES FOUR
BINGO (Bayesian Inference: Numerical, Graphical & Other)
FIRST BAYES
MCSim
XBAIES (Bayesian Analysis In Expert Systems)
BELIEF / GRAPHICAL-BELIEF
MBSN (Windows95 & NT) (Microsoft)

Appendix F

A review on some concrete developments

In this Appendix we provide some possible projects aimed at developing novel Bayesian methods of potential use within Official Statistics. We give a fairly detailed account of the first one (thanks to ideas from Prof. Giudici from Università di Pavia), whereas for the others only a shorter description is provided.

F.1 Data Mining techniques for the construction and interpretation of knowledge in very large data bases (VLDBs)

EPROS Section: Data Analysis and Knowledge Extraction (4.1)

Introduction Traditional data bases in the past decades are designed according to the Relational Model where the data are organised in tables and columns. Recently there has been a revolution in techniques for transaction processing and DBMS gave access to the data stored, the main features being:
–The data explosion: The continuous and successful operation of such systems in the last decades in several domains (such as banking, transactions, administration, statistical information systems) has resulted in tremendous masses of data that are residing in data warehouses. The data explosion issue has been facilitated by the increase in use of electronic data gathering devices e.g. point-of-sale, remote sensing devices etc. while Data storage became easier and cheaper with increasing computing power
–The On Line transaction Processing Systems (OLTPs). which are

good at putting data into databases but are poor in extracting interesting knowledge extraction.
Problems that have come up are:
–Data Base Management Systems gave access to the data stored but provided very limited capabilities of analysing the information.
–Analysis required to unearth the hidden relationships within the data i.e. for decision support
–Size of databases has increased e.g. VLDBs, need automated techniques for analysis as they have grown beyond manual extraction
The aforementioned issues meet the following obstacles in their solutions:
–the typical scientific user knew nothing of commercial business applications
–the business database programmers, knew nothing of massively parallel principles necessary to handle huge amounts of data.

Data Mining techniques Several definitions of the term " data mining" have been submitted. According to W.J Frawley, G. Piatetsky-Shapiro, C. J. Matheus data mining is " the non trivial extraction of implicit, previously unknown, and potentially useful information from data" .
Data mining is the analysis of data and the use of software techniques for finding patterns and regularities in sets of data. The computer is responsible for finding the patterns by identifying the underlying rules and features in the data. It is possible to 'strike gold' in unexpected places as the data mining software extracts patterns not previously discernible or so obvious that no-one has noticed them before.
In the data mining process there is a suite of techniques used aiming at extracting different 'knowledge artefacts'. A number of different technical approaches are encompassed, such as: clustering, classification, data summarisation, learning classification rules, analysing changes, and detecting anomalies.
The overall need that data mining aims at fulfilling needs of businesses looking for new ways to let end users find the data they need to: make decisions, serve customers and gain the competitive edge

Proposed Research Issues The Knowledge Discovery (KD) process aims at searching for interesting instances of patterns in data sets. A widely recognised requirement is that the patterns discovered must be valid and ultimately comprehensible. Another requirement addressed in KD process is the reveal and usage of uncertainty in the data mining tasks, i.e. in the clustering and classification processes and association rules extraction. In the context of the proposed project we aim at working on the reveal and handling of uncertainty in the data mining process using statistical data of substantial volume.

The issues that motivate our research are related to clustering, classification and association rules extraction.
More specifically the issues that motivate our research in this context are
The clusters are not overlapping.
The limits of clusters are crisp and each database value may be classified into at most one cluster. The result is that, in some cases "interesting" individuals fall out of the cluster limits so they are not be classified at all. This is unlikely to everyday life experience where a value may be classified to more than one categories. For instance a male person with height 182cm in Central Europe is considered as of " medium" height as well as " tall" to some degree.
The data values are treated equally in the classification process.
In traditional data mining systems database values are classified in the available categories in a crisp manner, i.e. a value either belongs to a category or not. The person with height 180cm is considered as tall and also another person with height 199cm is also considered tall. It is profound that the second person satisfies to a higher degree, than the first, the criterion of being " tall" . This piece of knowledge (the difference of belief that A is tall and also B is tall) cannot be acquired using the existing classification schemes.
Classification results may hide "useful" knowledge for our data set.
The most of classification methods define a model that is used to classify new instances to predefined categories. This assignment of data values in categories hides significant knowledge related to the information that is included in each category or in whole data set. As a consequence, we can exploit this knowledge for decision making as well as for the evaluation of our classification model. However, the most of traditional classification approaches don't introduce any well-established procedure for evaluation and reasoning related to information that is produced by classification process.

Objectives and results As it is clear from the above brief analysis, there is interesting knowledge that is not captured, due to the fact that uncertainty is not considered during data mining process as well as due to the ignorance of the information that is included in classification results.
Reveal and handling knowledge though the data mining process is a task we will accomplish within the proposed project. The statistical data of southern Europe countries, as the target data set of our study, will give a wide and sound requirements platform for i. revealing useful and comprehensible knowledge and ii. a comparative study of the results of the data mining process against the other methodologies that will employed in the proposed project.
More specifically we expect to have interesting results regarding the

statistical data of Southern Europe countries, in the form of:
Valid multidimensional clusters
In this project one of our goals is to define a clustering extraction scheme so as to support uncertainty. More specifically, we aim at working for an approach for clustering schemes evaluation. The schemes will be produced by well established clustering methodologies. The approach will select the best of them (i.e. the scheme with compact and well separated clusters) according to quality measures we define. Consequently, in order to handle uncertainty, mapping functions based on fuzzy logic are appropriately assigned to the clusters.
Flexible Classification schemes
A classification scheme will be defined that assigns non-categorical data to pre-specified categories maintaining the classification belief. We will use fuzzy logic in order to represent and manipulate this belief. Quality measures for the evaluation of above defined classification scheme based on fuzzy logic concepts. We can exploit classification belief based on these measures in order to support decision-making.
Association Rules
Based on the previous schemes, we will define an association rule extraction scheme that handles uncertainty.
Visualisation
We will employ existing methods of visualising multidimensional and large volume information in order to make the results of the previous steps more comprehensible to the end-users and decision makers.

F.2 Bayesian techniques for integrating data from different sources

EPROS Section: Data multi-sources integration and systemisation (5)

(Research proposal suggested by Mauro SCANU, ISTAT)
The following are two major issues with respect to record-linkage:
1))to match the same people between census and post enumeration survey data, for dual-system estimation of the census coverage
2))to integrate two or more administrative archives for creating a framework for enterprises surveys.
The Fellegi-Sunter methodology has been already implemented in ISTAT in the past years. Recently techniques based on the EM algorithm have been studied with the purpose of applying them to the census coverage estimation. Apart of the used linkage methodology, a very important issue is the assessment of the false match rates. In this context Belin and Rubin (1995) use transformed normal mixture models and identify " a full strategy" that can be considered a Bayesian approach.

It seems interesting to us to verify the possibility of application of either MCMC methodology or Bayesian mixture models both in linking records and estimating the false match rate.

F.3 Visualisation of information

EPROS Section: Innovative User-Interface and Visualisation Techniques (6.4)

Research is still needed to produce powerful, intuitive and economical visualisation techniques for navigation through large multidimensional data sets, such as those that will be available after the 2000-2001 Round of Censuses.
Self-organising maps (SOM) – such as the Kohonen one - are becoming increasingly appreciated for this task (particularly in connection with various projects related to WWW document organization and retrieval). Indeed, the standard SOM has been shown to provide a wealth of information with regard to both the cluster analysis and outlier detection goals.
While the trained SOM structure can not be inspected directly, Sammon's mapping (1969) has proved invaluable for sensible 2-D projection. Interactive user-friendly interfaces are beginning to flourish.
Overall, much remains to be done about progressive refinement of the SOM information display via user-controlled, 2-D or 3-D projections of the map.
Another area of research refers to the dynamic shaping of the map. This idea was originally implemented via ad-hoc methods. Again, reversible-jump Bayesian Markov chain Monte Carlo (BMCMC) methods (Green, 1995) provide a principled alternative worth examining in detail.

F.4 Bayesian techniques for typology identification

EPROS Section: Data Analysis and Knowledge Extraction (4.1)

Once again, the large data sets obtained for the Round of Censuses will give the opportunity to research into innovative methods of automatic knowledge extraction. In particular, problems such as finding typologies of households or dwellings will be a test.
Research is ongoing about the BYPASS system and its ability to uncover general patterns in classification problems with many explanatory variables. This system follows a population-based, evolutionary approach to inference and can be seen as a flexible alternative to stan-

dard recursive-splitting (or tree-oriented) methods. BYPASS rules are indeed comparable to terminal nodes in classification trees, but they may overlap. Overlapping rules obviously provide the opportunity for co-operation, which is particularly attractive when one wishes to focus on general patterns and their associated, relatively flat predictive distributions. It is thus interesting to study the extent to which this co-operative overlapping may signify an improvement, in large-scale mining tasks, over conventional trees fitted traditionally or according to Bayesian principles.

F.5 Bayesian techniques for missing data imputation

EPROS Section: Nascent Techniques (4.2)

Missing data imputation is a classical problem in Official Statistics, being one of the main activities of the Data Editing phase. The problem of assigning a value for an unknown variable (due to non response, for instance) can use nascent techniques like Neural Networks (Statistics Finland has tried them occasionally).
A more unified mathematical framework for neural networks has been declared a research priority. Bayesian modelling has proved extremely useful in augmenting the available body of fitting techniques for standard feed-forward nets. Furthermore, Bayesian methods have opened the way for (1) rigorous treatment of delicate design issues, and (2) new probabilistic approaches where only heuristic fitting algorithms were available.
As regards (1), Bayesian modelling has provided a principled way to tackle inference about the number of units in the hidden layer of a feed-forward net. Traditionally, this matter is decided upon by trial-and-error. While the underlying posterior distributions are rather intractable, successful inference is accomplished via Bayesian Markov chain Monte Carlo (BMCMC) methods.
Point (2) refers by and large to the self-organising map (SOM) neural architecture, where some authors have advocated the use of mixtures of multivariate normal distributions as sampling distributions for the SOM model.
These ideas have a huge potential since they lead naturally to alternative fitting algorithms; in contrast, the standard SOM algorithm is very fast but well-known for escaping rigorous mathematical analysis. Indeed, we know intuitively what this algorithm does, yet it is hard to pinpoint exactly what defines the state-of-affairs in the resulting structure. Therefore, the idea of characterising that state-of-affairs as, say, a local posterior mode arising from a suitable prior distribution is a very appealing one.

It remains to be seen whether these advantages apply also to alternative neural models. For example, BMCMC methods may be put to work in Kohonen nets, whereas the Fuzzy ARTMAP architecture may be amenable to Bayesian modelling as well.

F.6 Further Developments

The following lines summarise additional areas which require Bayesian developments and might contribute to its wider acceptance within NSIs. Indeed, we could argue that we could set up a research program for each important official statistics problem, by comparing current non-Bayesian solutions with Bayesian ones:

- *Software development.* There is a clear need to develope suitable software allowing routine implementation of Bayesian methods, much as classical methods are implemented. This would involve creating default Bayesian procedures for the large variety of situations where classical procedures now exist. On the other hand, software to fully support a Bayesian analysis with proper priors would be needed as well. Appendix C outlines Bayesian software available.
- *Prior elicitation.* Much more work is required to determine ways to elicit prior information, perhaps through some interactive assessment methods. When no such information is available, we turn to noninformative default priors, and more generic derivation of such priors is required, specially for hierarchical and nonlinear models.
- *Model diagnosis, checking and choice.* This is an expandinga area of research. However, many competing methods have appeared and there is a clear need to develop theory and compare those methods.
- *Tools for sensitivity analysis.* There is a need to provide better methods to check sensitivity to the prior, model and loss, in particular how to communicate and present the results of a sensitivity analysis, and, as a consequence, refine a Bayesian analysis.
- *Bayesian methods for large data sets.* This is of special interest for Official Statistics. Topics include development of multiscale Bayesian methods capable of dealing with huge amounts of data, including Bayesian Data mining and Bayesian methods for neural netowrks: the analysis of missing data in large databases by using multiple imputation, detection of outliers and robust methods appropriate in large databases.
- *Bayesian computation.* In spite of great developments, there is still important work to do in computing, especially to learn when chains are not necessary and test whether an MCMC has converged.

– *Sampling and Design.* There is a need to further develop Bayesian methods in these areas.

Appendix G

Bibliography

Here we provide the results of the literature search on Bayesian Analysis and Official Statistics, as described in the main text. Entries are grouped according to the NORIS terminology.